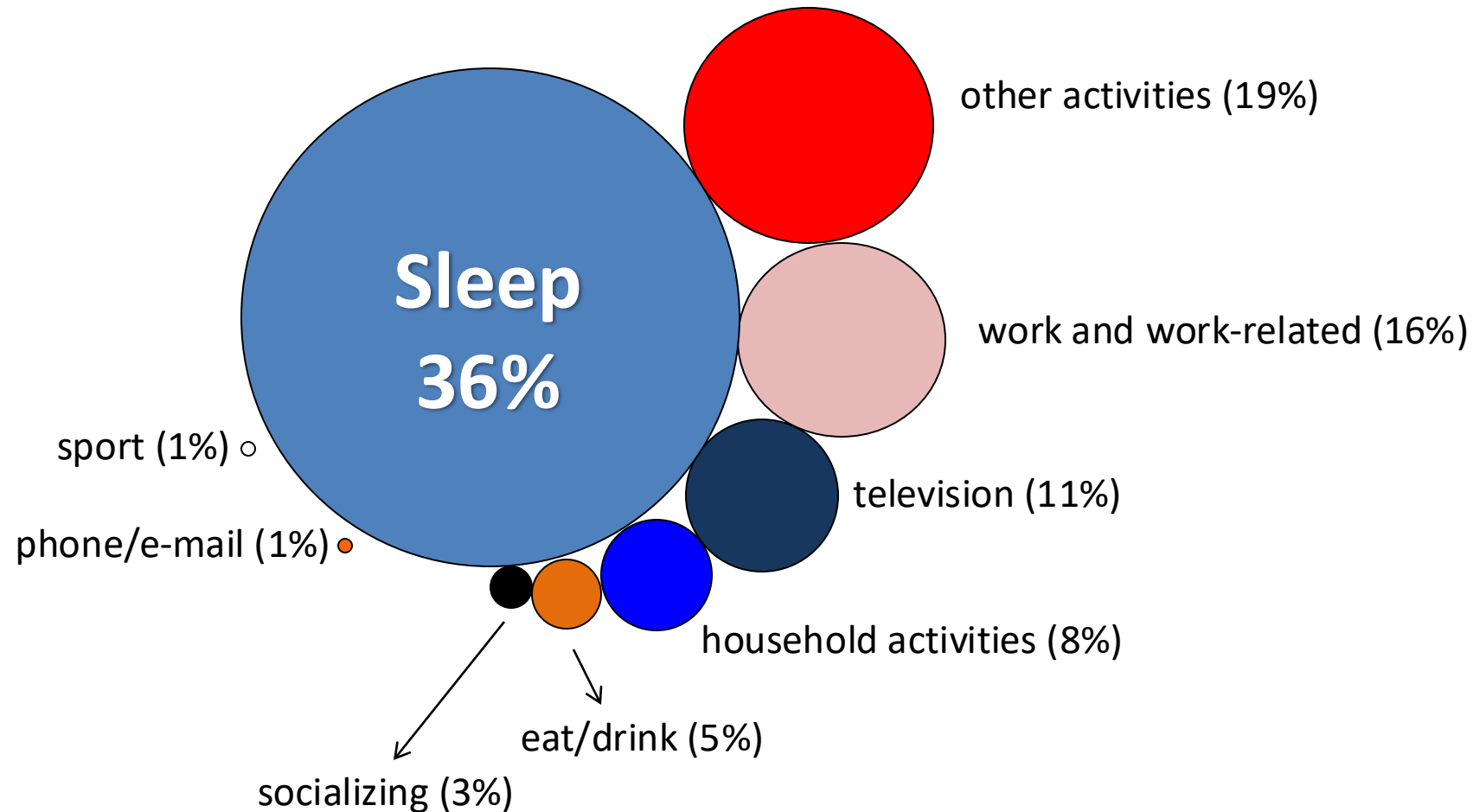


Vigilance and alertness



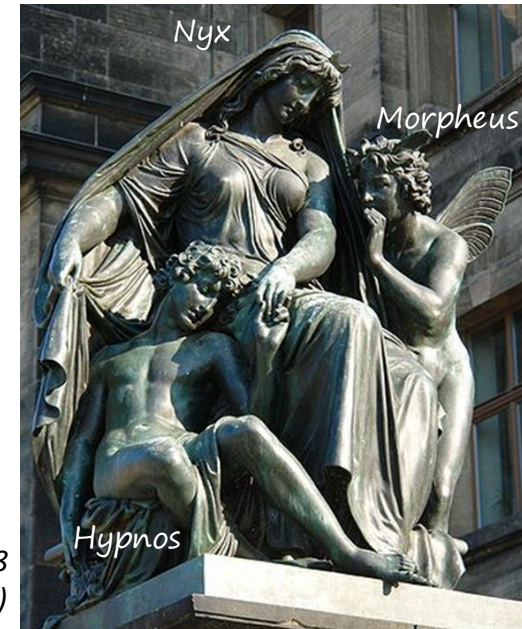
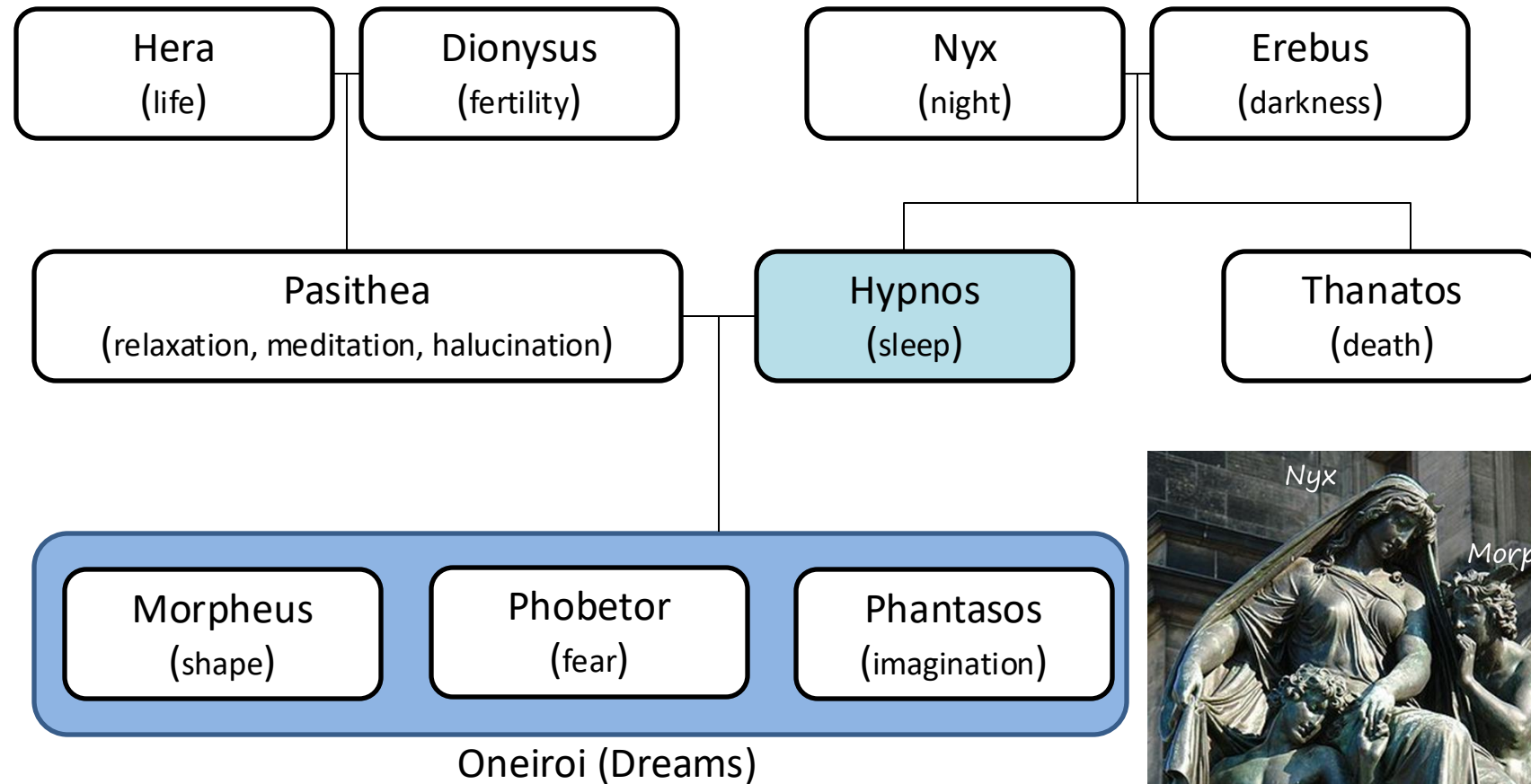


Sleep



By the age of 80, a person will have spent more than 25 years sleeping.

Hypnos and Thanatos



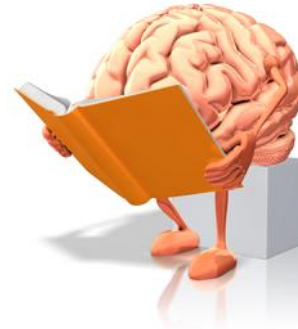
Nyx, Johannes Schilling, 1868
(Dresden, Germany)



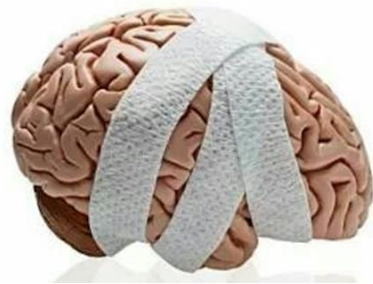
The functions of sleep...



Metabolite Clearance



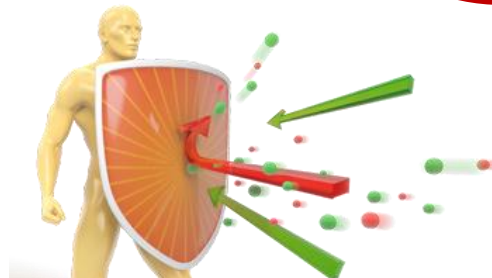
Memory processing



Tissue repair



Metabolism



Immune System

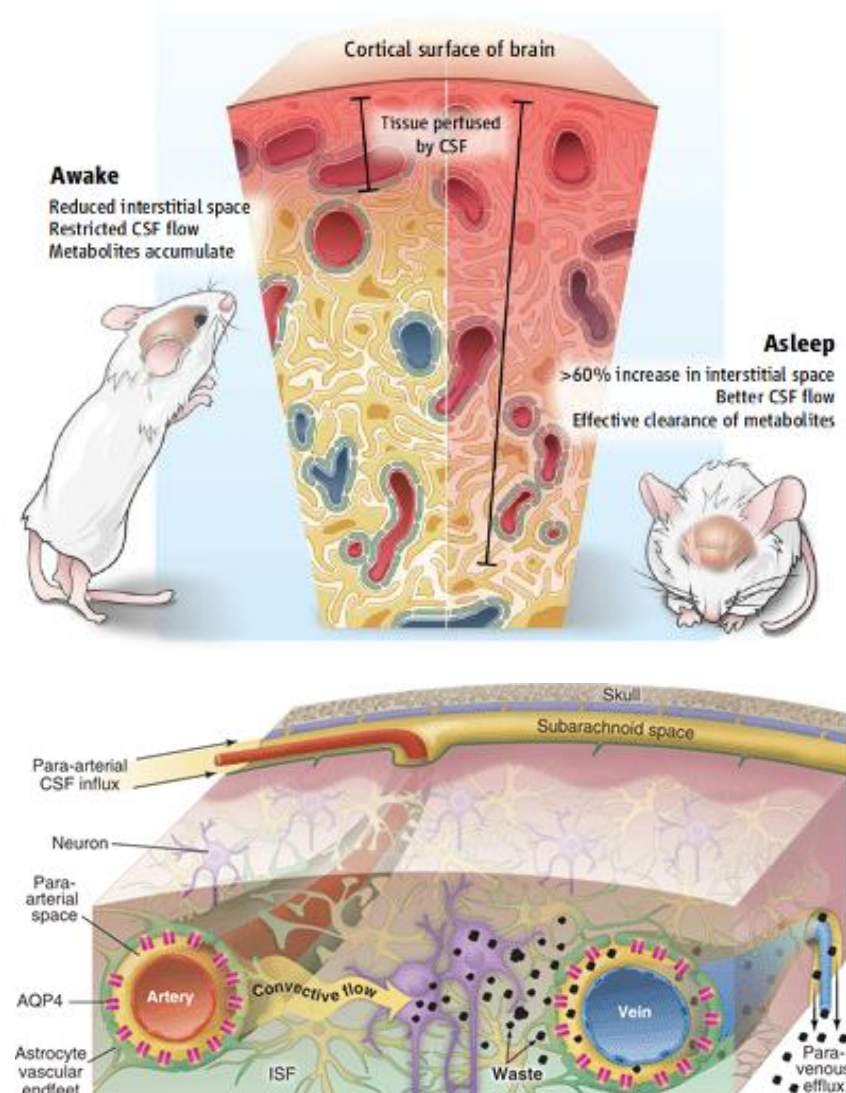


Mood

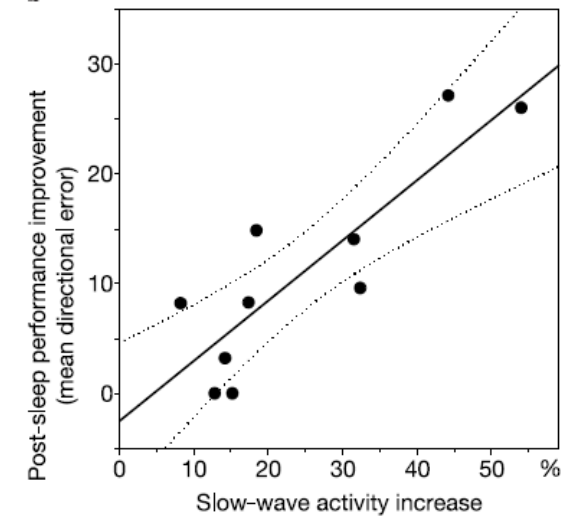
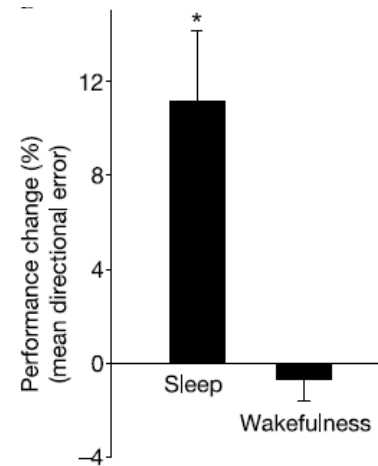
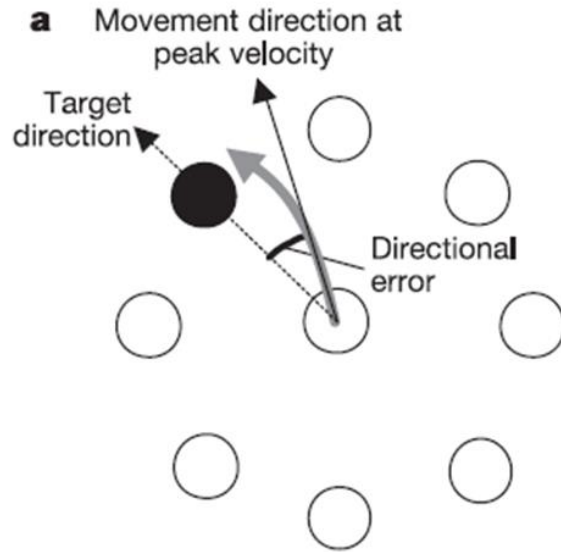
Functions of Sleep – Clearance of Metabolites

The **glymphatic system** is a functional waste clearance pathway that consists of a para-arterial influx route for cerebrospinal fluid (CSF) to enter the brain parenchyma, coupled to a clearance mechanism for the removal of interstitial fluid (ISF) and extracellular solutes from the interstitial compartments of the brain and spinal cord.

Exchange of solutes between the CSF and the ISF is driven by arterial pulsation and regulated during sleep by the expansion and contraction of brain extracellular space.

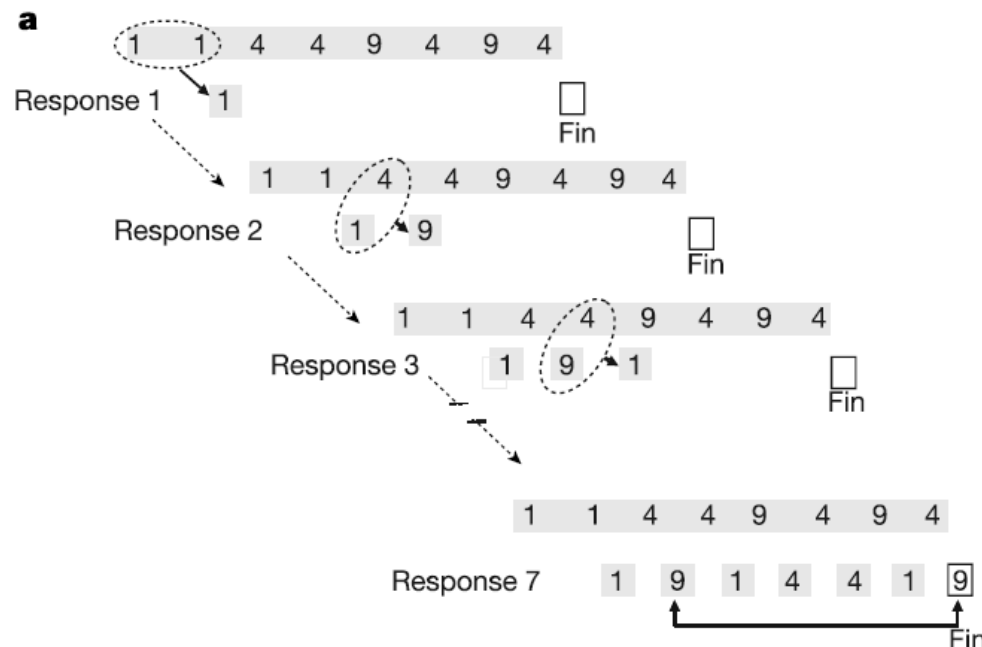


Functions of Sleep – Memory and Learning



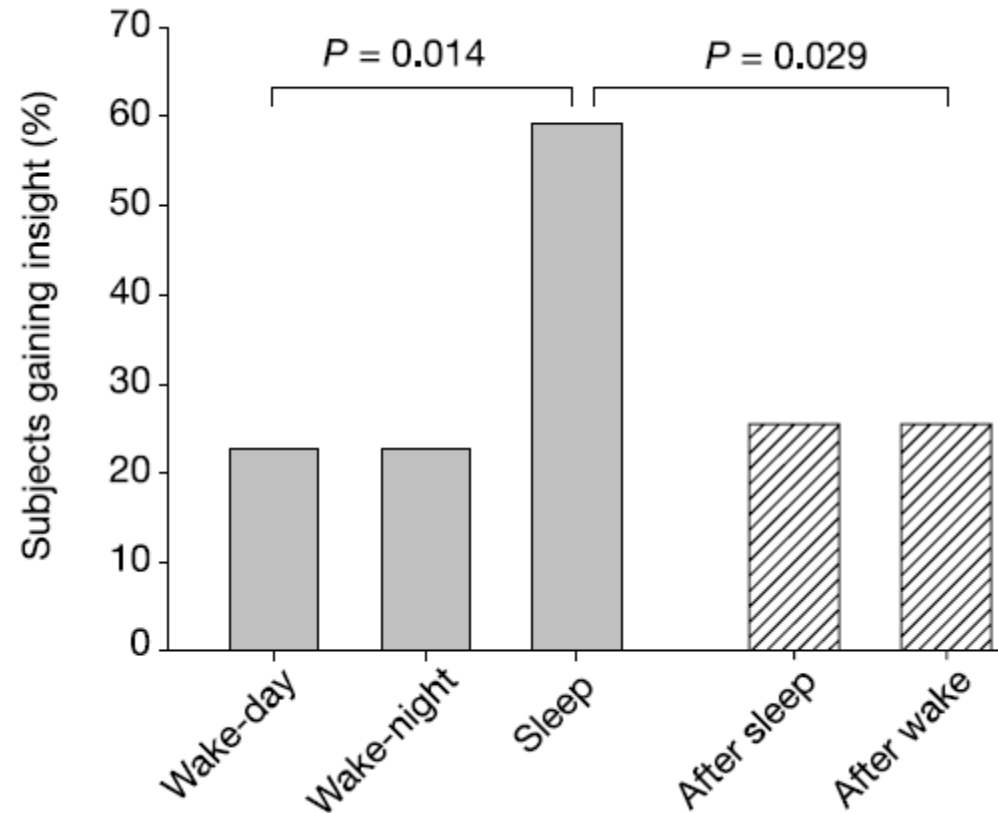
Memory processes have been shown to be stabilized and enhanced by nocturnal sleep and even daytime naps. Certain sleep stages have been suggested to improve specific subtypes of memory. Generally, declarative memories are enhanced by slow-wave sleep, while non-declarative memories are enhanced by rapid eye movement (REM) sleep (there are inconsistencies among experimental results!).

Functions of Sleep – Learning and Insight



Number Reduction Task (NRT). On each trial, a different string of eight digits was presented. Each string was composed of the digits '1', '4', and '9'. For each string, subjects had to determine a digit defined as the 'final solution' of the task trial (Fin). This could be achieved by sequentially processing the digits pairwise from left to right according to two simple rules. The seventh response indicates the final solution. Not mentioned to the subjects, the second response coincided with the final solution. Subjects who gain insight into this hidden rule abruptly cut short sequential responding by pressing the solution key immediately after the second response.

Functions of Sleep – Learning and Insight



Effects of sleep and wakefulness on the occurrence of **insight**. Columns indicate percentage of subjects gaining insight into the hidden rule in the three experimental conditions of the main experiment.

Functions of Sleep – Physical Health

Sleep is essential for the healing and repair of cells and also helps to maintain the balance of hormones in the body, such as ghrelin and leptin, which regulate feelings of hunger and fullness. Other hormones such as insulin, also change during sleep deprivation, leading to an increase in blood sugar level.



Chronic sleep deficiency is linked to a higher risk of **obesity, cardiovascular disease, stroke, diabetes** and **kidney disease**.

Sleep is necessary for normal brain functioning



Randy Gardner (born 1947) is the holder of the scientifically documented record for the longest a human has intentionally gone without sleep not using stimulants of any kind.

In 1964, Gardner (at the time high school student in San Diego, California) stayed awake for 264.4 hours (**11 days 24 minutes**).

The SD led to serious cognitive and behavioral changes:

- Alterations of mood,
- Problems with concentration,
- Problems with short term memory,
- Paranoia,
- Hallucinations.

Sleep is necessary for normal brain functioning



On the eleventh day, when he was asked to subtract seven repeatedly, starting with 100, he stopped at 65. When asked why he had stopped, he replied that he had forgotten what he was doing.

Sleep is necessary for ~~normal brain functioning~~ LIFE



Marie de Menaceine, 1843-1903

First Experimental Study Of SD (1894): 10 puppies, 2-4 months old “irreparably lost” after SD Of 96-120 h (4-5 days). Older dogs were more resistant.

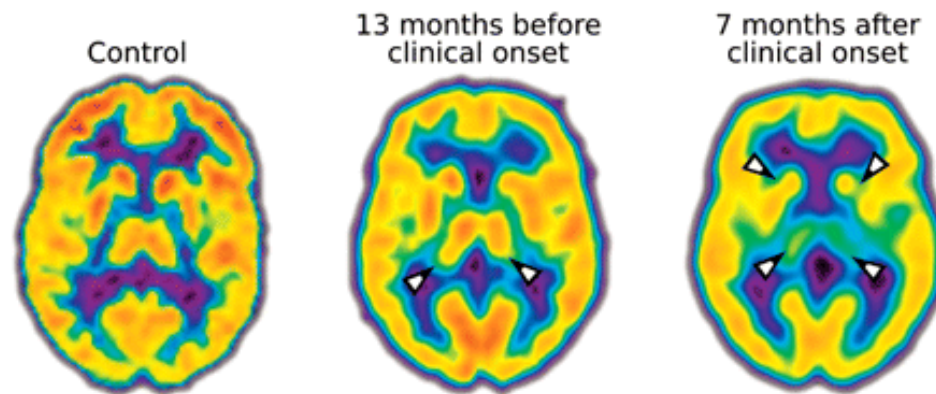
“Fat degeneration” In many brain “ganglia” (histological study, only macroscopic examination). “The brain was the site of predilection of the most severe and most irreparable changes” (different from starvation, where the brain is spared)

Sleep is necessary for ~~normal brain functioning~~ LIFE

Fatal familial insomnia (FFI) is an extremely rare **autosomal dominant** inherited prion disease of the brain. It is usually caused by a mutation to the protein PrPC, but can also develop spontaneously in patients with a non-inherited mutation variant called **sporadic fatal insomnia** (sFI).

FFI has no known cure and involves progressively worsening **insomnia**, which leads to **hallucinations**, **delirium**, **confusional states**, and eventually, **death**.

The average survival time for patients diagnosed with FFI after the onset of symptoms is **18 months**.



Sleep is necessary for ~~normal brain functioning~~ LIFE



2005: "A 28 year old man in Korea has died after sitting in an Internet cafe for nearly 50 hours straight, playing Starcraft online."

2012: "A Chinese football fan died after going 11 nights without sleep as he tried to watch every single kick of the ball of Euro 2012."



PEANUTS



Sleep is universal



Sleep is universal



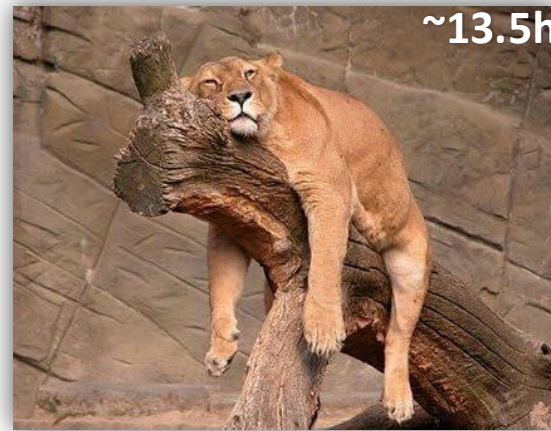
Sleep is universal



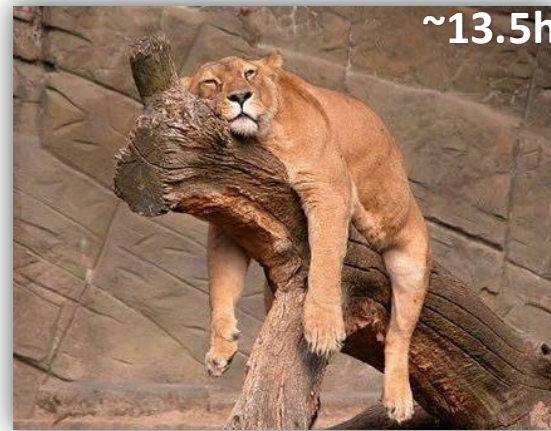
Sleep is universal



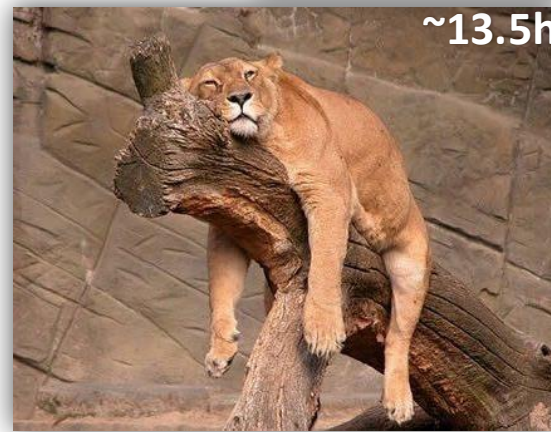
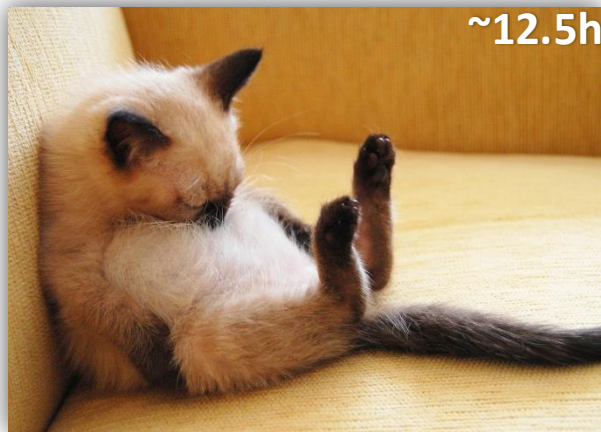
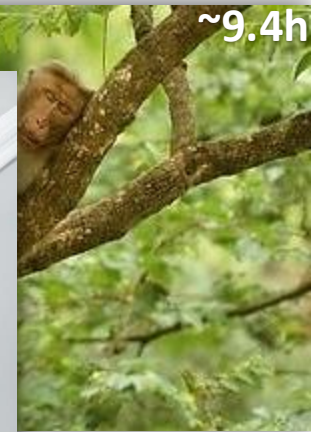
Sleep is universal



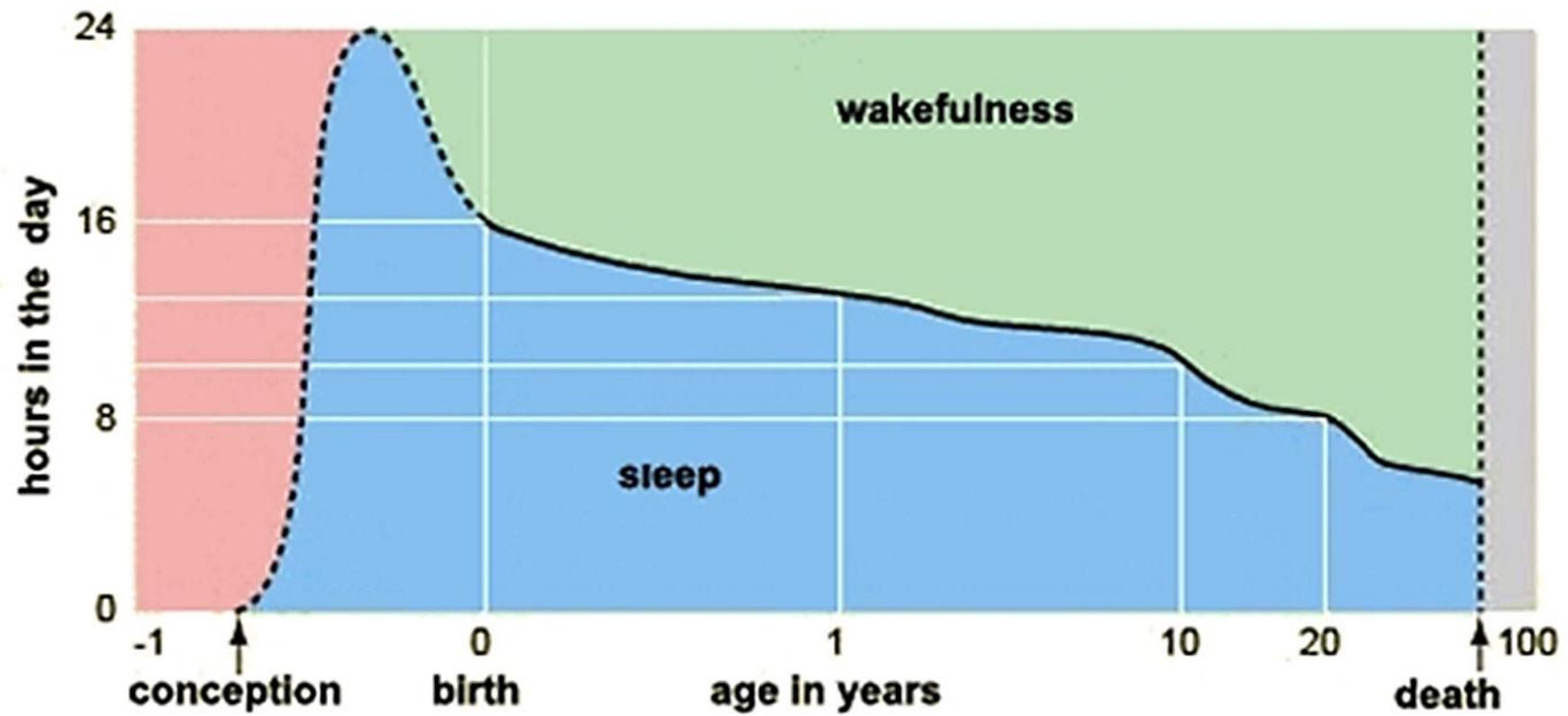
Sleep is universal

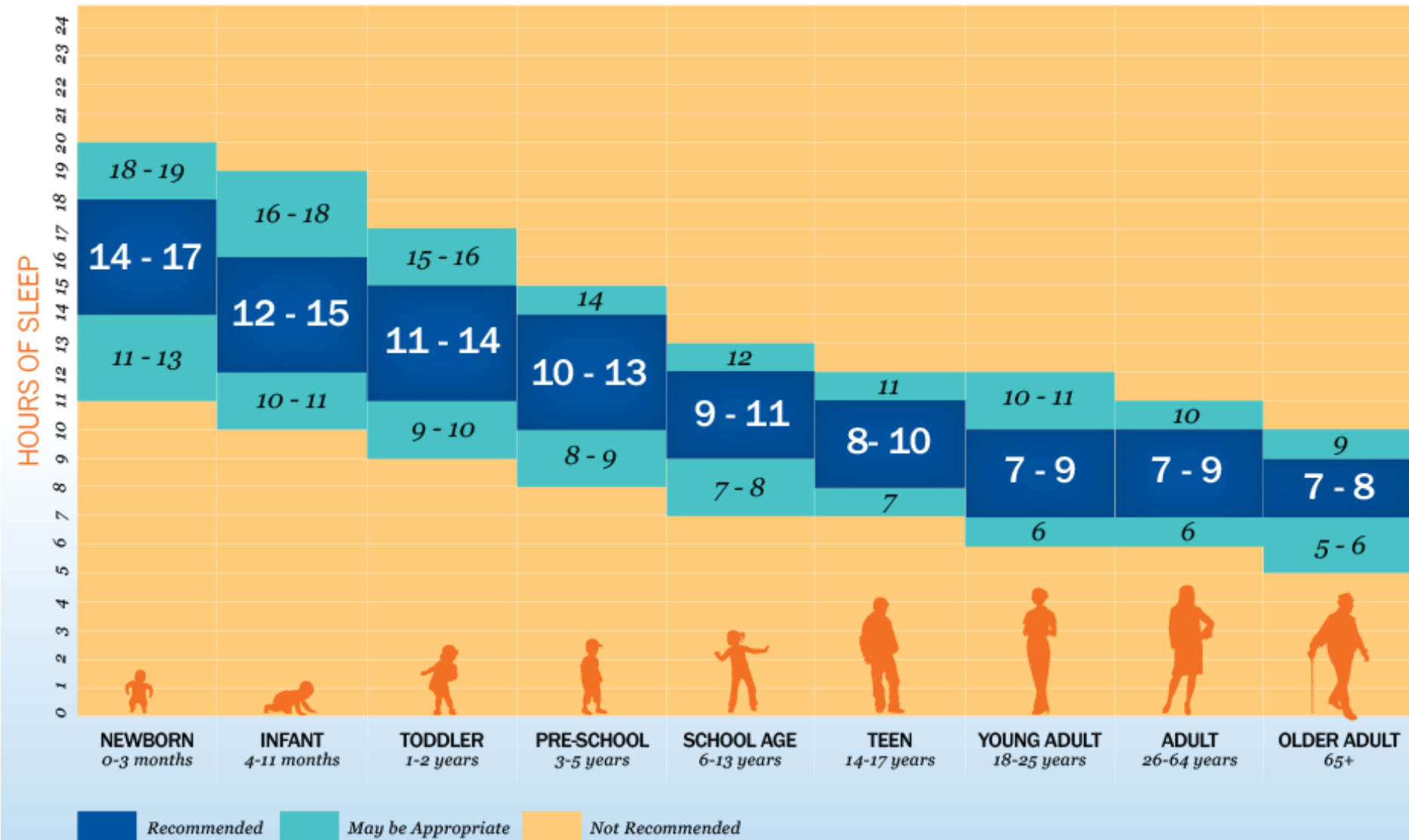


Sleep is universal



Sleep in Humans



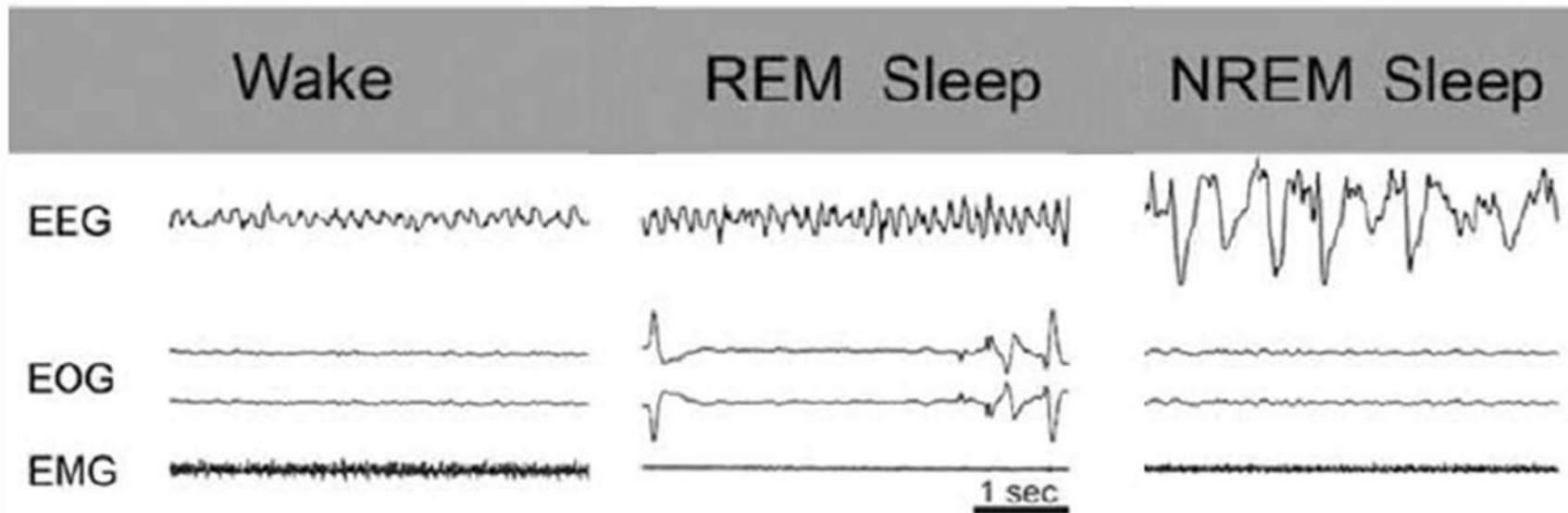




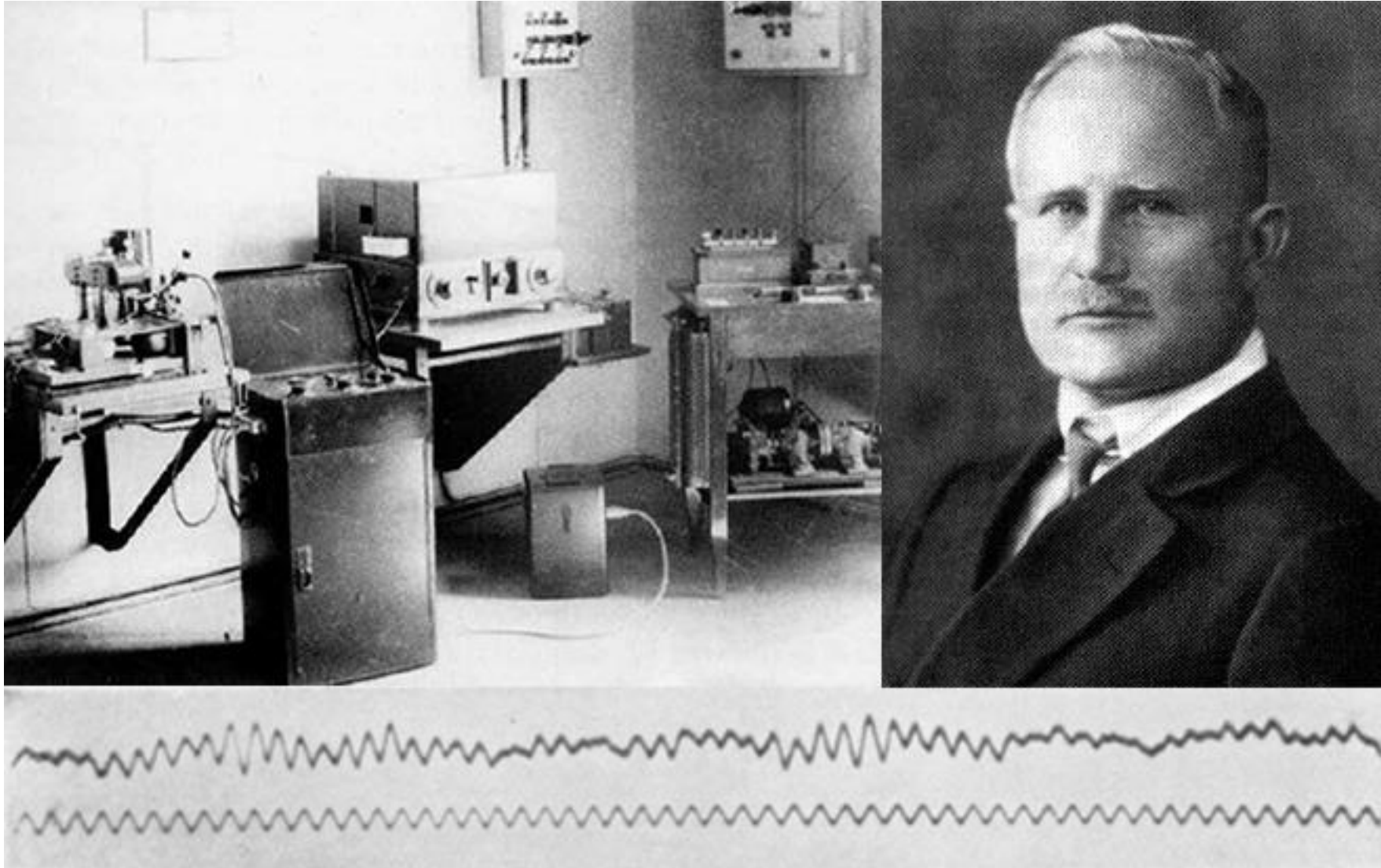
Definition of Sleep

Behavioral: Behavioral state with characteristic **immobile posture** and diminished but readily reversible **sensitivity to external stimuli**.

Electrophysiological :



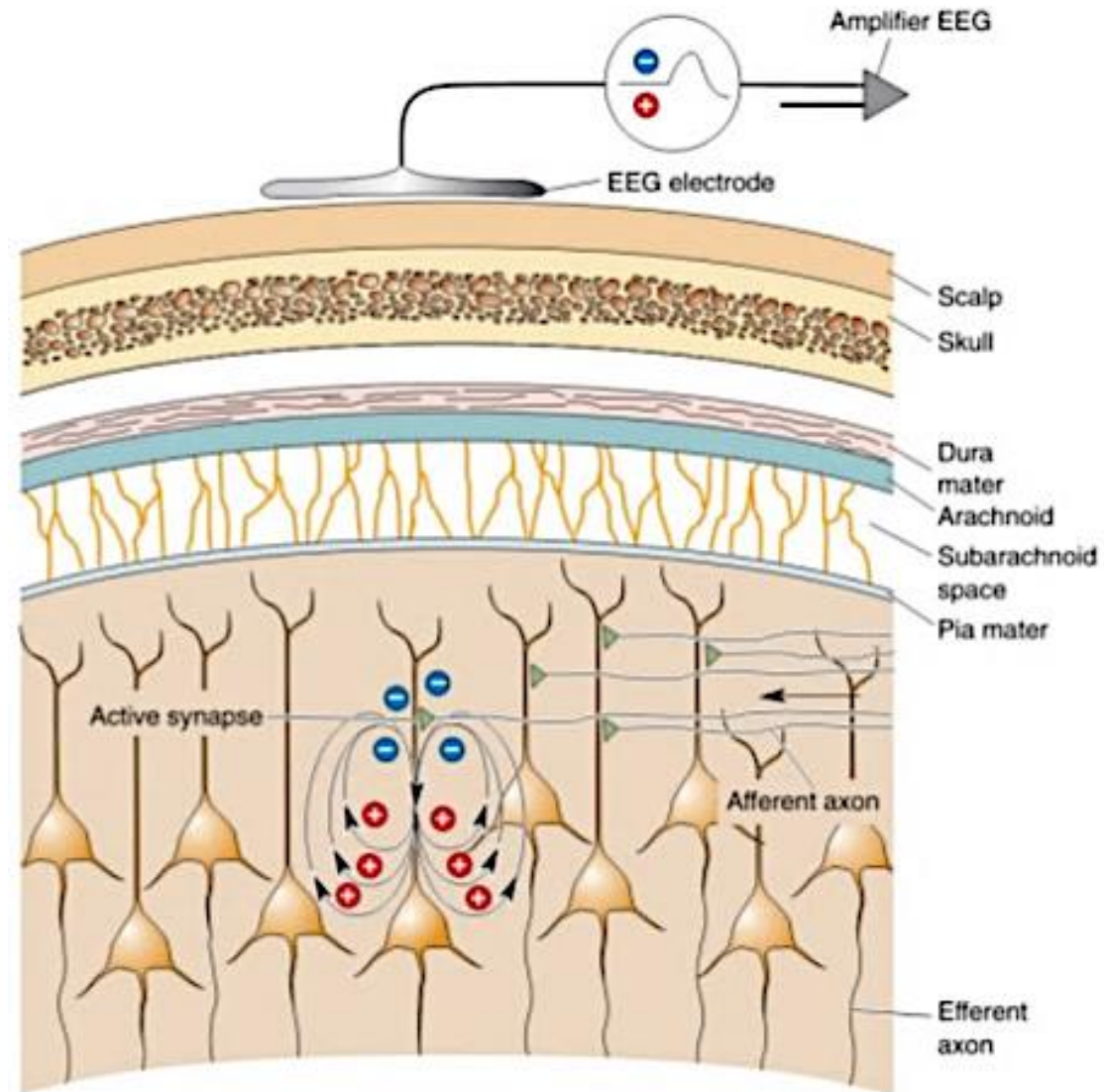
Hans Berger and the first electroencephalogram (1928)



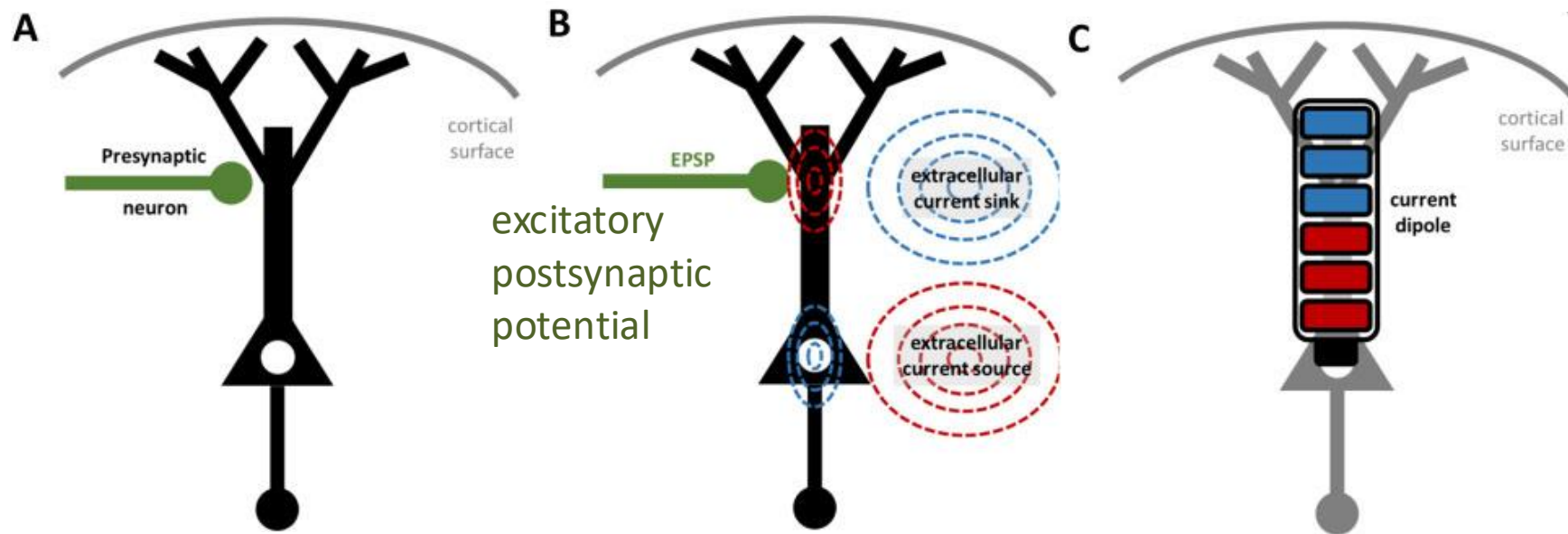
The EEG signal

The EEG signal reflects the generation of very small electrical fields by synaptic currents in *pyramidal cells*.

Only if thousand of cells contribute their small voltage is the signal large enough to reach the scalp surface.

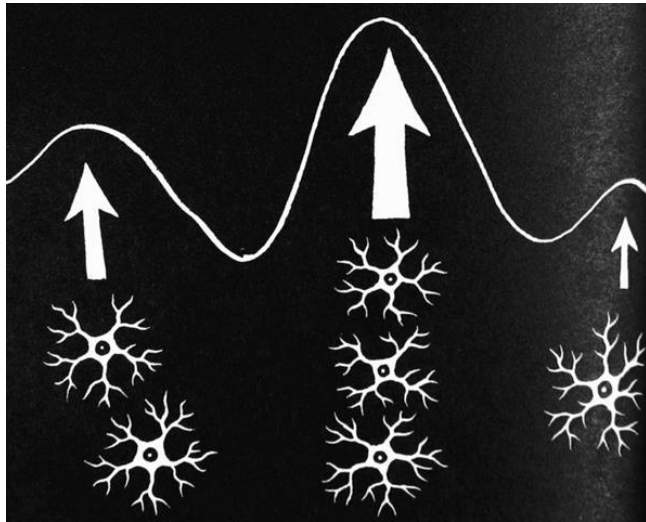


The EEG signal



EEG measures postsynaptic currents in extracellular space.

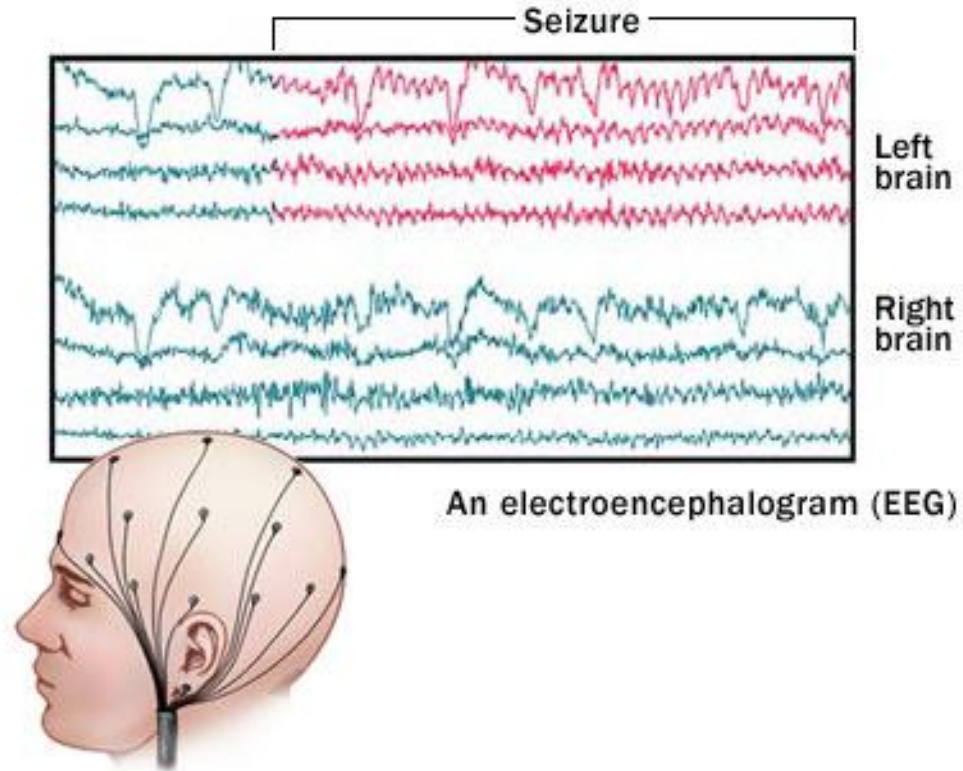
The EEG signal



- ✓ The amplitude of EEG oscillations correlates with the activity of cortical neurons
- ✓ Peaks correspond to moments of maximum synchronicity within oscillating neuronal populations
- ✓ Periods of synchronization/de-synchronization correspond to changes in local cortical activities
- ✓ Waves coordinated between distant electrodes can indicate underlying brain networks and information flow

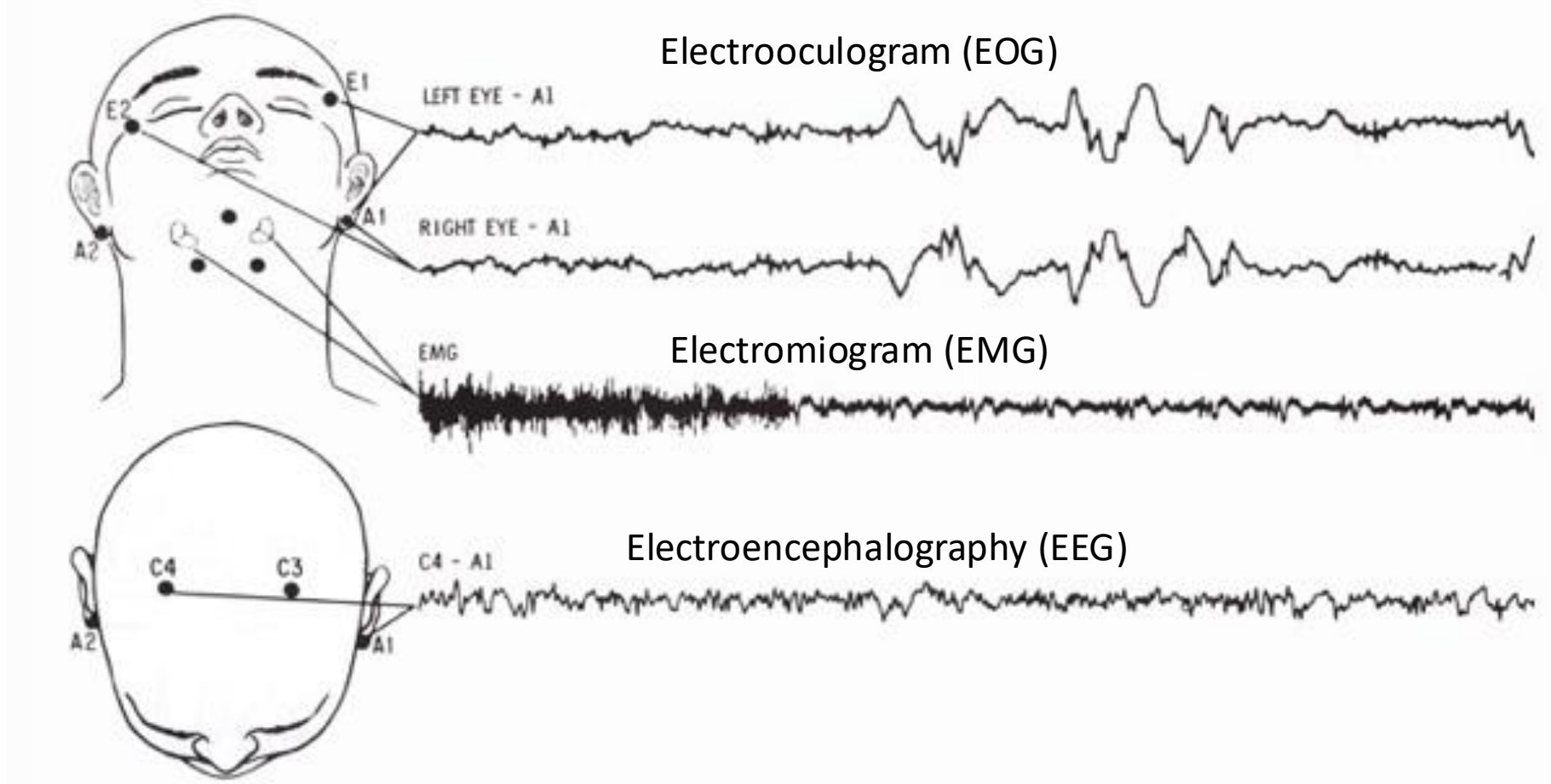
Clinical EEG

Typically based on scalp EEG recording from up to 21 electrodes (2-8 sensors are commonly used in polysomnographic [PSG] recordings).



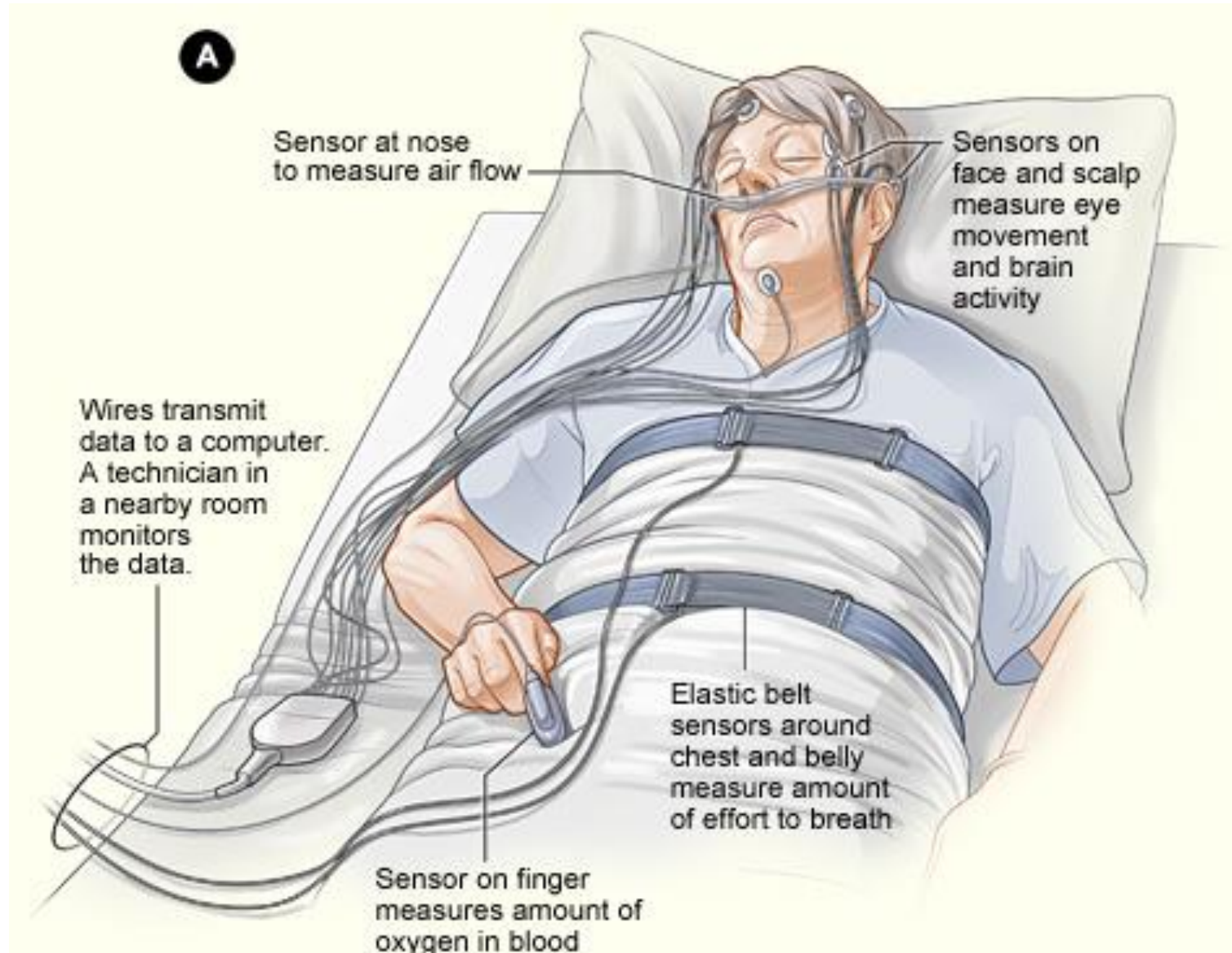
Routine EEG is typically used for the diagnosis and evaluation of epilepsy and/or for the diagnosis and evaluation of sleep-related clinical conditions.

Electrophysiological Characteristics of Human Sleep

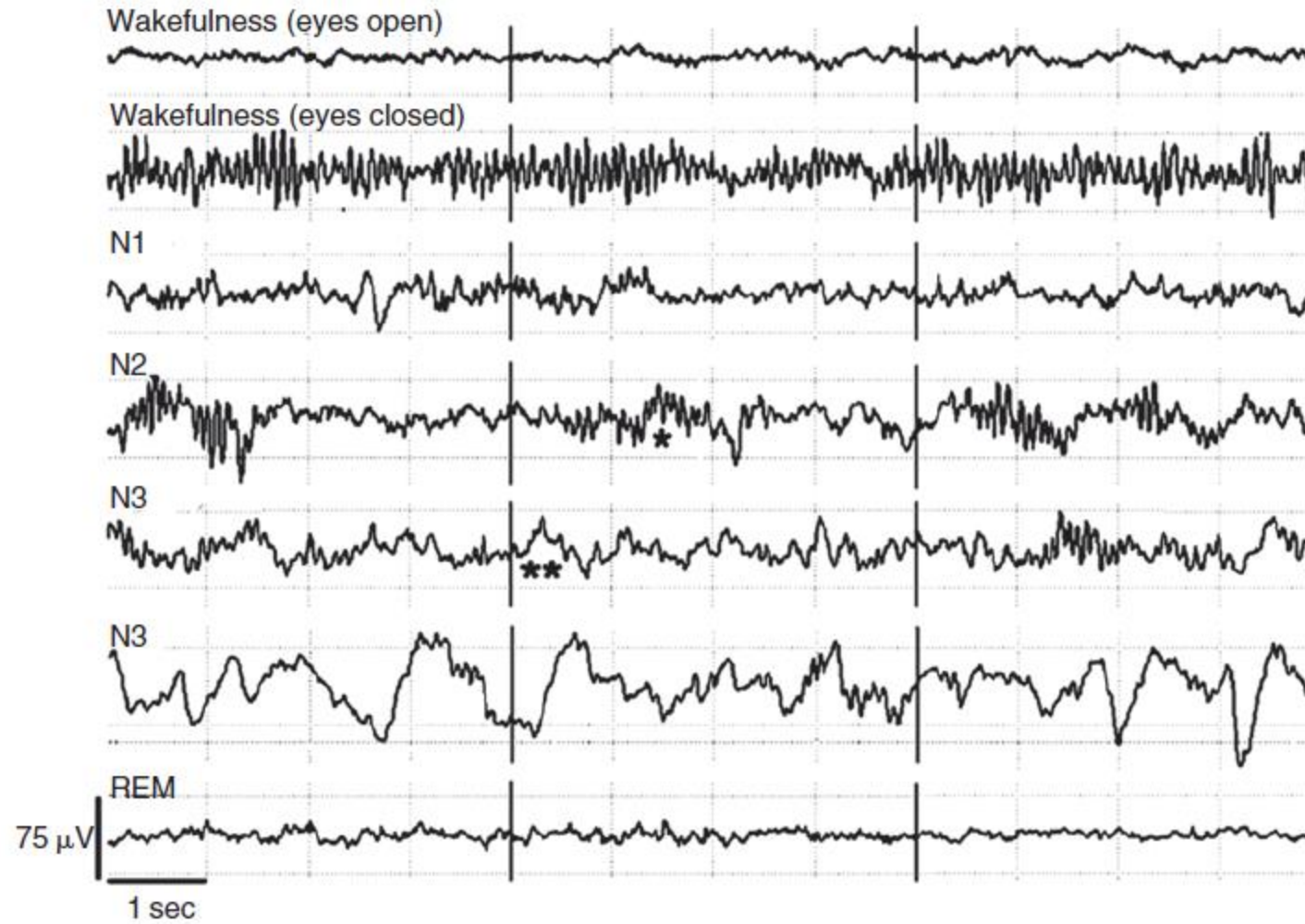


In clinical use (polysomnography) also electrocardiogram (EKG), respiration and body position are measured.

Electrophysiological Characteristics of Human Sleep

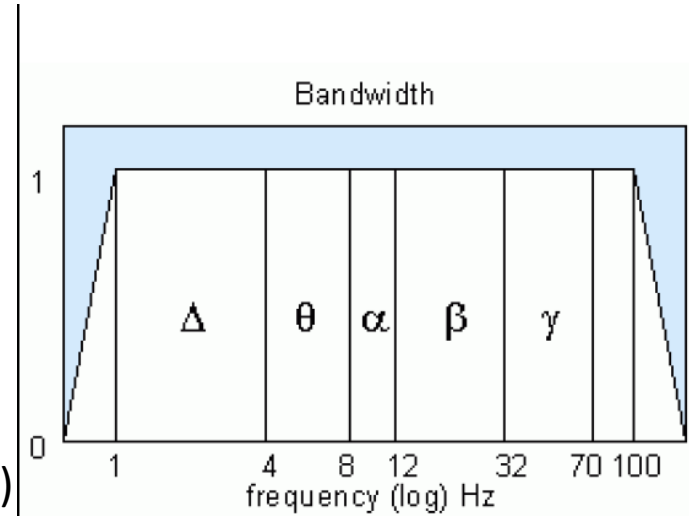
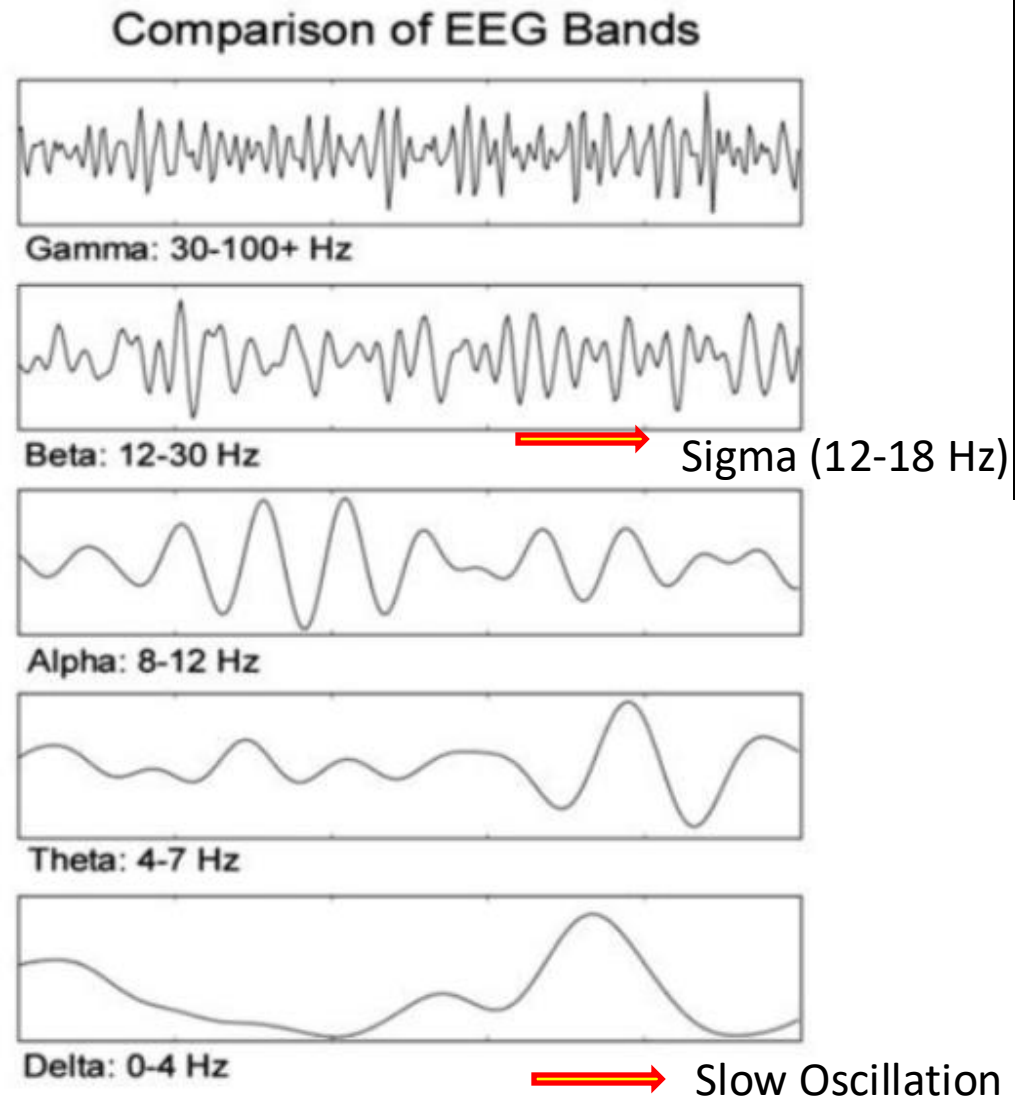


The Human EEG in Wakefulness and Sleep



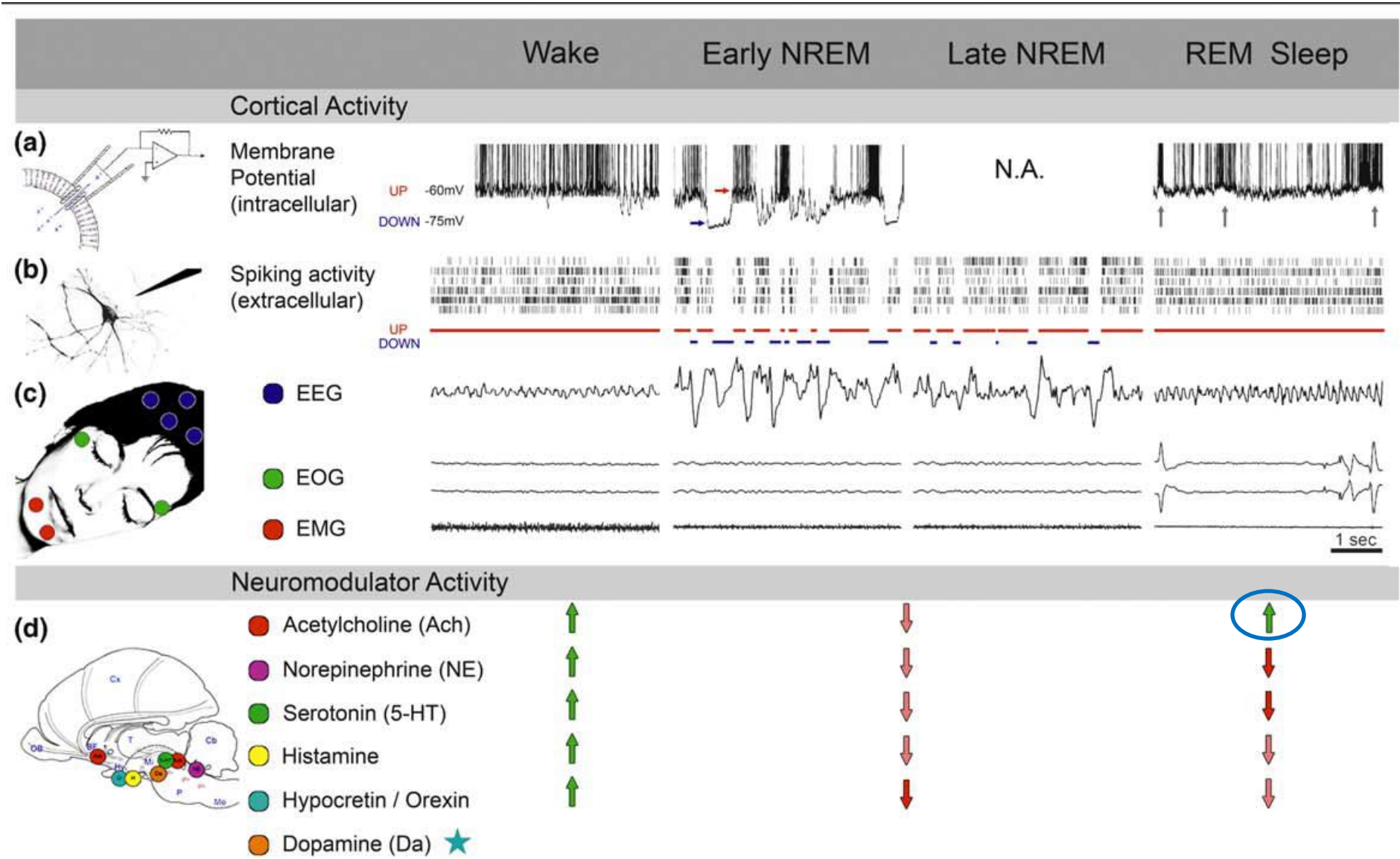
Sleep is traditionally categorized into non-rapid eye movement (NREM) sleep and REM sleep. Human NREM sleep, in turn, is divided into stages N2 and N3.

The Human EEG in Wakefulness and Sleep

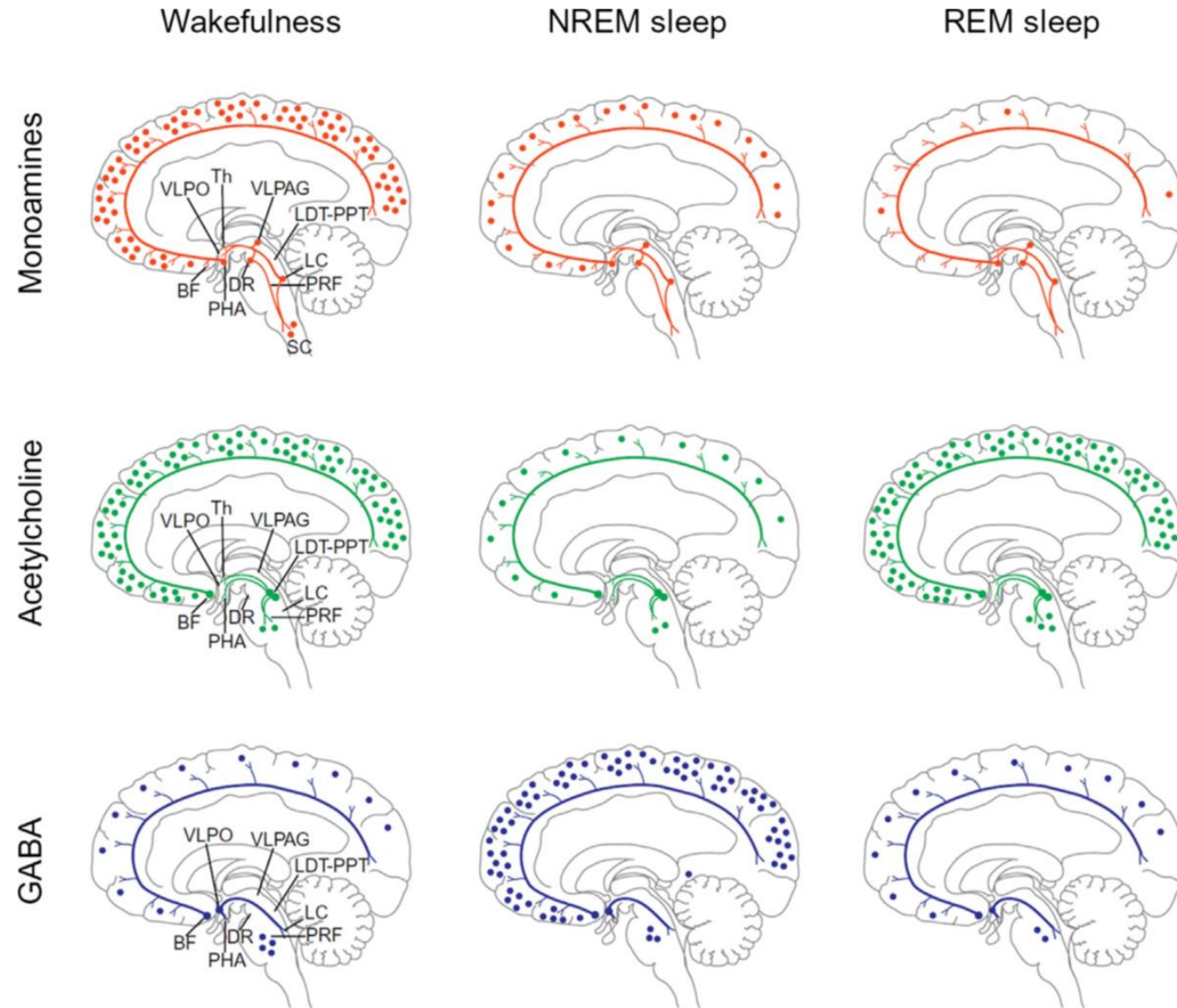


- Delta (1–4 Hz)
- Theta (4–8 Hz)
- Alpha (8–13 Hz)
- Beta (13–30 Hz)
- Gamma (36–44 Hz)

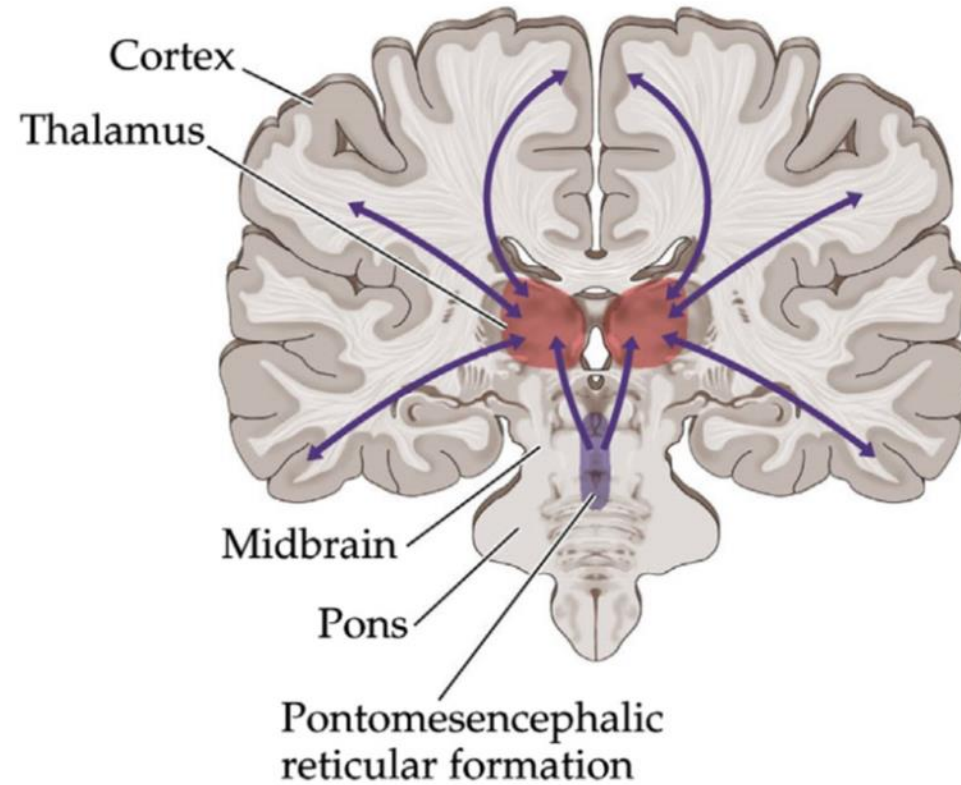
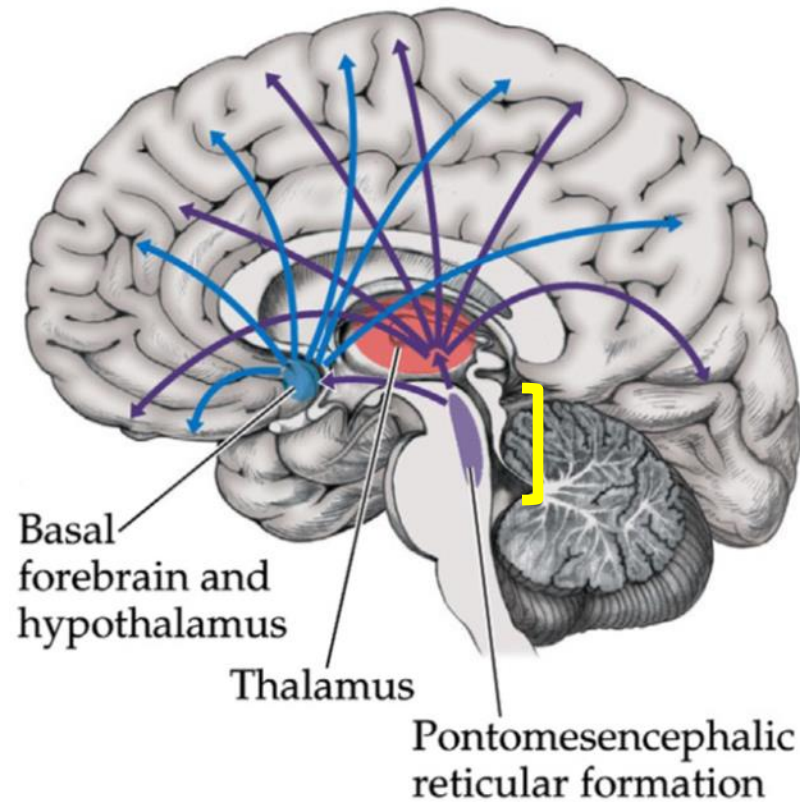
Changes in Neuromodulation During Sleep



Changes in Neuromodulation During Sleep



Arousal Circuits



The **subcortical arousal systems** consist of multiple parallel neurotransmitter systems and pathways. In the brainstem, the **reticular formation** begins in the upper pons and extends to the midbrain.

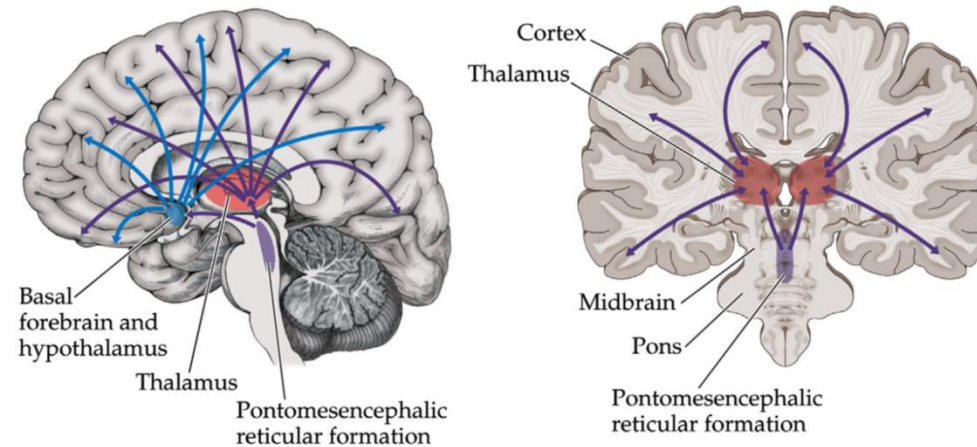
Arousal Circuits

The subcortical arousal systems have three main targets:

- ***Glutamatergic*** neurons from the **reticular formation** and ***cholinergic*** neurons from the **pedunculo pontine tegmental nucleus** and **laterodorsal tegmental nucleus** project mainly to the thalamus (*thalamic intralaminar nuclei*) which, in turn, increase cortical arousal.
- Other neurons project to the **nucleus basalis** and **hypothalamus**, which again relay arousal influences to the cortex.
- The ***monaminergic neurotransmitter systems*** (*norepinephrine, dopamine, serotonin*) project directly to the entire forebrain including the cortex and subcortical structures.

Glutamatergic and Related Arousal Systems

Glutamate is the most prevalent excitatory neurotransmitter of the central nervous system.



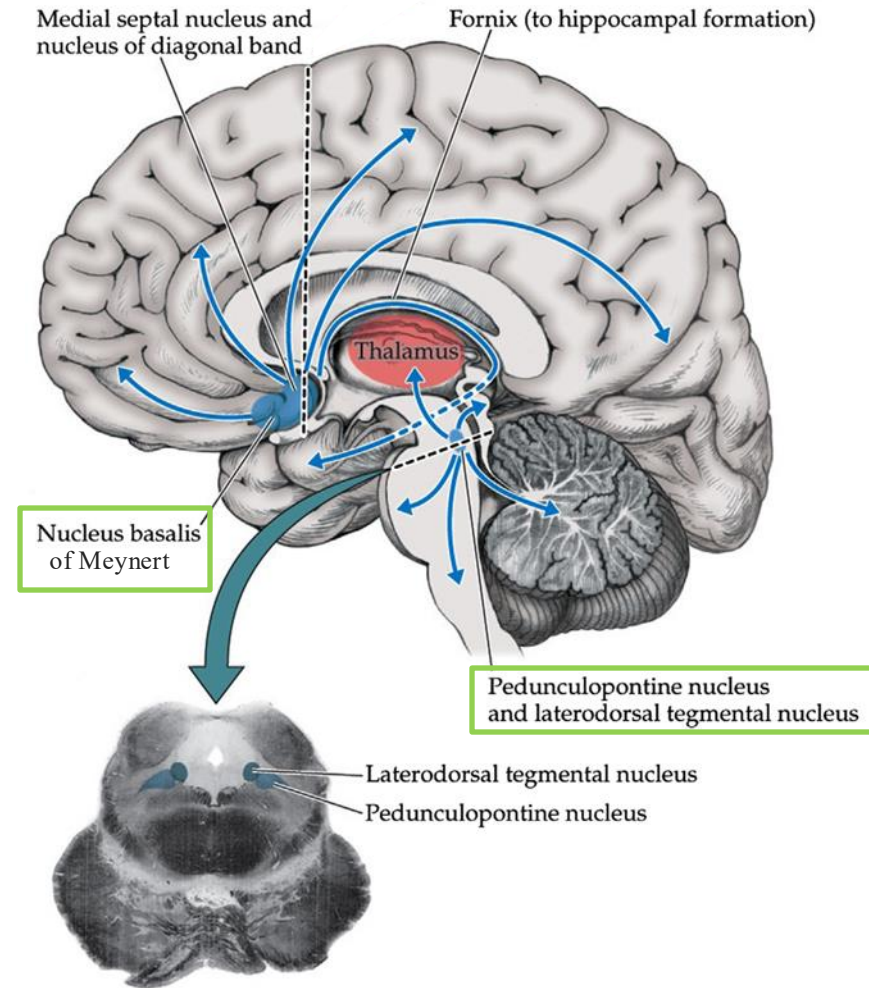
Arousal system pathways mediated by glutamate include those arising from the **midbrain** and upper **pontine reticular formation** that project to the *thalamus* and *basal forebrain*, as well as the widespread projections from the **thalamic intralaminar nuclei** to the *cortex*.

Cholinergic Arousal Systems

The two main sources of cholinergic projections neurons in the central nervous system lie in the brainstem **pontomesencephalic reticular formation** and in the **basal forebrain**.

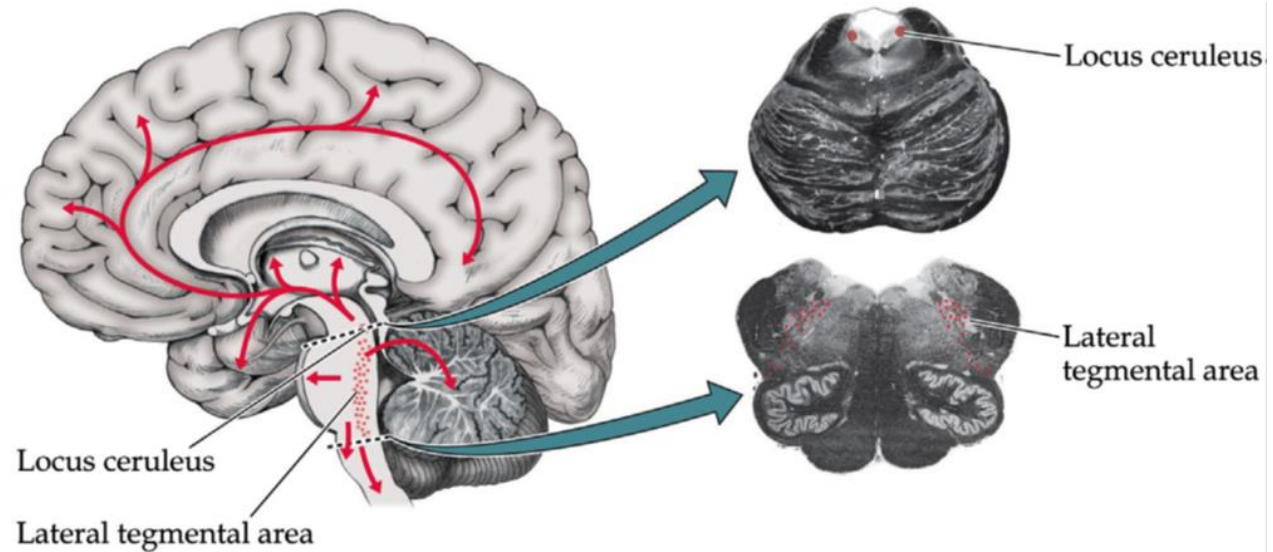
The brainstem has very few direct cholinergic projections to the cortex (effects are mediated by the Thalamus). The major source of cholinergic input to the cortex is the basal forebrain.

Selective damage to cholinergic neurotransmission does not produce coma.



Noradrenergic Arousal Systems

Norepinephrine-containing neurons are located in the **locus ceruleus** (in the rostral pons, adjacent to the 4th ventricle), and in the **lateral tegmental area**.



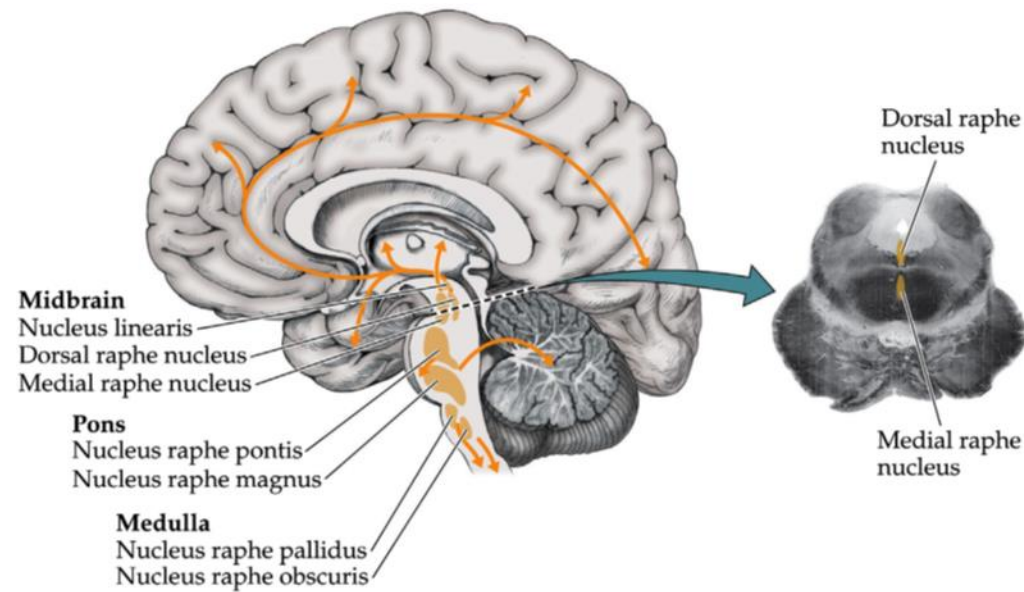
Ascending noradrenergic projections reach the cortex, thalamus and hypothalamus to regulate sleep-wake cycles, attention, and mood, while descending projections to the brainstem, cerebellum, and spinal cord modulate autonomic function and gating of pain.

Selective blockade of noradrenergic neurons does not produce coma.

Serotonergic Arousal Systems

Serotonergic neurons are found predominantly in the **midline raphe nuclei** of the midbrain, pons, and medulla.

The more rostral serotonergic neurons in the midbrain and upper pontine raphe nuclei project to the entire forebrain, participating in sleep-wake regulation.



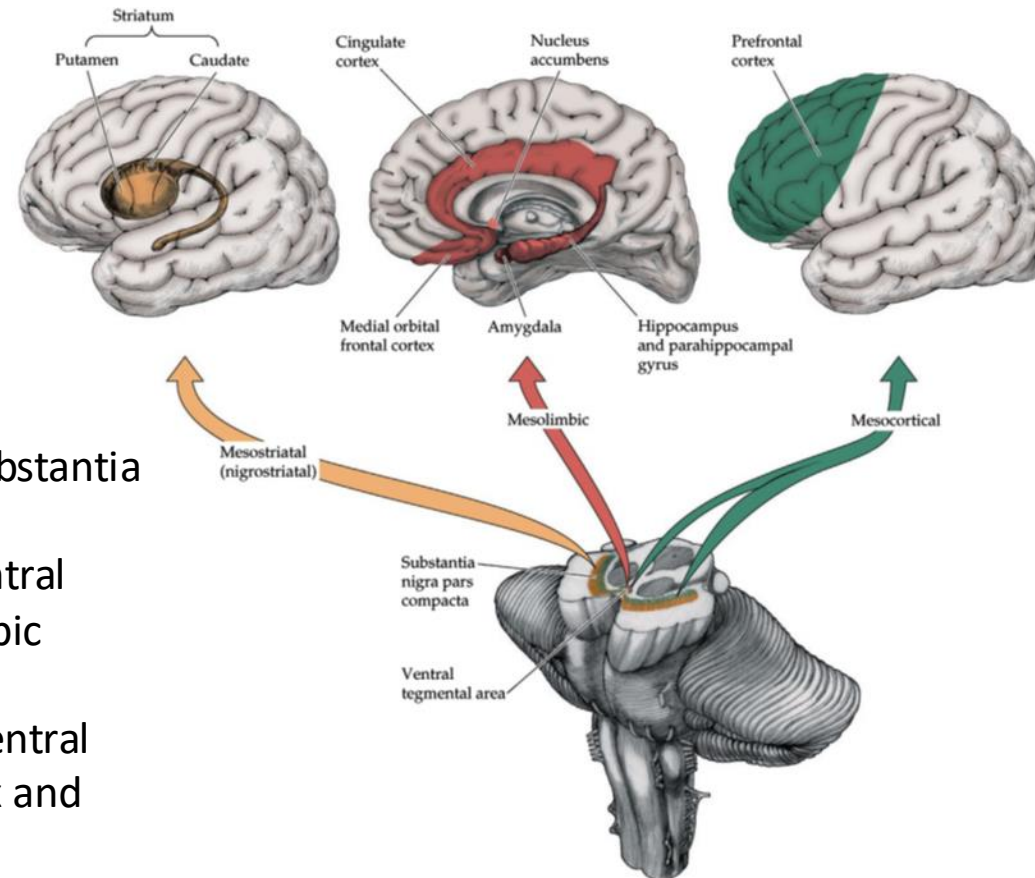
The role of serotonergic neurons in arousal is complex, possibly because the large diversity of serotonin receptors leads to effects that either promote or inhibit arousal in different brain regions.

Dopaminergic Arousal Systems

Most dopaminergic neurons are located in the **ventral midbrain**, either in the **substantia nigra** pars compacta or in the adjacent **ventral tegmental area**.

3 dopaminergic projection systems:

1. **Nigrostriatal Pathway**, from the substantia nigra to the caudate and putamen;
2. **Mesolimbic Pathway**, from the ventral tegmental area and projects to limbic structures;
3. **Mesocortical Pathway**, from the ventral tegmental area to prefrontal cortex and thalamus.

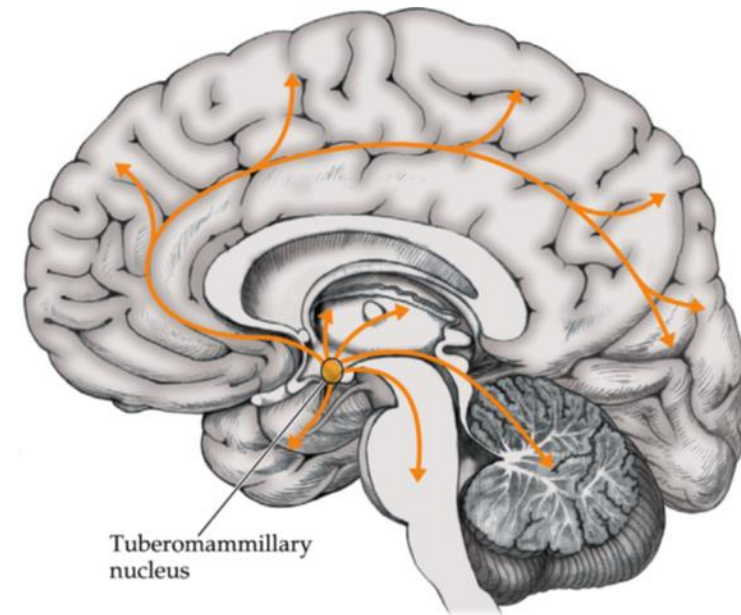


Effects of dopamine on the thalamus and cortex can be either excitatory or inhibitory. Impaired dopaminergic transmission to the prefrontal cortex has been proposed to be related to states of markedly reduced motivation and initiative.

Histaminergic Arousal Systems

Histamine-containing neurons are found mainly in the **tuberomammillary nucleus** of the **posterior hypothalamus**.

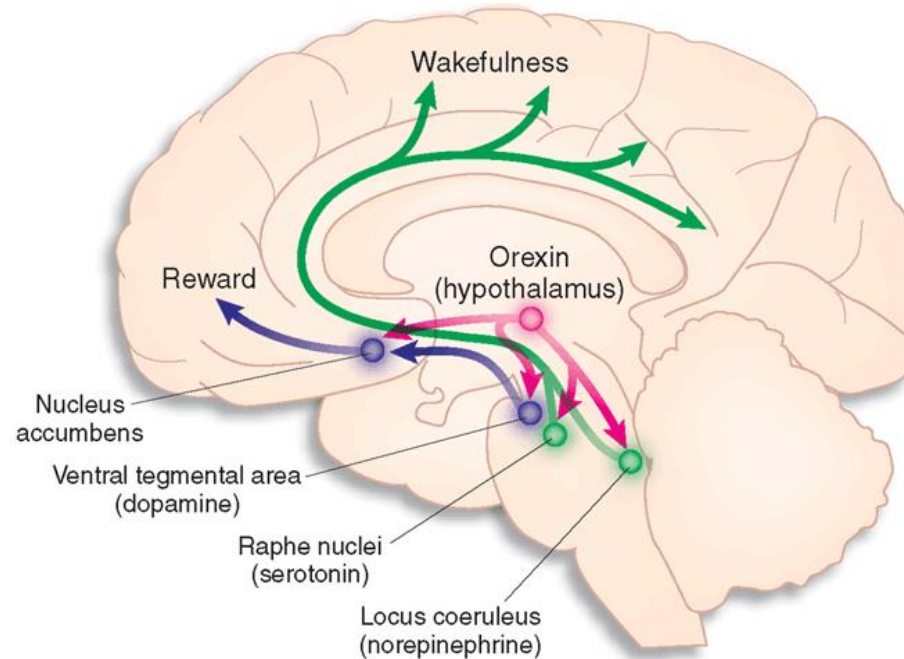
Widespread ascending projections of histaminergic neurons reach nearly the entire forebrain including cortex and thalamus.



Effects of histamine are receptor-dependent as activation of H_1 receptors promotes wakefulness, whereas H_3 -receptors appear to have the opposite role.

Orexinergic Arousal Systems

Orexin (hypocretin) is a peptide produced in neurons of the perifornical, lateral, and posterior **hypothalamus**, which project to both cortex and virtually all subcortical arousal systems to promote the awake state.



Abnormalities of the orexin system are thought to play a role in **narcolepsy**, a disorder characterized by excessive daytime sleepiness and pathological transitions into rapid eye movement sleep.

GABAergic Arousal Systems

Found in local **inhibitory interneurons** throughout the cortex and subcortical structures, **GABA** (gamma-aminobutyric acid) is the most prevalent inhibitory neurotransmitter in the central nervous system and plays a major role in regulating arousal.

Several long-range GABAergic projection systems also contribute to controlling arousal (with heterogeneous effects, depending on several factors...).

For instance, some GABAergic neurons in the basal forebrain are thought to promote arousal because these inhibitory neurons in turn project to cortical inhibitory interneurons.

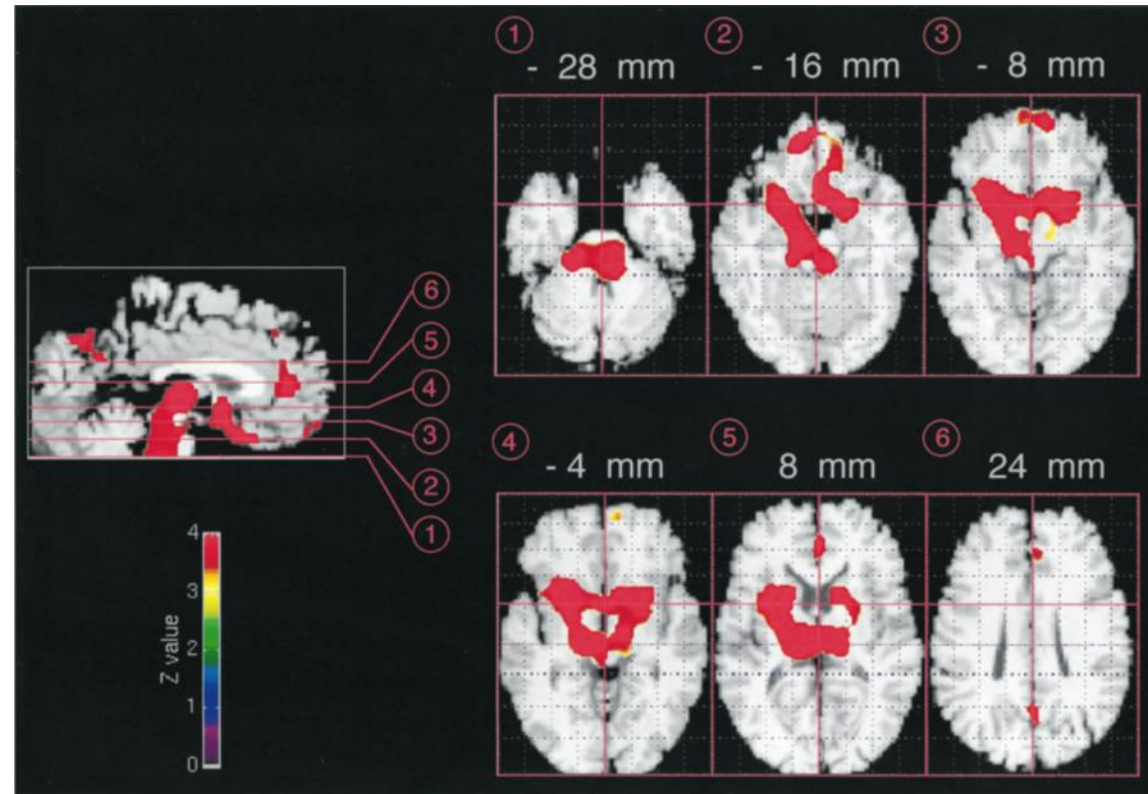


GABAergic Arousal Systems

Alcohol is an *indirect GABA agonist*. When bound to a GABA receptor on a neuron, alcohol allows either the influx of negative (or efflux of positive) ions, giving the cell a more negative charge. Thus, the neuron's attempt to fire an action potential is thwarted.



Changes in brain activity during NREM sleep

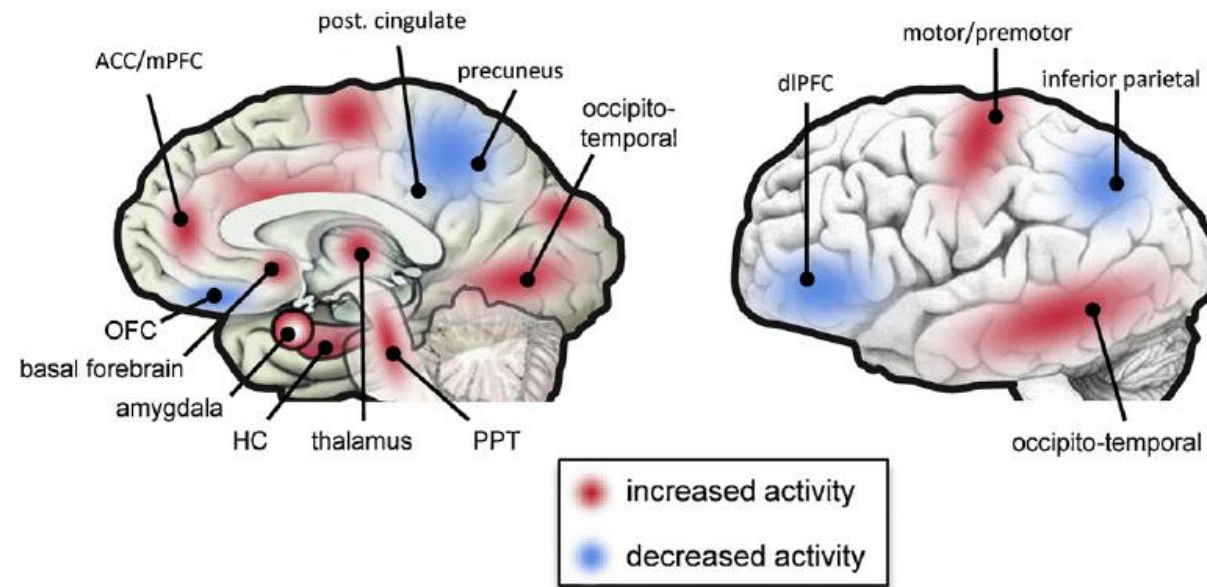


During deep **NREM sleep**, **metabolic activity** can be **reduced as much as 40% relative to resting wakefulness**. Activation is especially reduced in the **thalamus** due to its profound hyperpolarization during NREM sleep. In the cerebral cortex, activation is reduced in **dorsolateral prefrontal cortex**, **orbitofrontal** and **anterior cingulate cortex** (slow waves are especially prominent in these regions).

Changes in brain activity during REM sleep

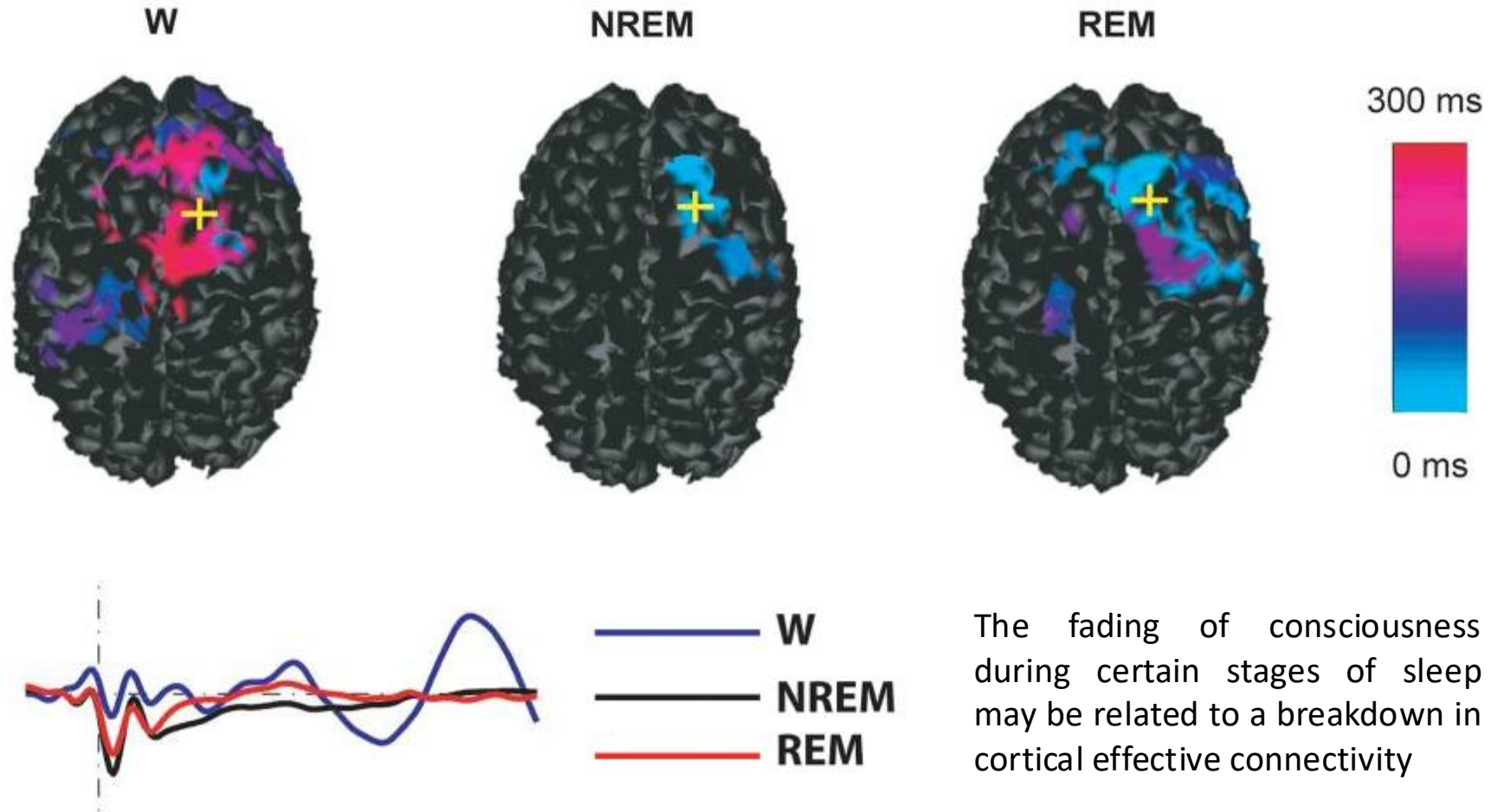
During **REM sleep**, absolute levels of **blood flow** and **metabolic activity** are high, reaching levels similar to those seen during wakefulness.

Some areas appear to be **more active in REM sleep than in wakefulness**: e.g., **amygdala**, **parahippocampal cortex**, **anterior cingulate**, **parietal lobule** and **extrastriate areas**.

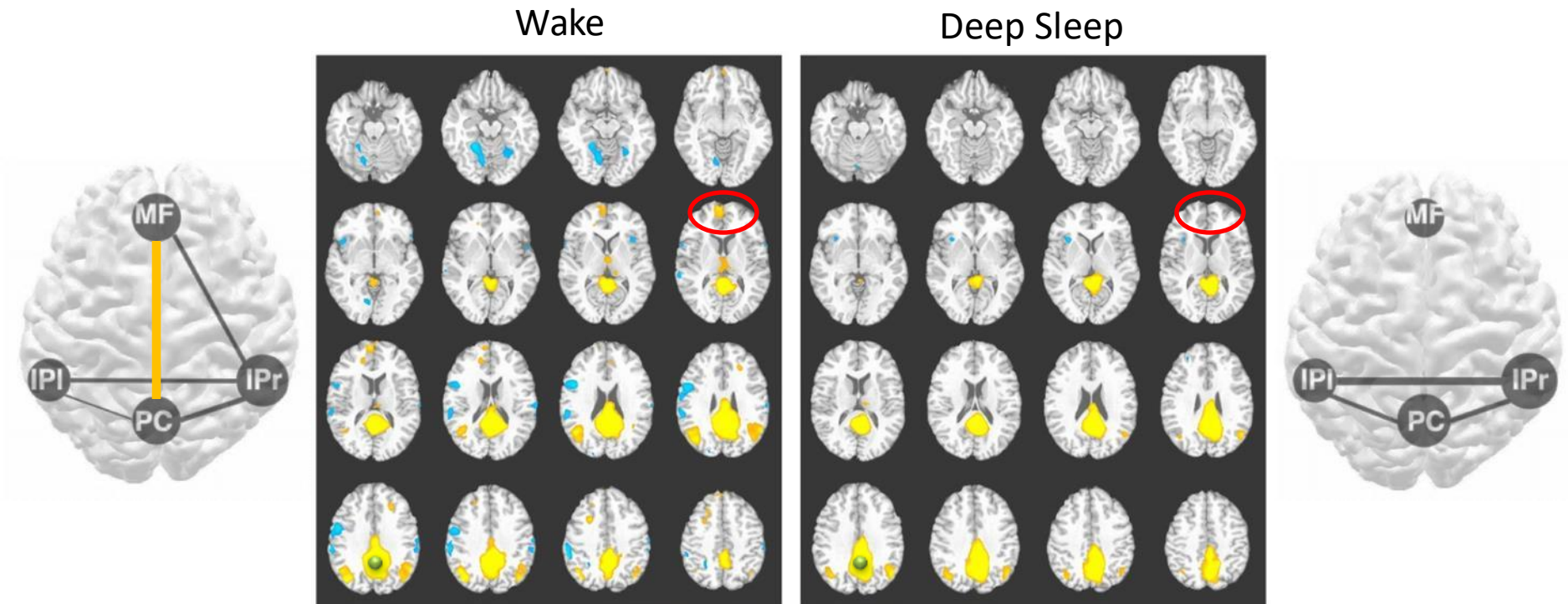


By contrast, the rest of parietal cortex, precuneus, posterior cingulate cortex and dorsolateral prefrontal cortex are **relatively deactivated**.

Breakdown of Cortical Connectivity During Sleep

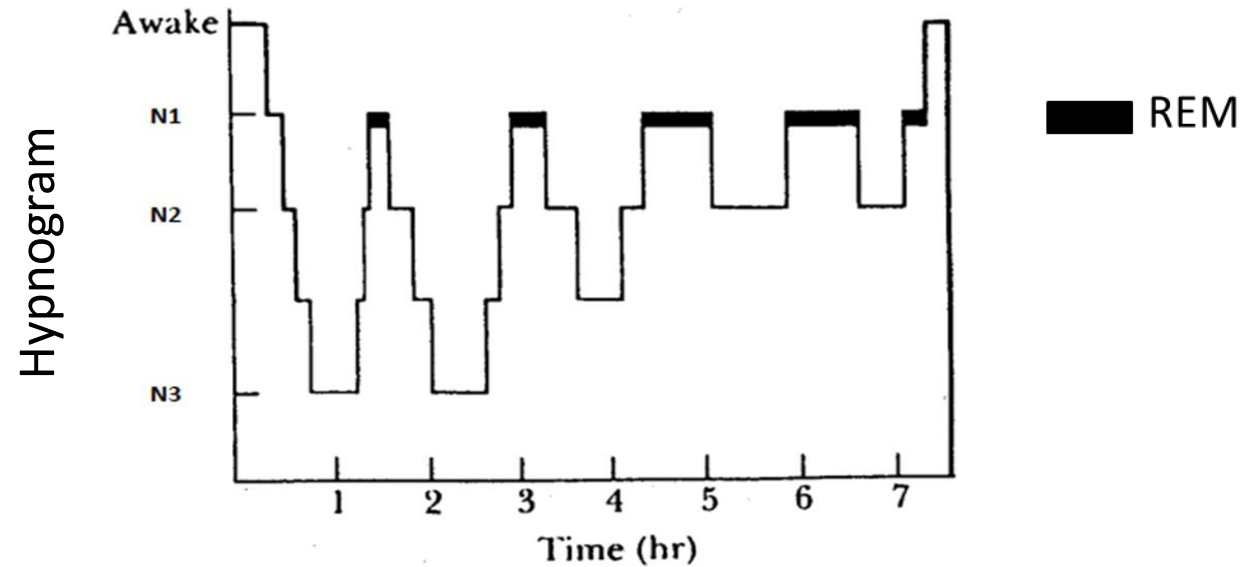


Changes in brain connectivity during NREM sleep



Sleep-induced changes in consciousness are reflected in **reduced correlations between frontal and posterior areas of the DMN** during deep sleep.

Sleep Cycles



The succession of NREM sleep stages followed by an episode of REM sleep is called a **sleep cycle**, and lasts **approximately 90–110 minutes** in humans. There are a total of **4–5 cycles per night**.

Slow wave sleep (N3) is **prominent early in the night**, especially during the first sleep cycle, and diminishes as the night progresses. As slow wave sleep wanes, periods of REM sleep lengthen and show greater phasic activity.

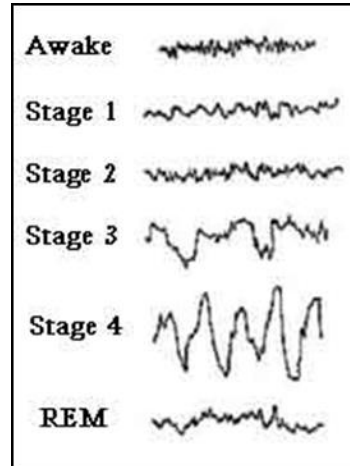
A healthy young adult typically spends about **5%** of the sleep period in stage **N1**, about **50%** in stage **N2**, **20–25%** in stage **N3**, and **20–25%** in **REM** sleep.

Fluid Boundaries Between Wake and Sleep

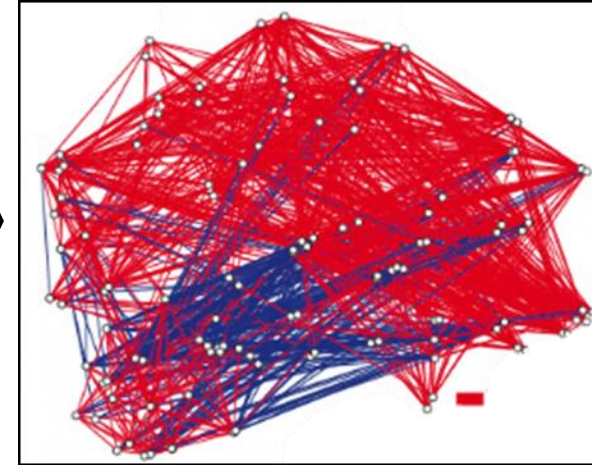
Simultaneous EEG-fMRI recordings



Sleep Scoring



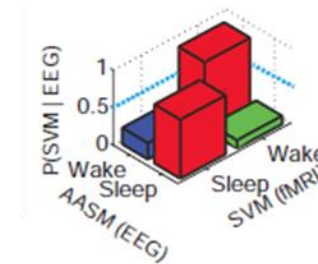
SVM Classifier



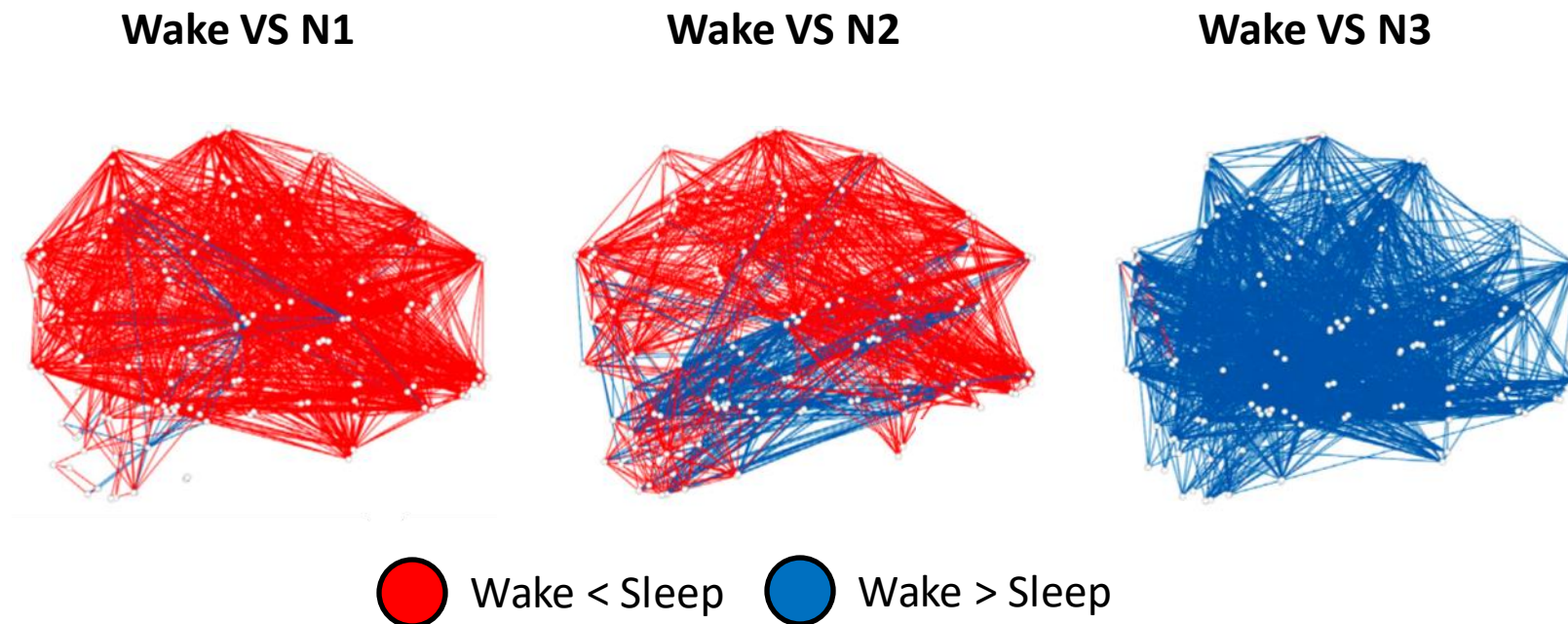
Based on fMRI connectivity matrices

Accuracy corresponding to 75%

Sleep classification was conservative



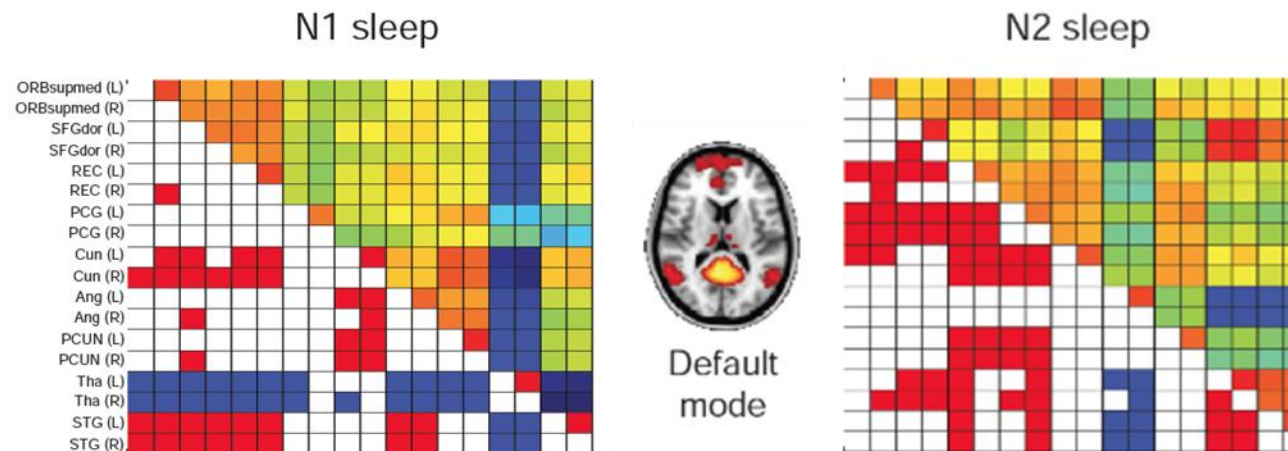
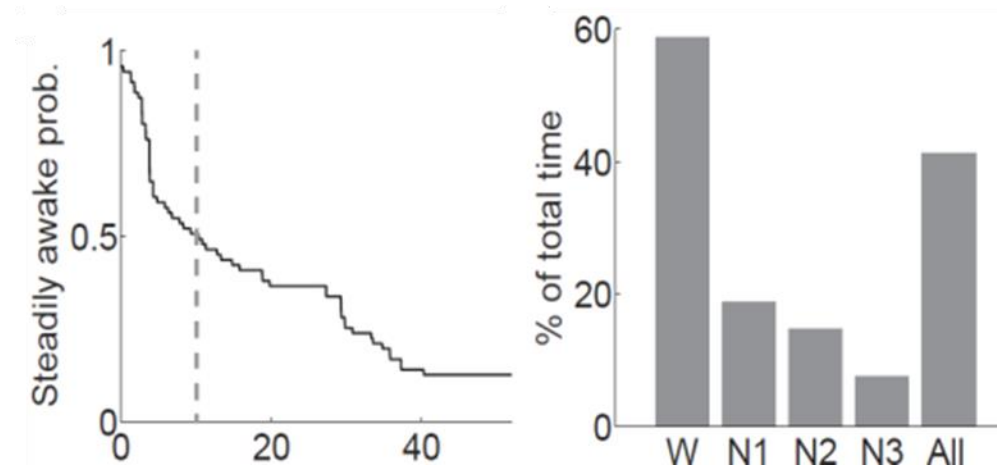
Fluid Boundaries Between Wake and Sleep



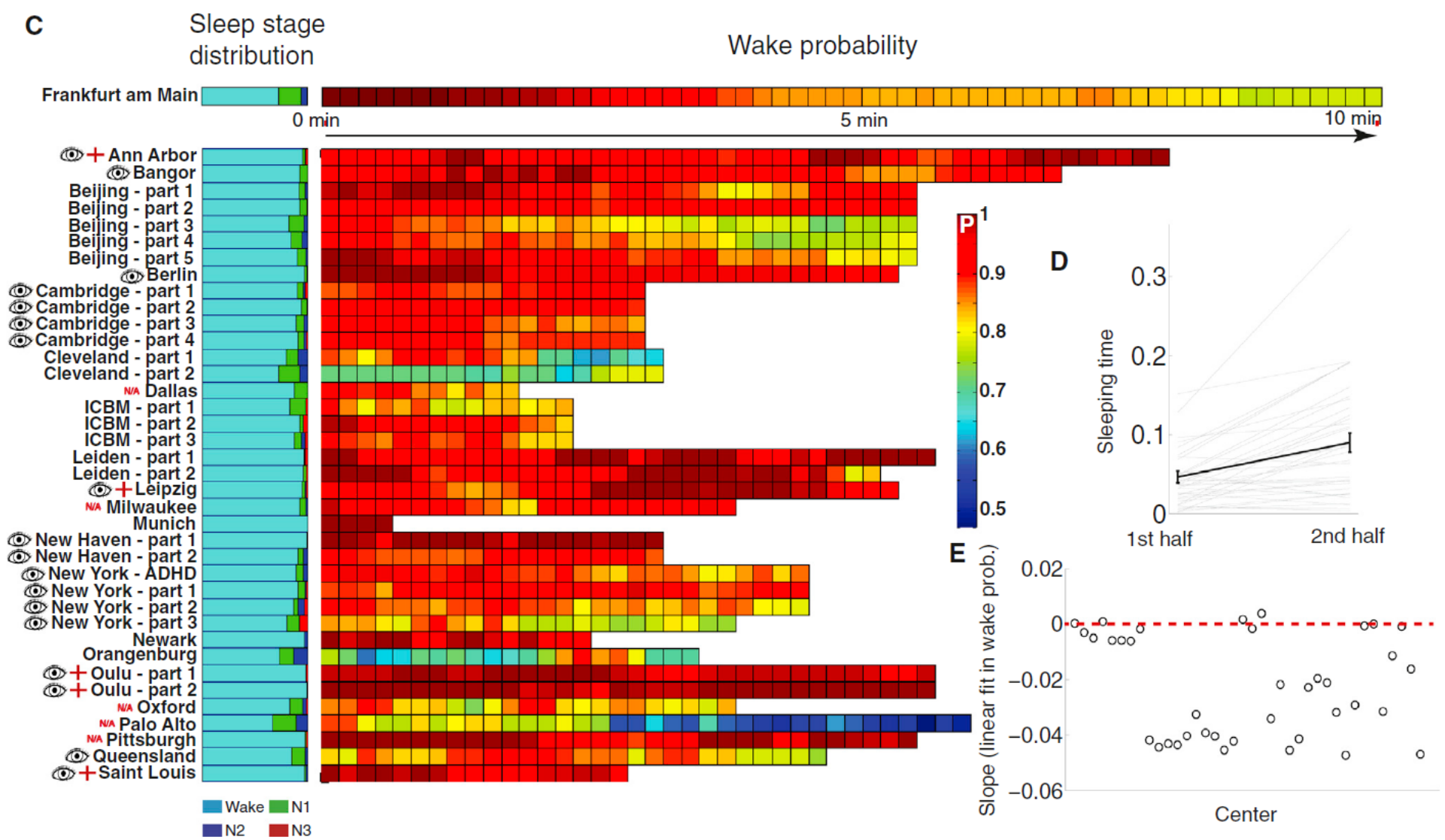
The net reduction in connectivity observed during deep sleep reflect a reduction in brain interregional cross talk, i.e., a loss of global integration well in line with a phenotypically concomitant reduction in consciousness. Modularity, a spatial measure of functional segregation, generally increased across the brain with deepening sleep.

Fluid Boundaries Between Wake and Sleep

- ✓ EEG-fMRI data collection
- ✓ Classifier trained to detect sleep
- ✓ 1147 fMRI datasets Analyzed
- ✓ Transition to sleep in $\frac{1}{3}$ within 3'

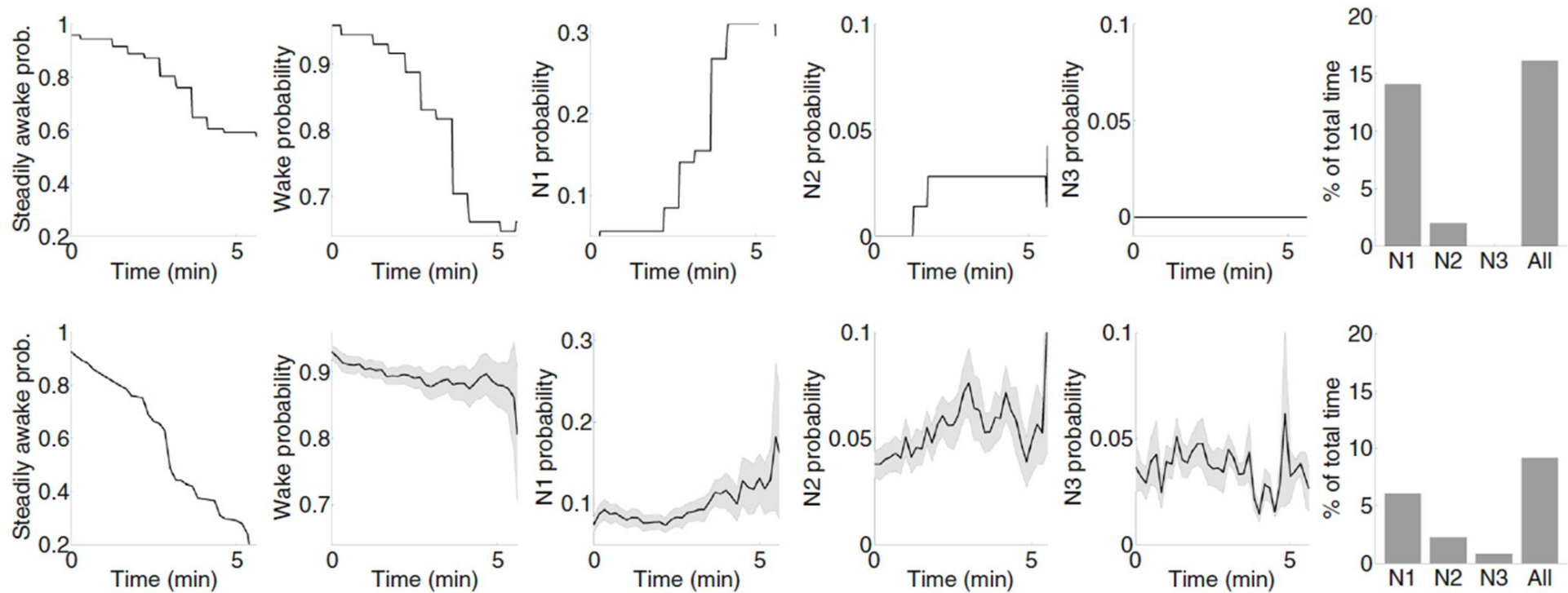


Fluid Boundaries Between Wake and Sleep



Fluid Boundaries Between Wake and Sleep

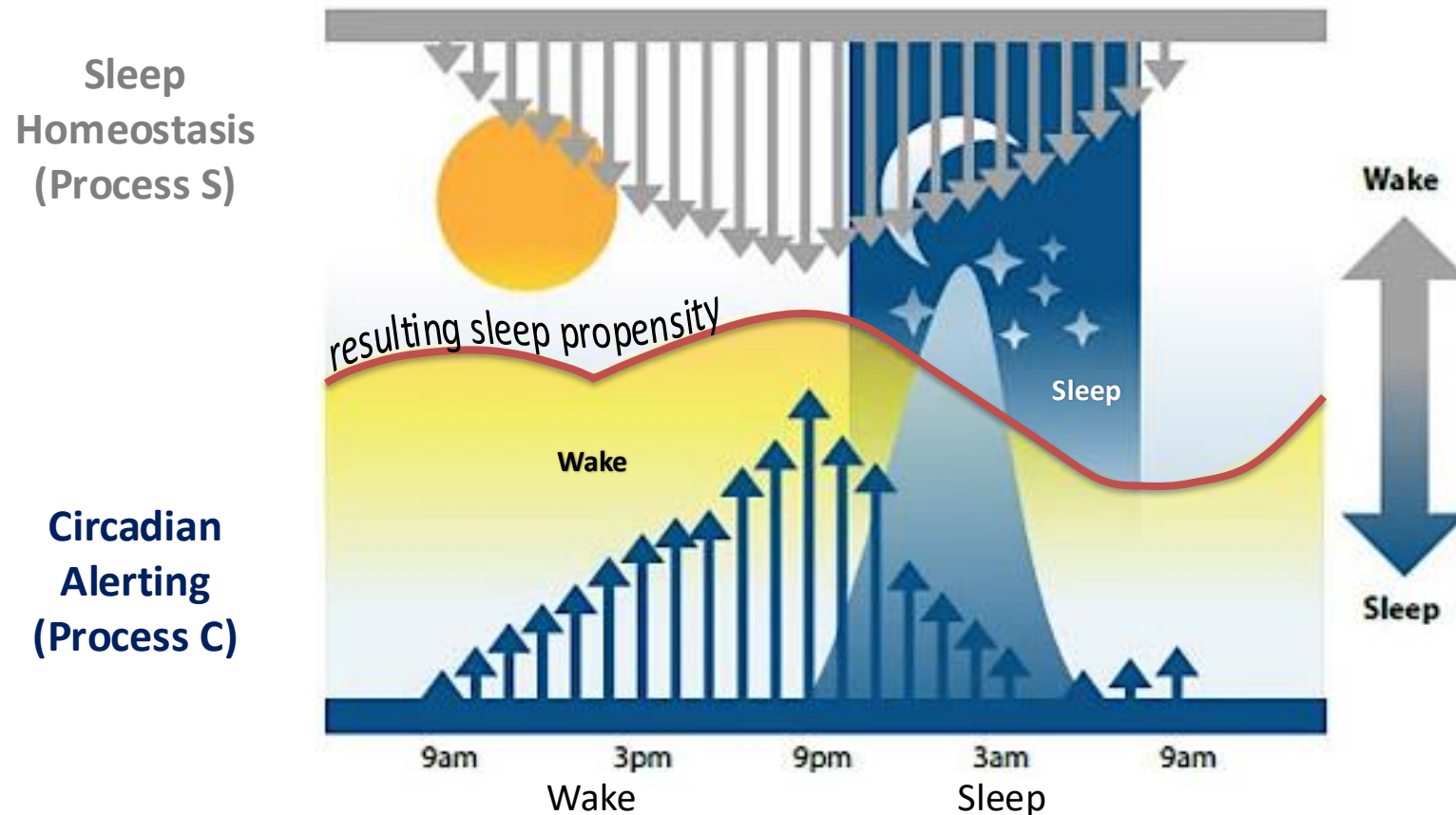
Frankfurt Dataset



Connectome Dataset

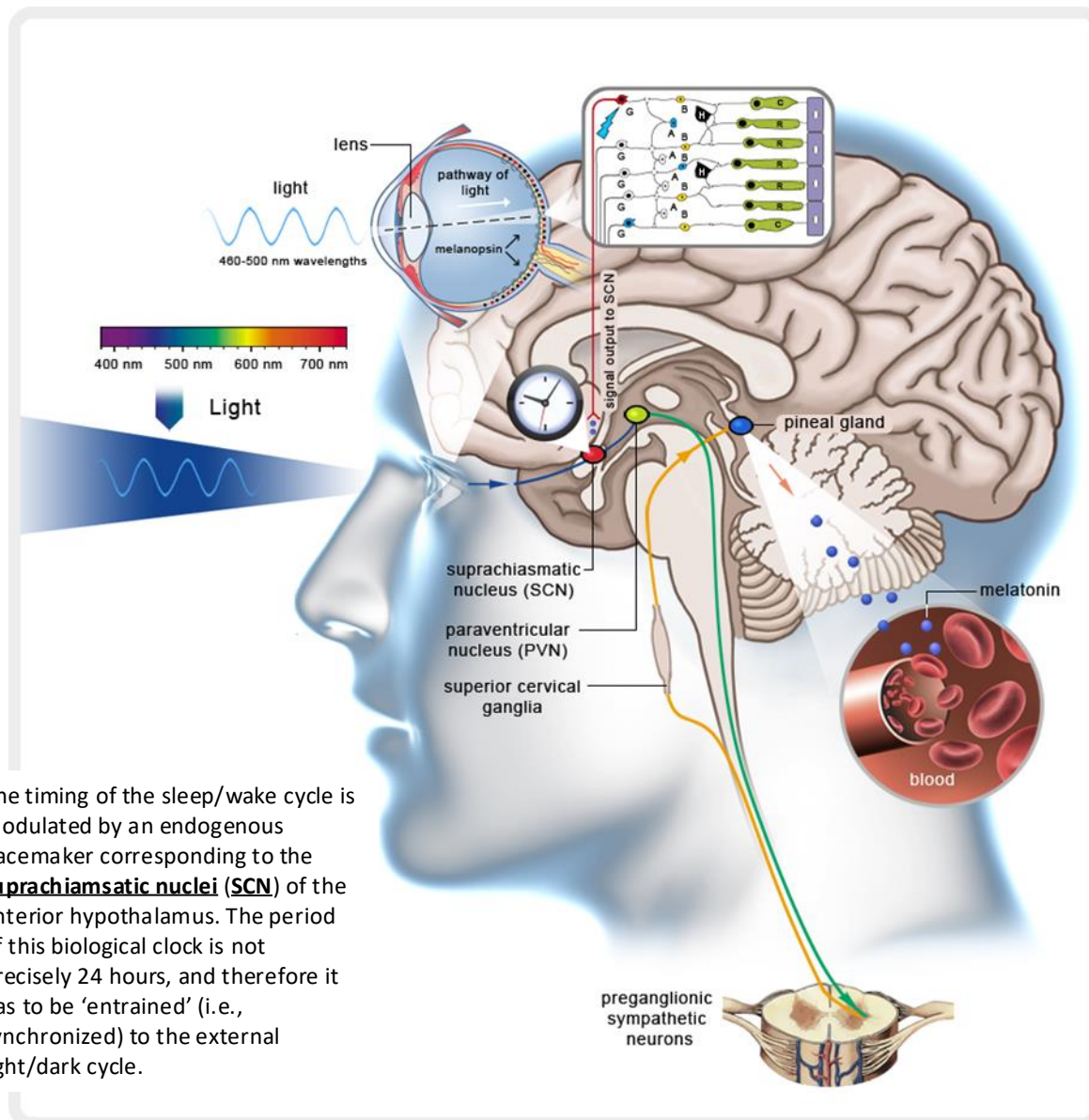
A Two Process Model of Sleep Regulation

The two-process model of sleep regulation posits that the interaction of its two constituent processes a **homeostatic Process S** and a **circadian Process C** determines the timing of sleep and waking.



Borbély et al., Hum. Neurobiol., 1982

Circadian Regulation of Wakefulness and Sleep



The timing of the sleep/wake cycle is modulated by an endogenous pacemaker corresponding to the **suprachiasmatic nuclei (SCN)** of the anterior hypothalamus. The period of this biological clock is not precisely 24 hours, and therefore it has to be 'entrained' (i.e., synchronized) to the external light/dark cycle.

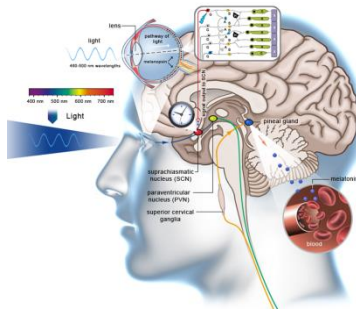
Light information is captured by specialized **retinal photosensitive ganglion cells** and transduced to the **SCN** via the **retinohypothalamic tract**. In turn, the SCN regulates the release of the hormone **melatonin** from the **pineal gland** (normally occurring only during **dark periods**), which determines **drowsiness** and a **reduction in the core body temperature**.

Zeitgeber (time-giver): external cue that entrains or synchronizes an organism's biological rhythms to the 24h light/dark cycle.

Circadian vs. Diurnal Rhythms

circadian rhythm

Biological process that displays an endogenous, entrainable oscillation of about 24 hours



diurnal rhythms

Processes with 24-hour oscillations (unconfirmed endogenous nature)



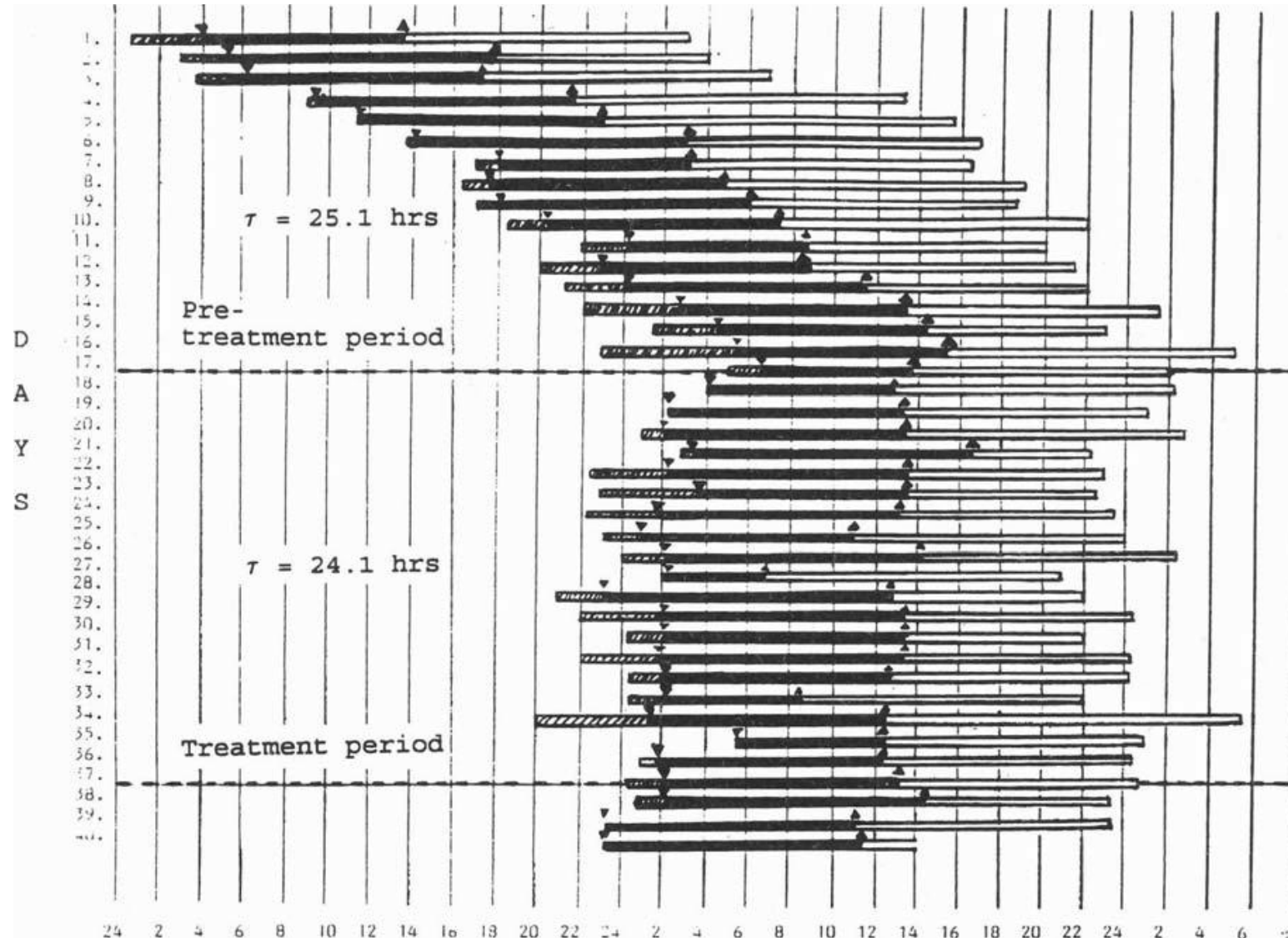
chronobiology

Biological temporal rhythms, such as daily, tidal, weekly, seasonal, and annual rhythms

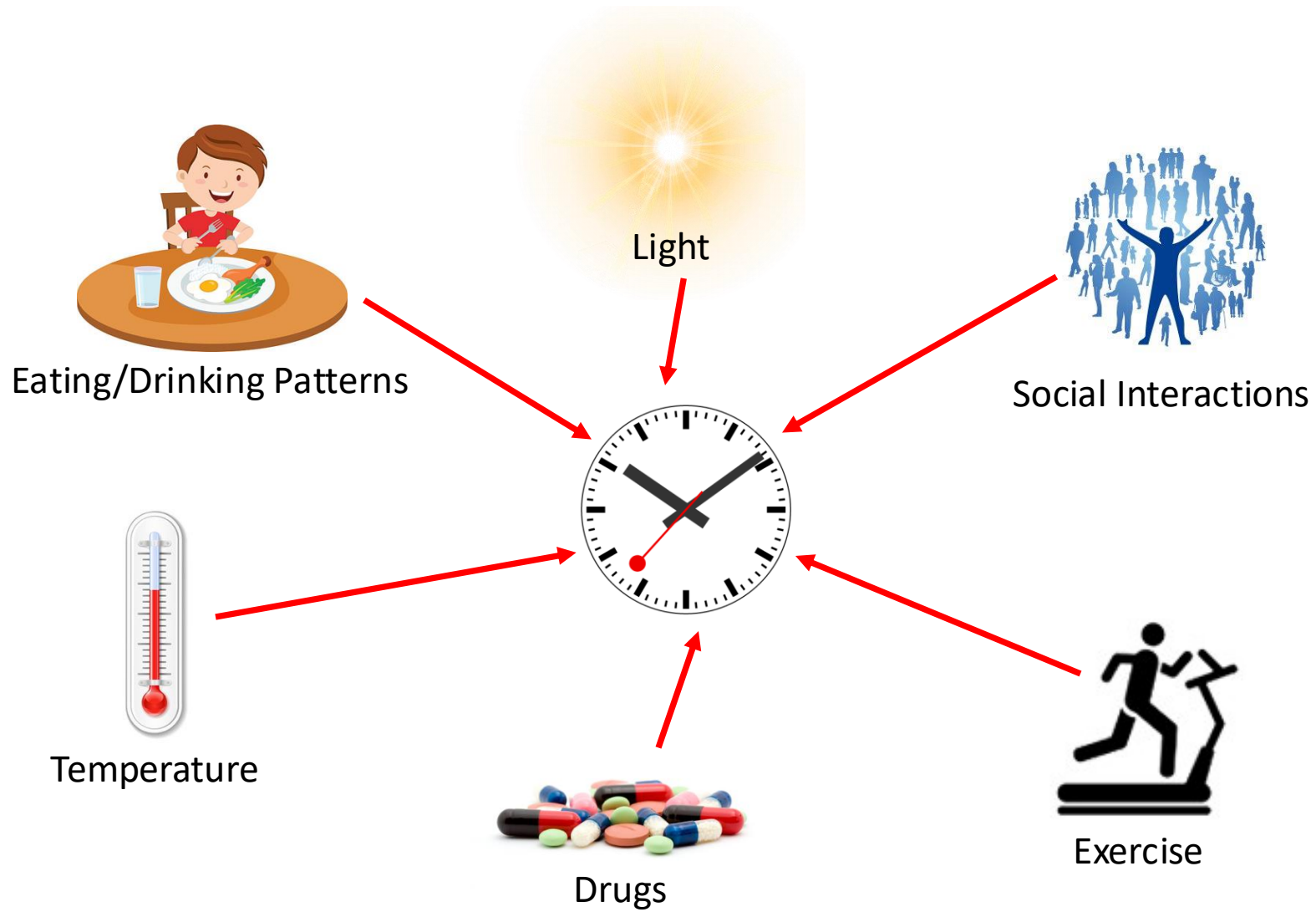


- The rhythm has an endogenous **free-running** period that lasts approximately 24 hours.
- The rhythms are entrainable (**zeitgeber**).
- The rhythms exhibit **temperature** compensation (maintained circadian periodicity).

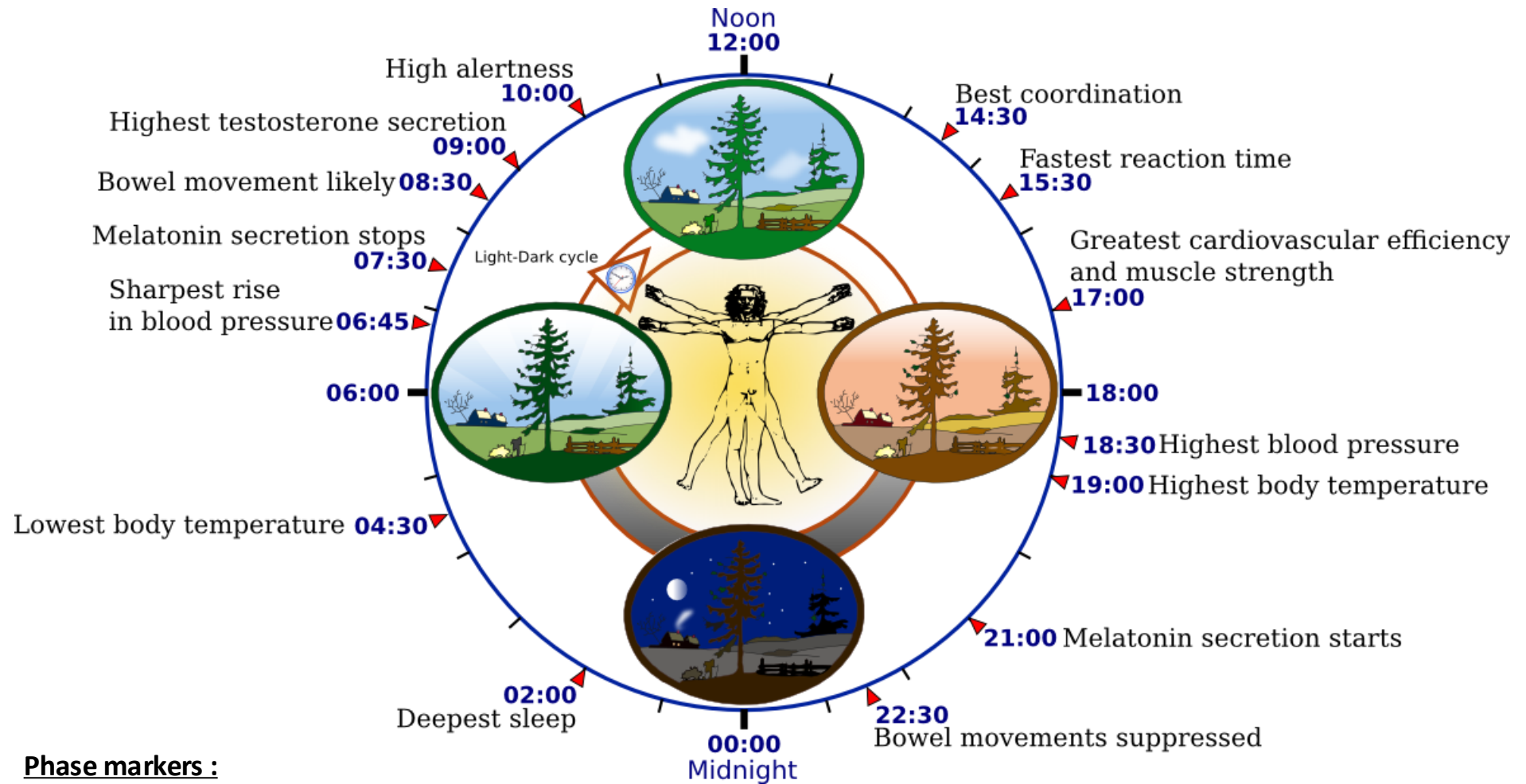
Implications for neuroimaging and behavioral studies



Zeitgebers



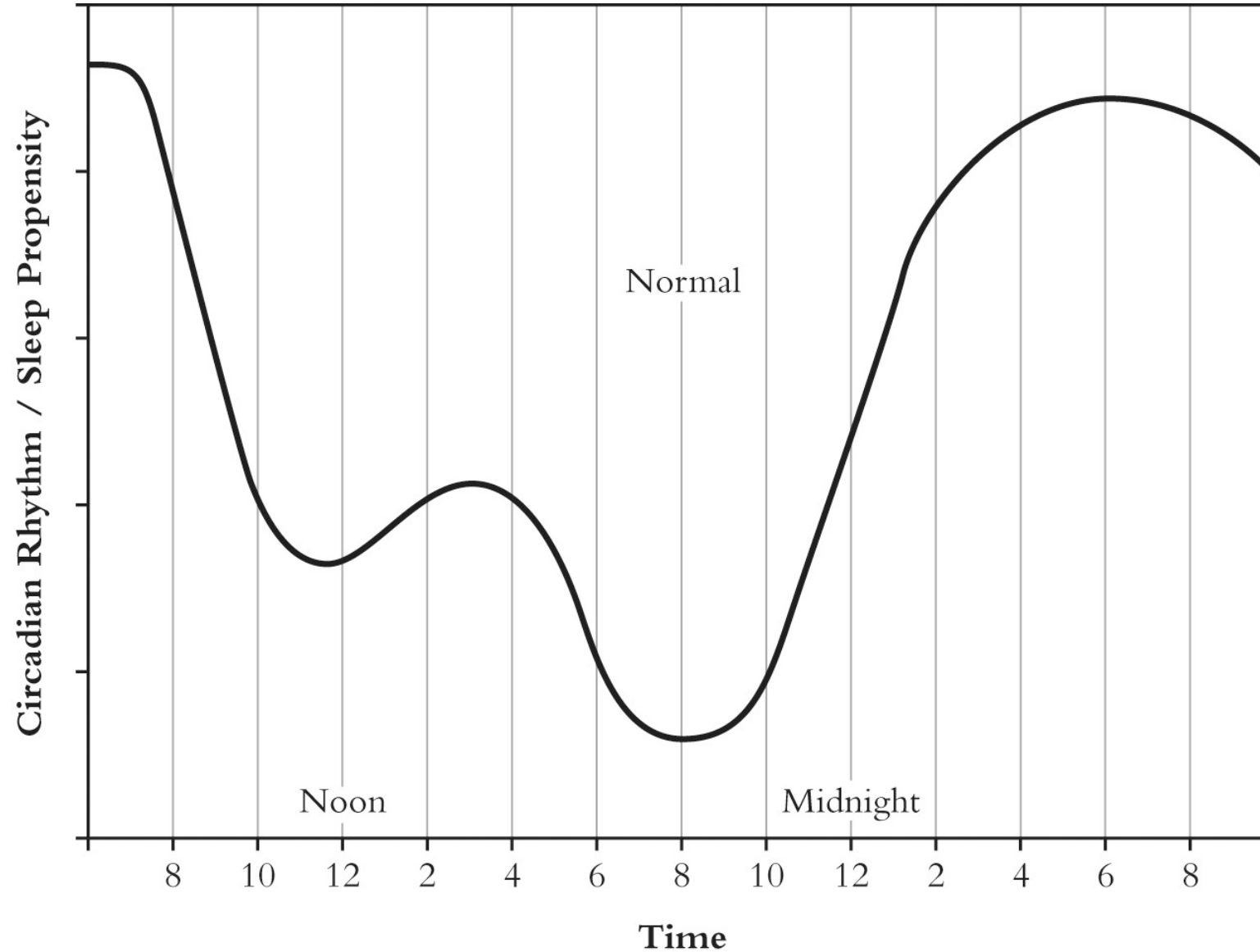
Circadian Regulation



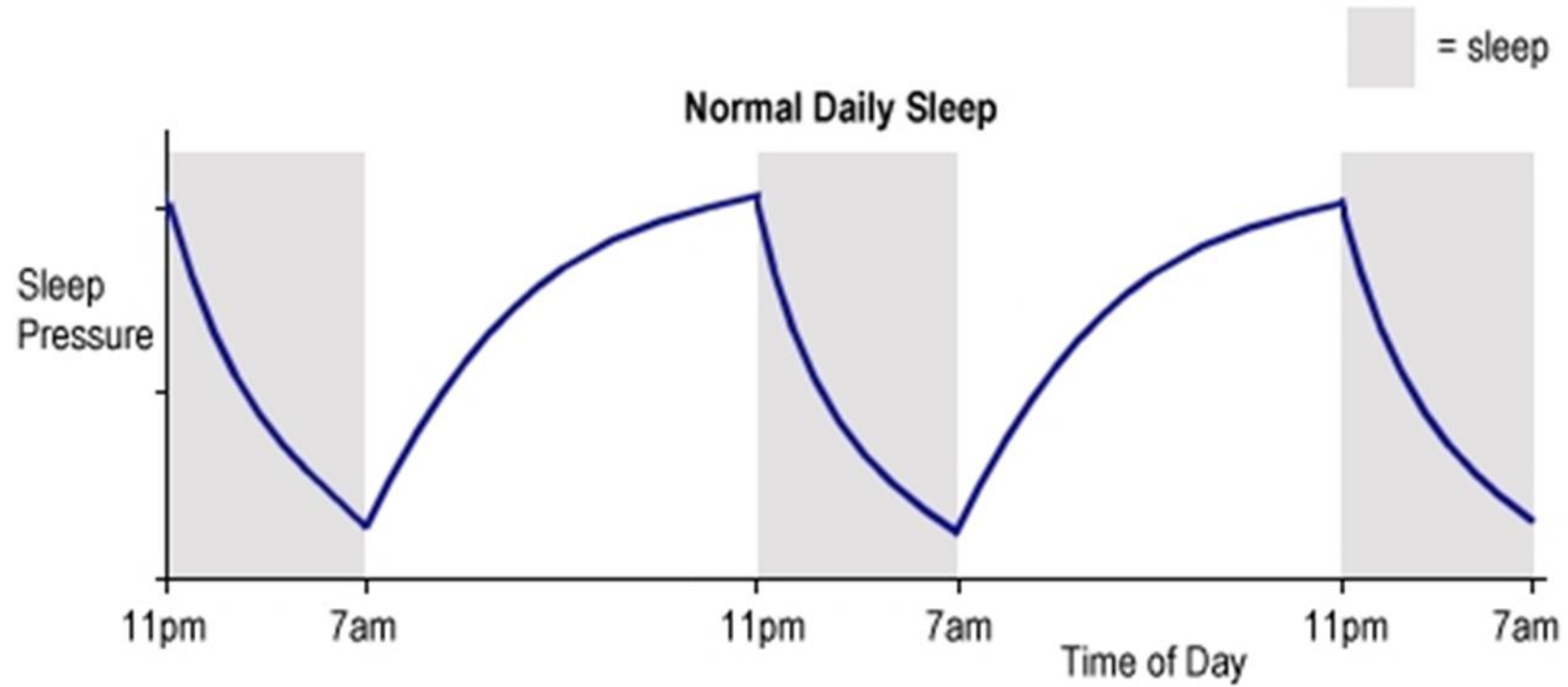
Phase markers :

- *melatonin secretion by the pineal gland*
- *core body temperature minimum*
- *plasma level of cortisol*

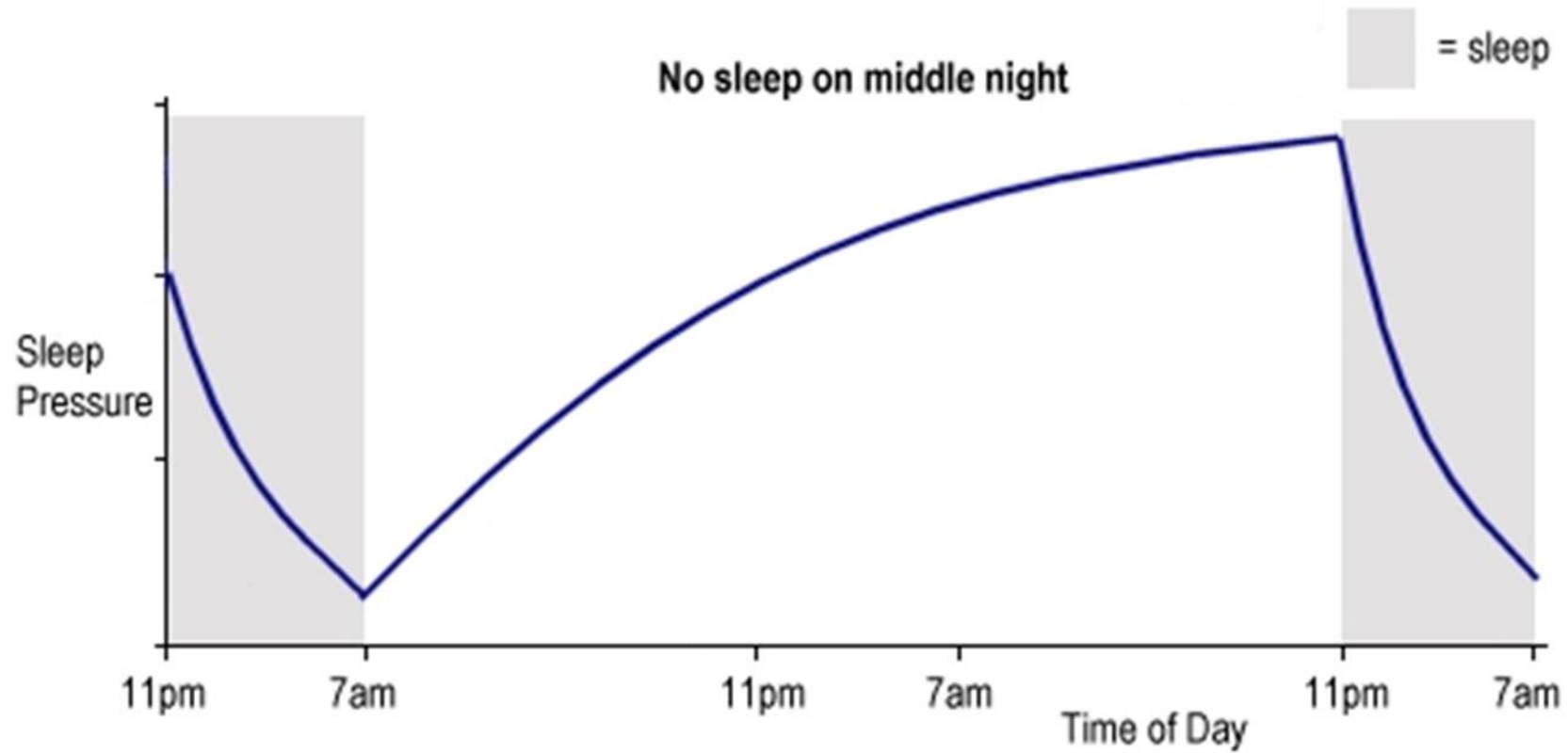
Circadian Regulation of Wakefulness and Sleep



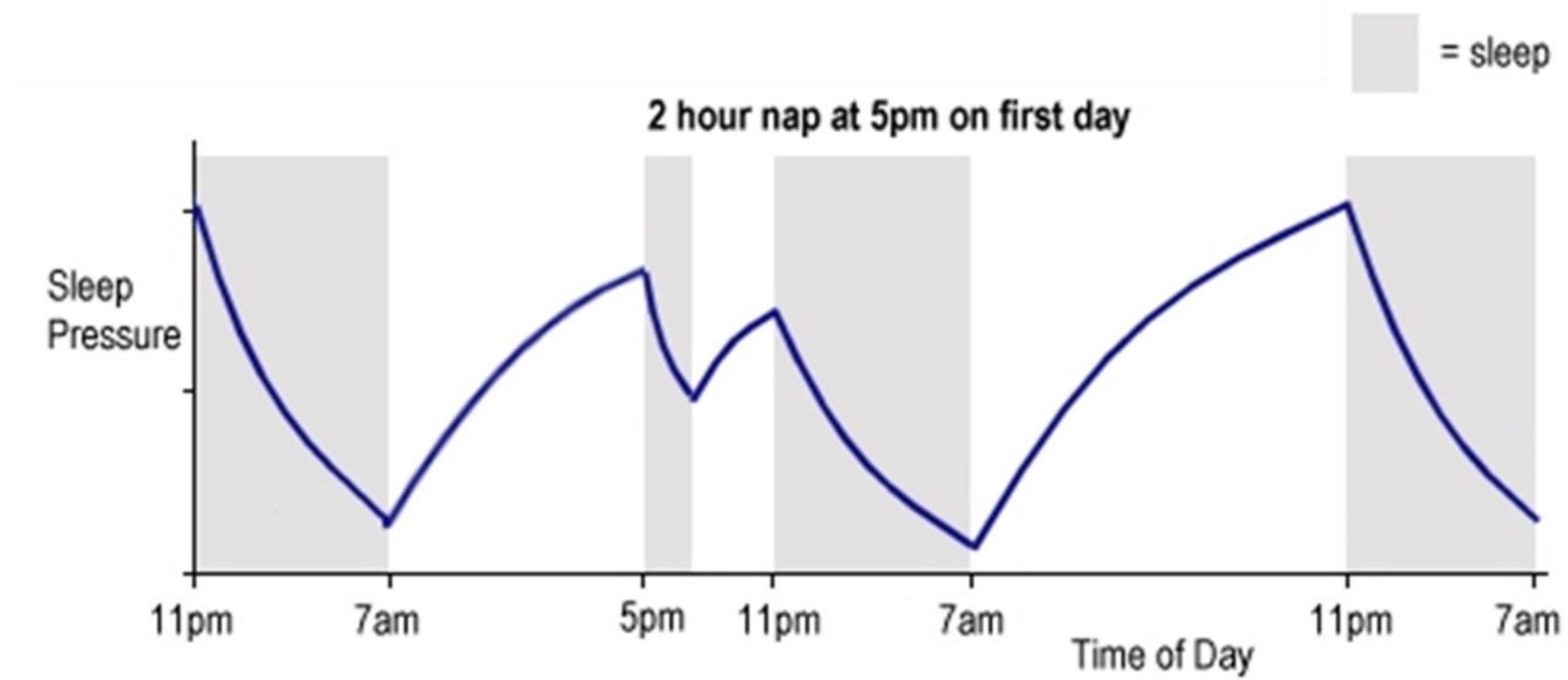
Homeostatic Sleep Regulation



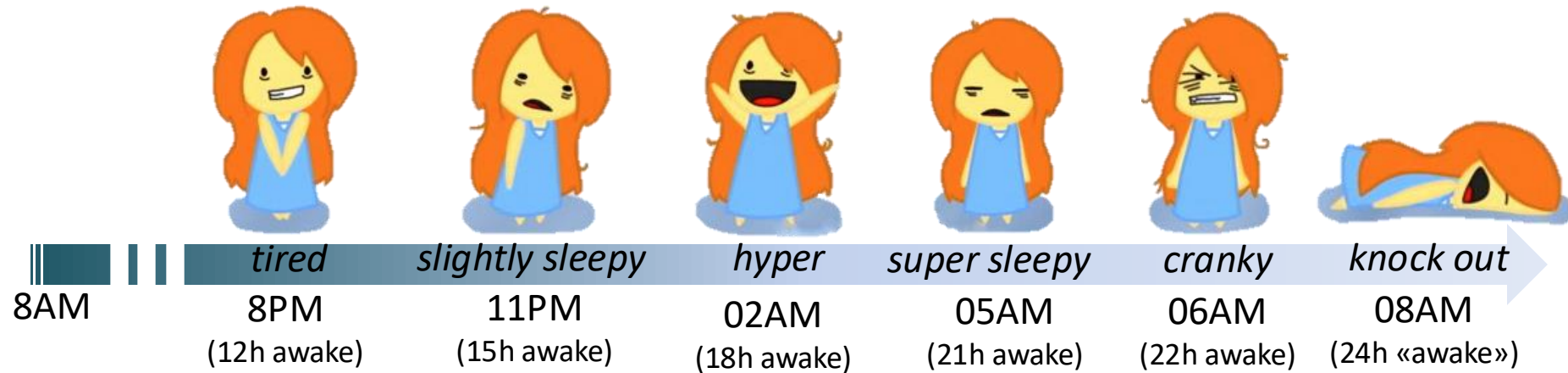
Homeostatic Sleep Regulation



Homeostatic Sleep Regulation



Changes in behavior during (prolonged) Wakefulness



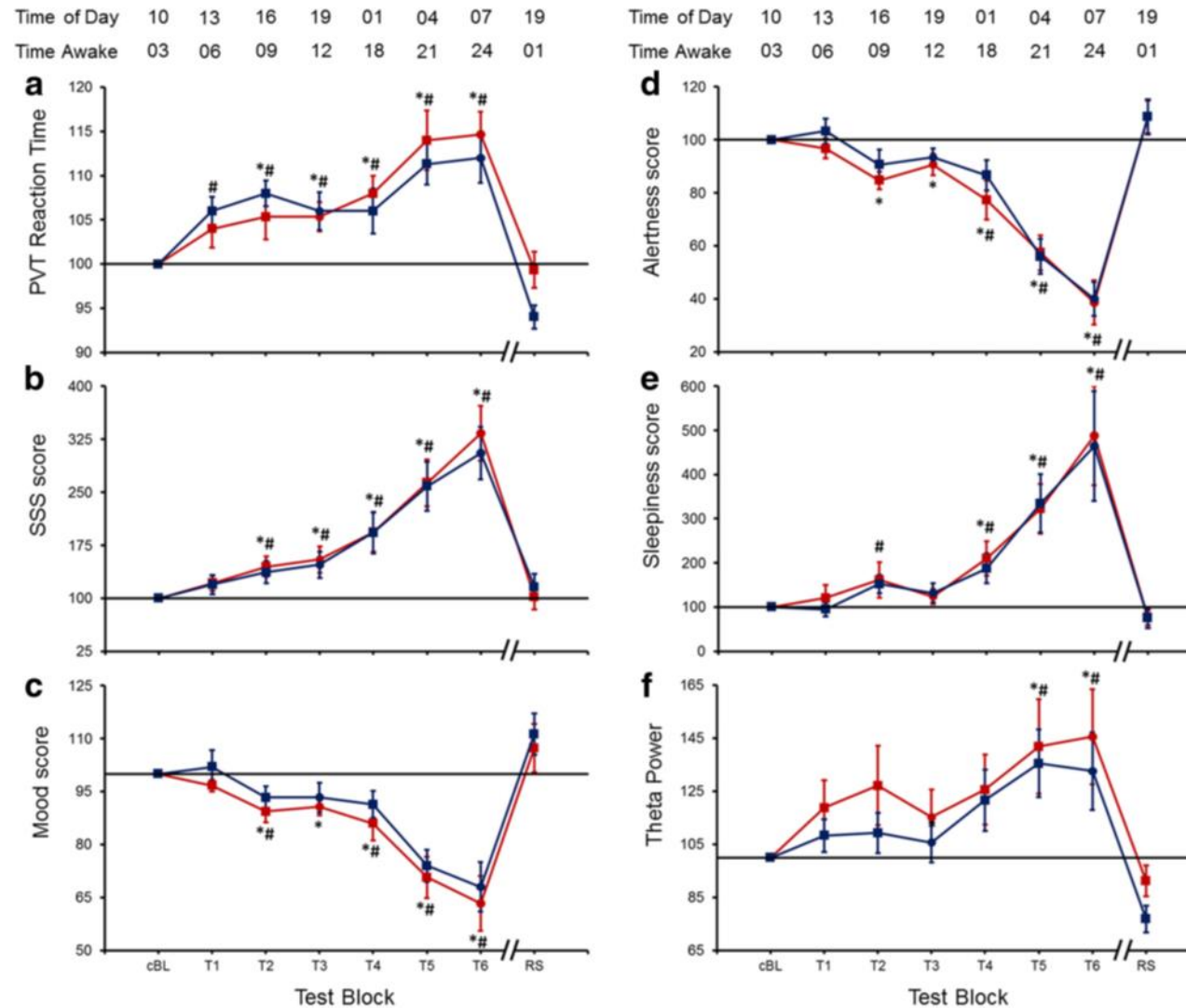
CONFUSION **MEMORY LOSS** **IRRITABILITY**

YAWNING **MEMORY LAPSES** **COGNITIVE IMPAIRMENT**

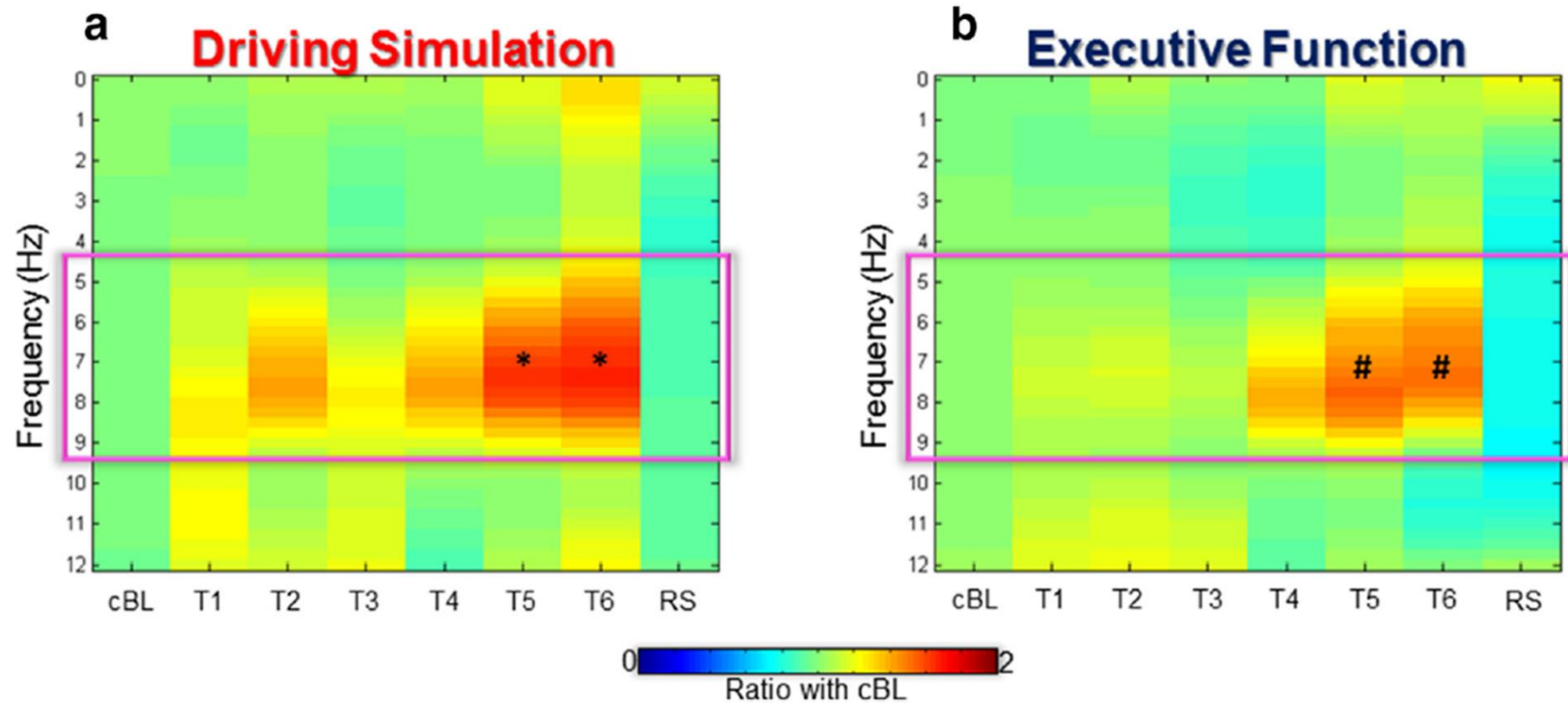
ATTENTION LAPSES **HEADACHES**

IMPAIRED MORAL JUDGMENT **HALLUCINATIONS**

Changes in behavior during (prolonged) Wakefulness

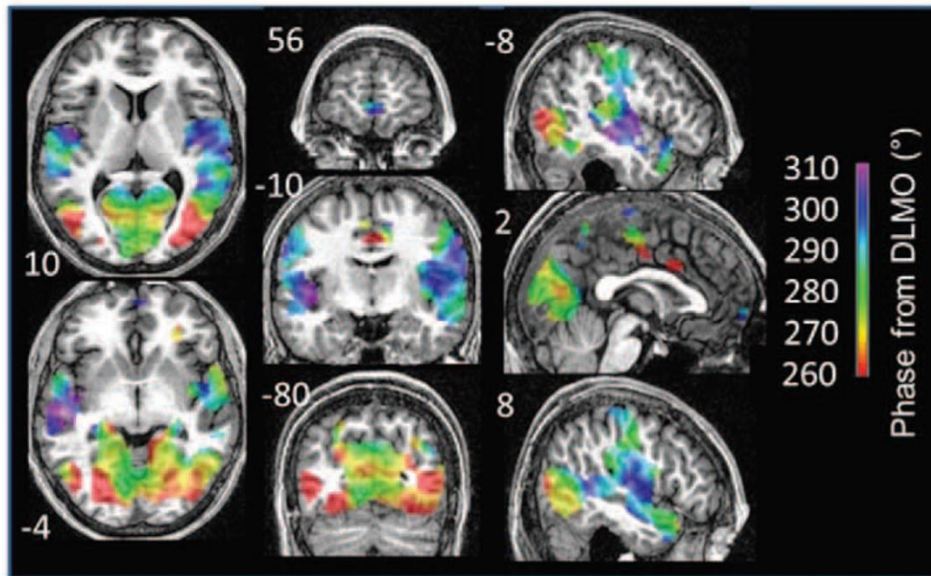


Changes in behavior during (prolonged) Wakefulness

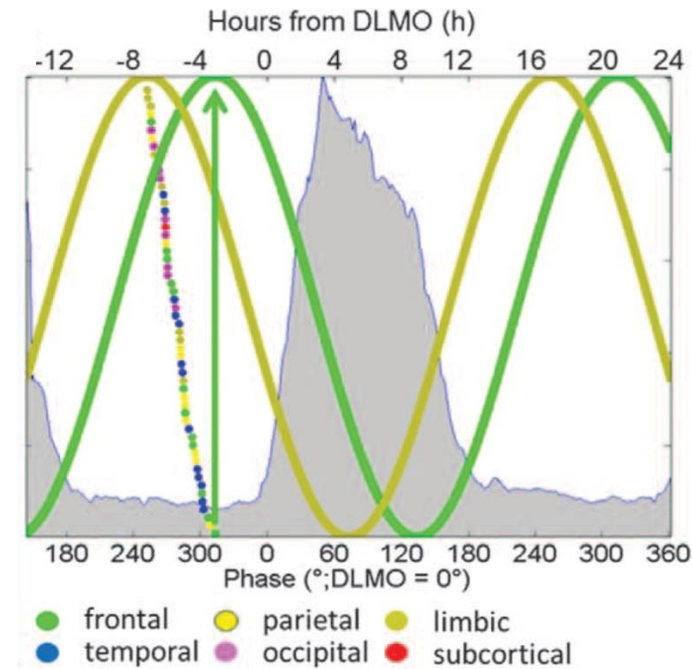


Changes in Brain Activity During Wakefulness

The timing of brain imaging-assessed peak activity varies with brain region and different brain regions are differentially affected by sleep loss and circadian rhythms.

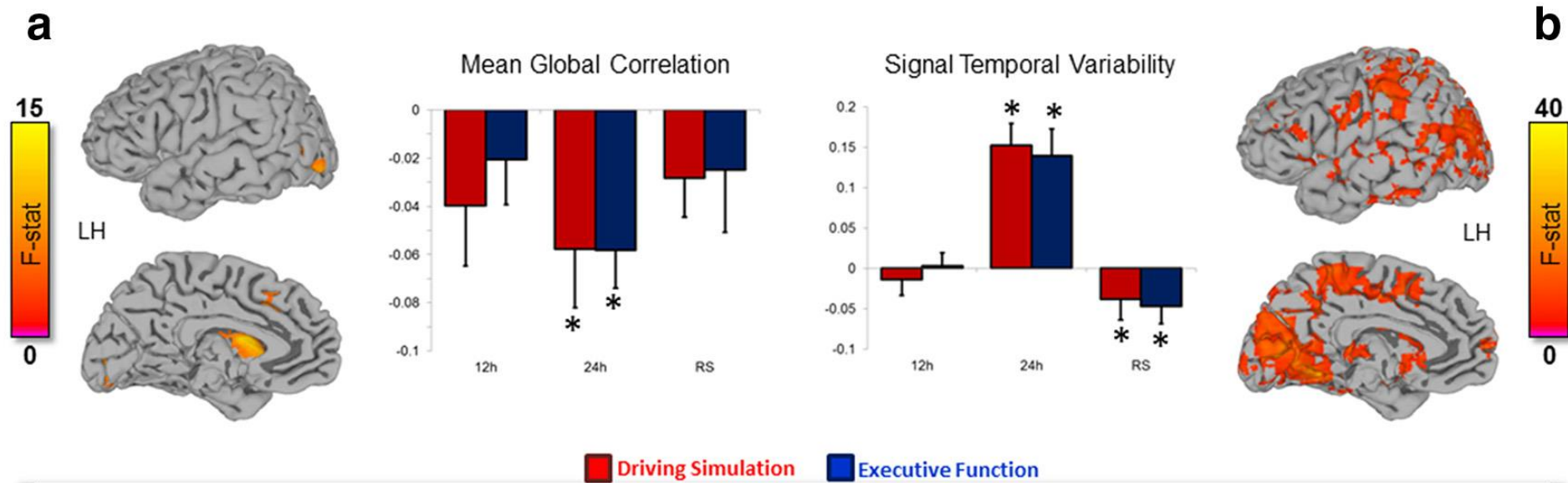


Circadian phase map of brain responses to PVT. The local response phase is displayed according to the color scale (melatonin secretion onset, DLMO = 0°).



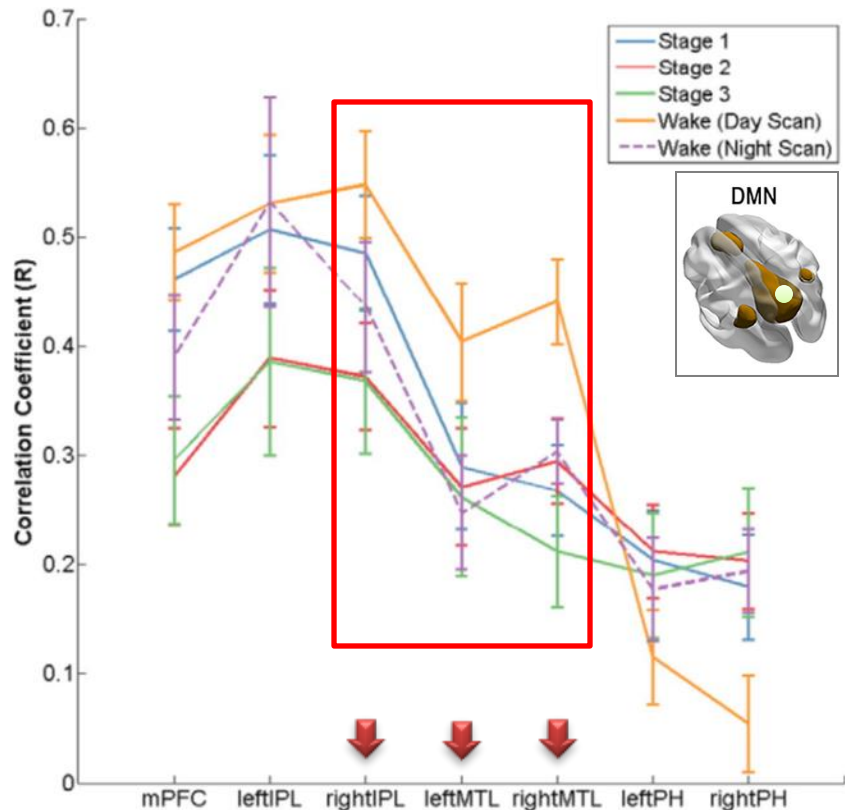
Time courses of 24-hour period responses expressed as phase and approximate hours from DLMO. Mean melatonin profile is shown in gray. Sine waves illustrate the earliest (beige, amygdala) and latest (green, inferior frontal gyrus) response timing.

Changes in Brain Activity During Wakefulness



Prolonged wakefulness is associated with a decrease in the mean strength of coupling among brain areas. The effect was particularly strong in the striatum, an area previously linked to the regulation of the sleep/wake alternation.

Changes in Brain Activity During Wakefulness

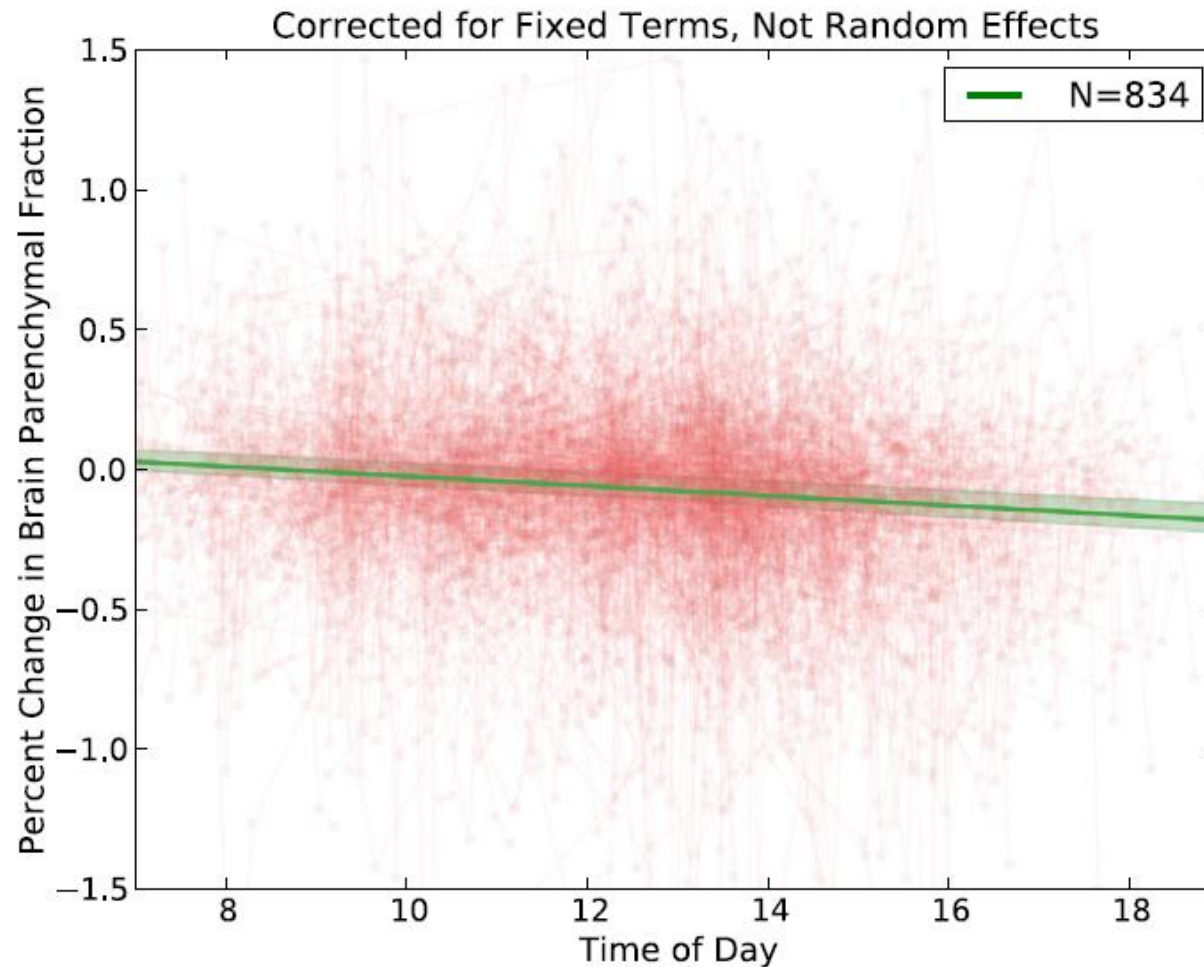


The graph shows the effect of sleep stage on the group average functional connectivity (calculated over multiple 30sec epochs, no overlap) between the PCC and all DMN target ROIs.

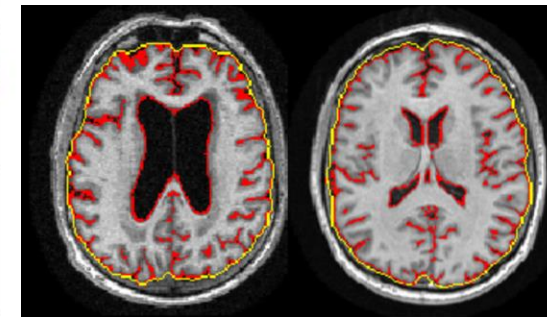
FC of PCC in both periods of wake (day and night) has the greatest temporal variability, with more variability during daytime wake, whereas this is markedly reduced in sleep stage N3 for all regions within the DMN.

1 MRI session during the day
1 MRI session at night (between 22.00-00.00)

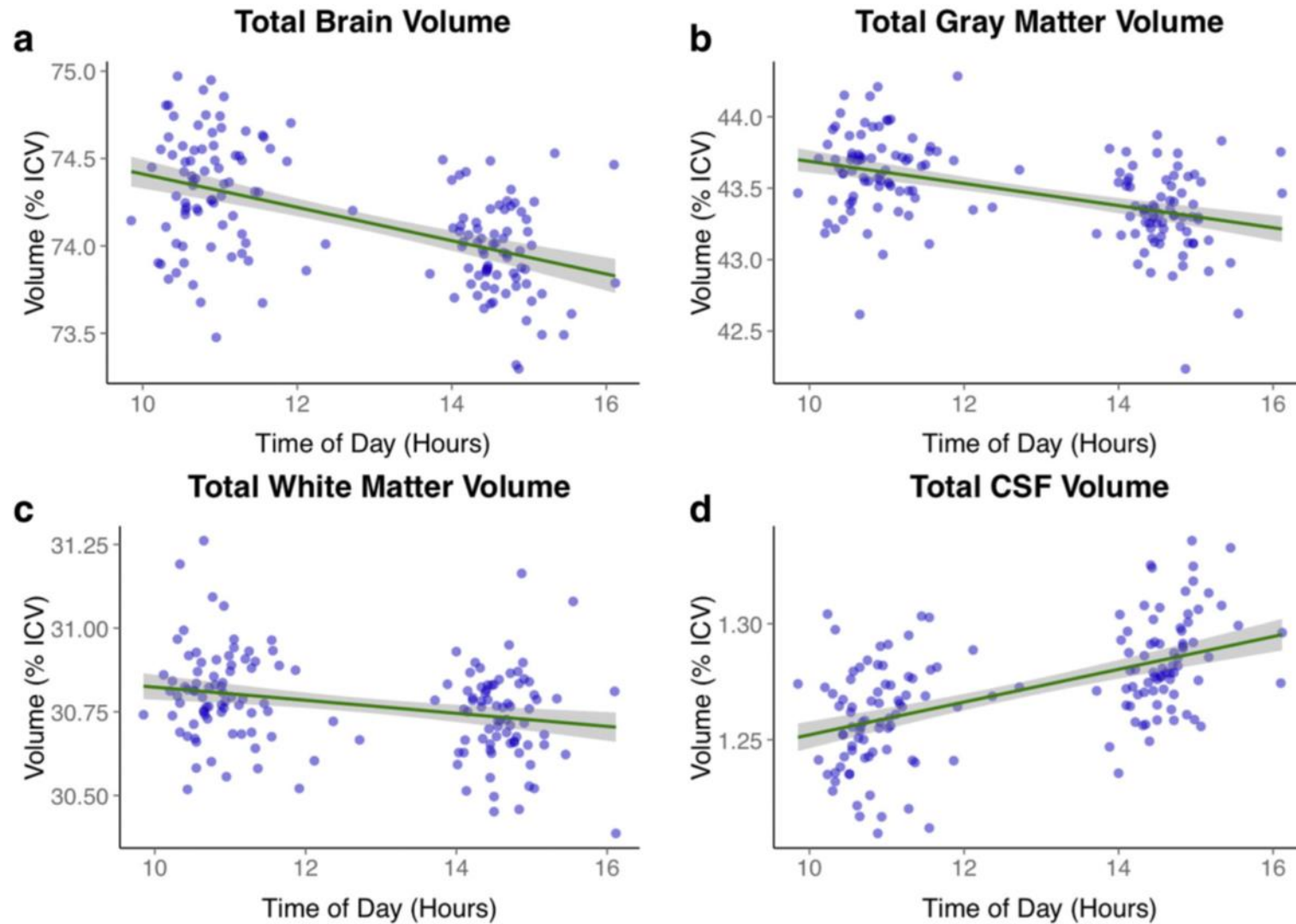
Changes in Brain Structure During Wakefulness



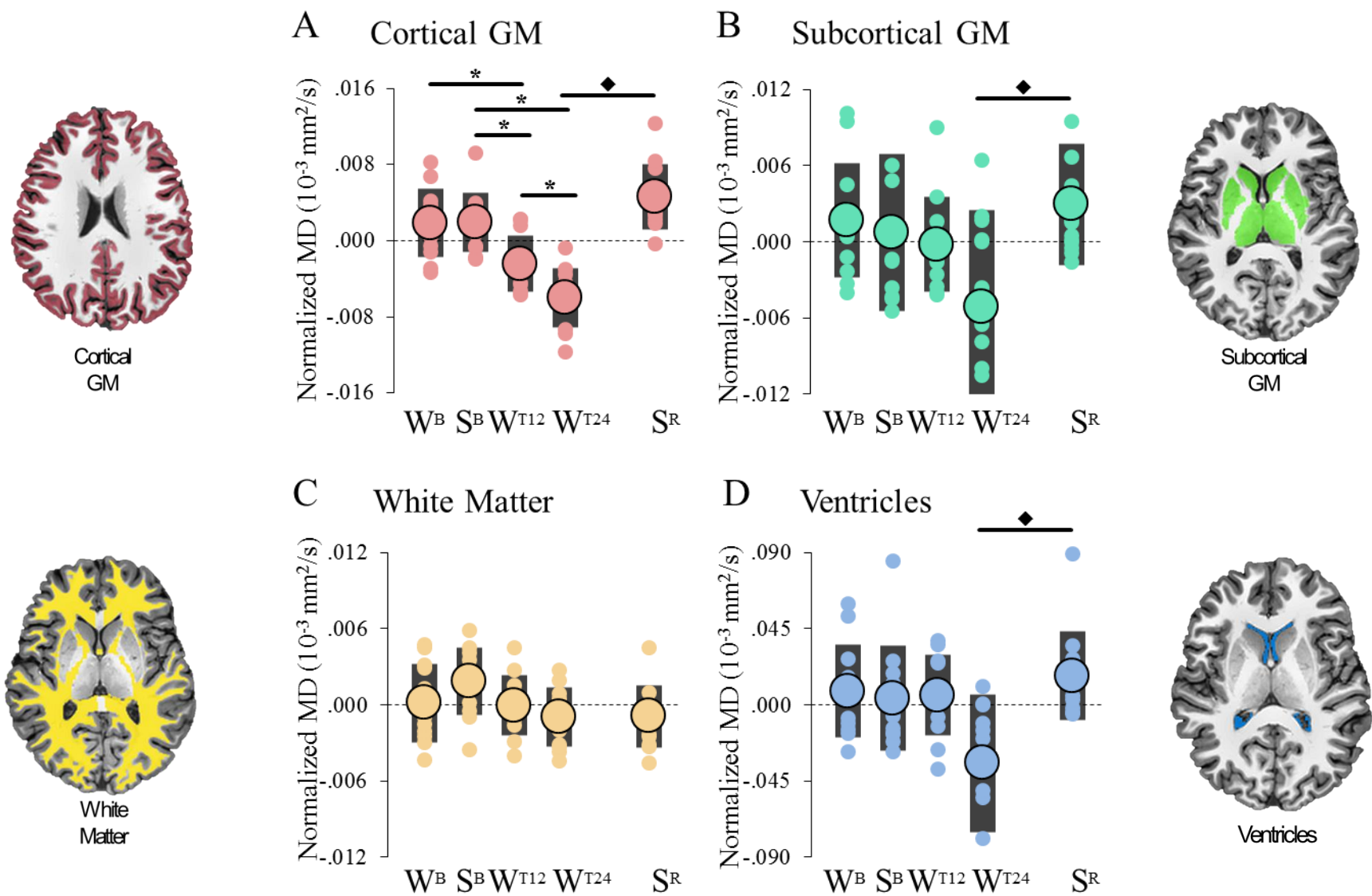
ADNI Dataset
(Alzheimer's
Disease
Neuroimaging
Initiative)



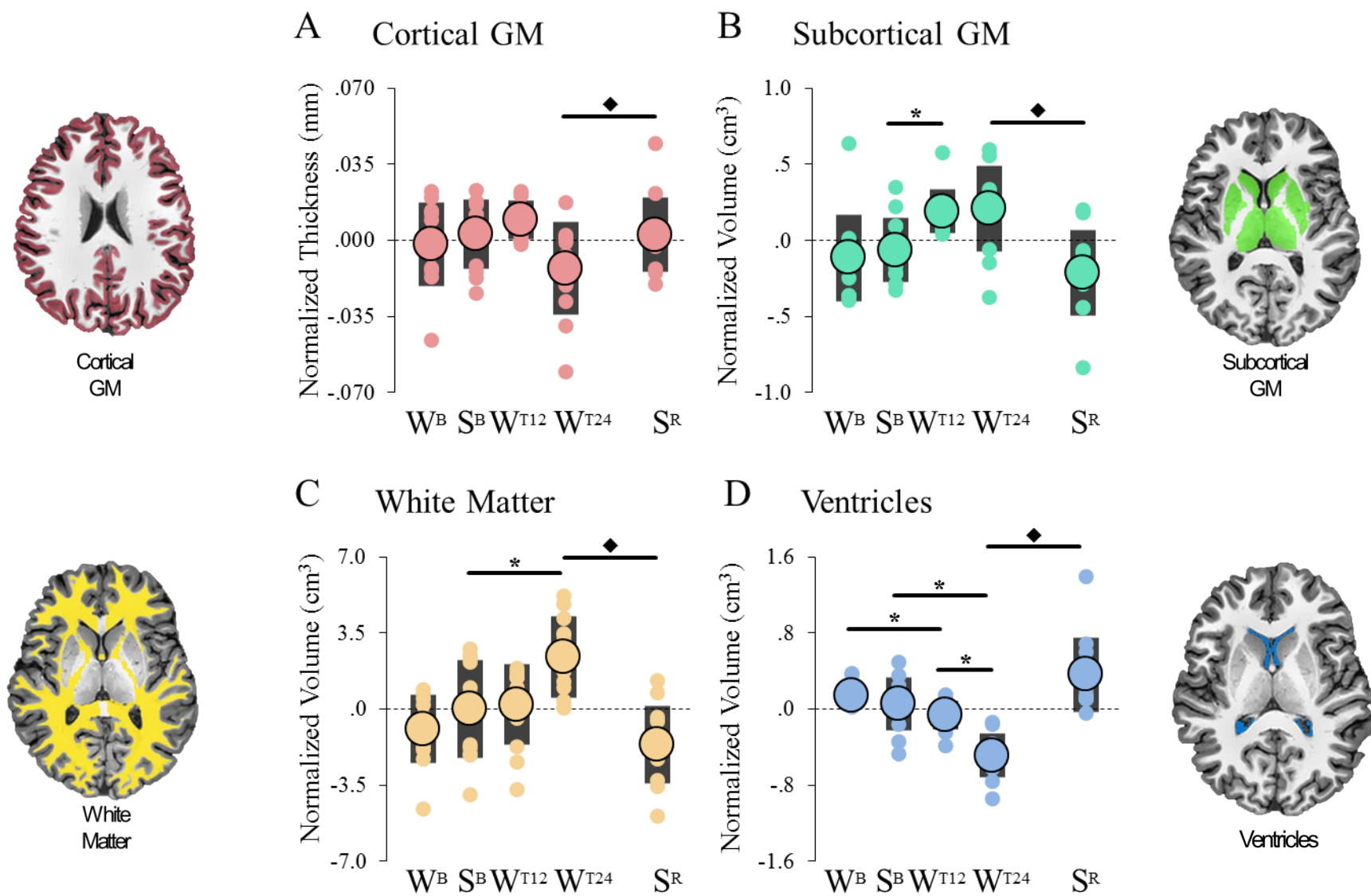
Changes in Brain Structure During Wakefulness



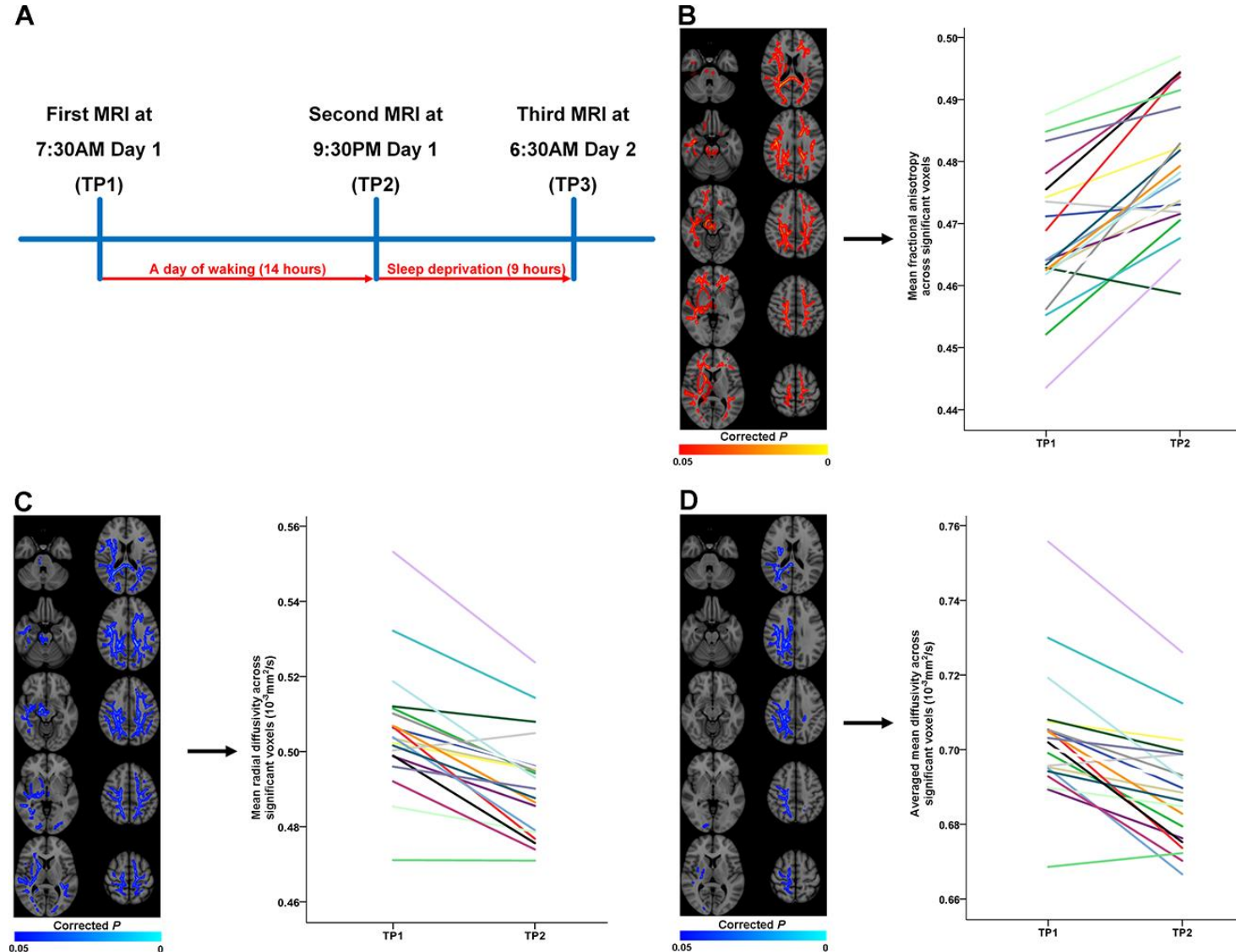
Changes in Brain Structure During Wakefulness



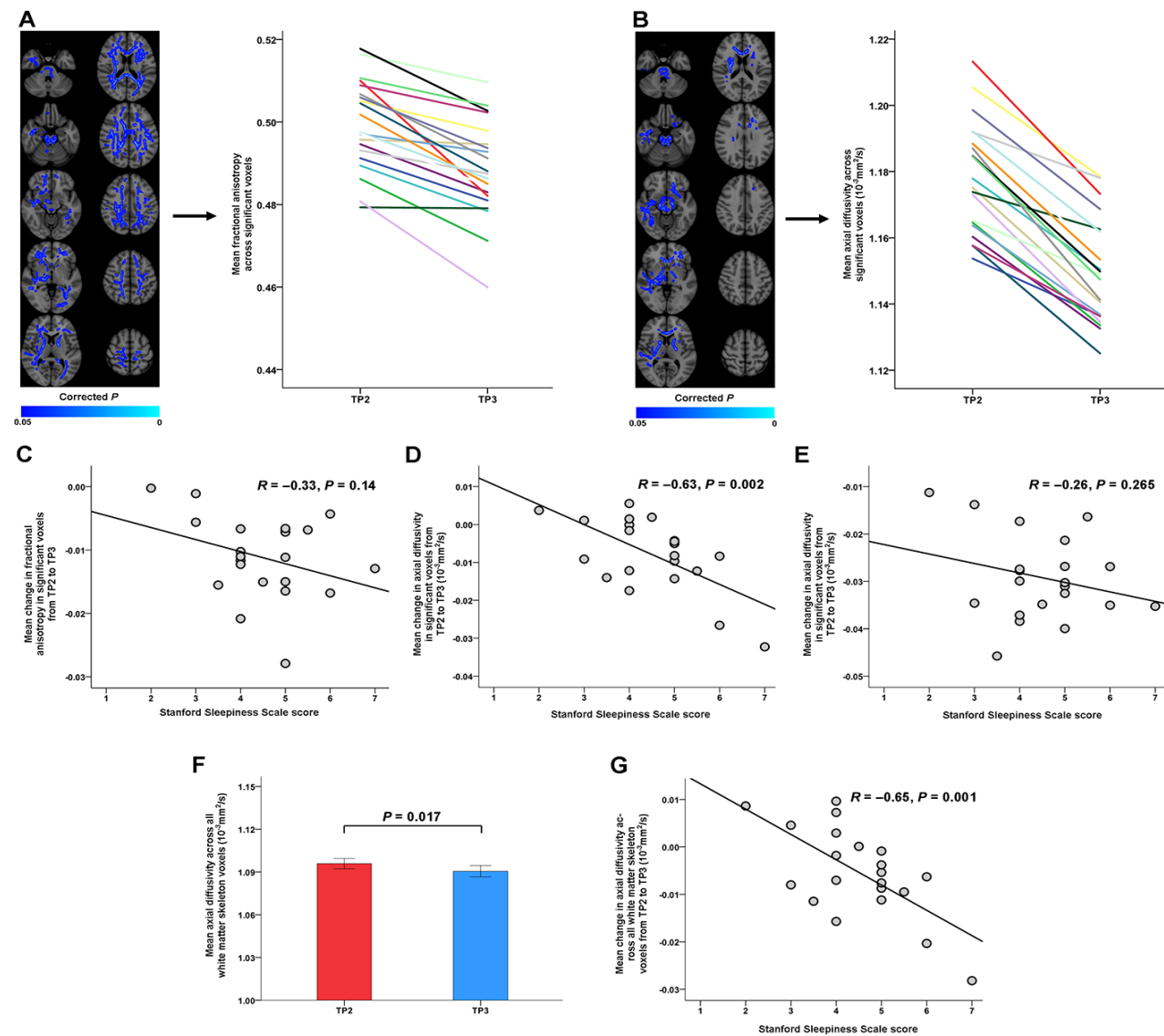
Changes in Brain Structure During Wakefulness



Changes in Brain Structure During Wakefulness



Changes in Brain Structure During Wakefulness



Implications for neuroimaging and behavioral studies

Implications for neuroimaging and behavioral studies

Systematic biases in data collection time may lead to spurious differences between conditions or groups!

Even a difference as small as 3-4 hours may lead to significant effects!

Example

In a study investigating differences between patients and control subjects, patients are examined during the morning, while control subjects are examined during the evening.

Implications for neuroimaging and behavioral studies

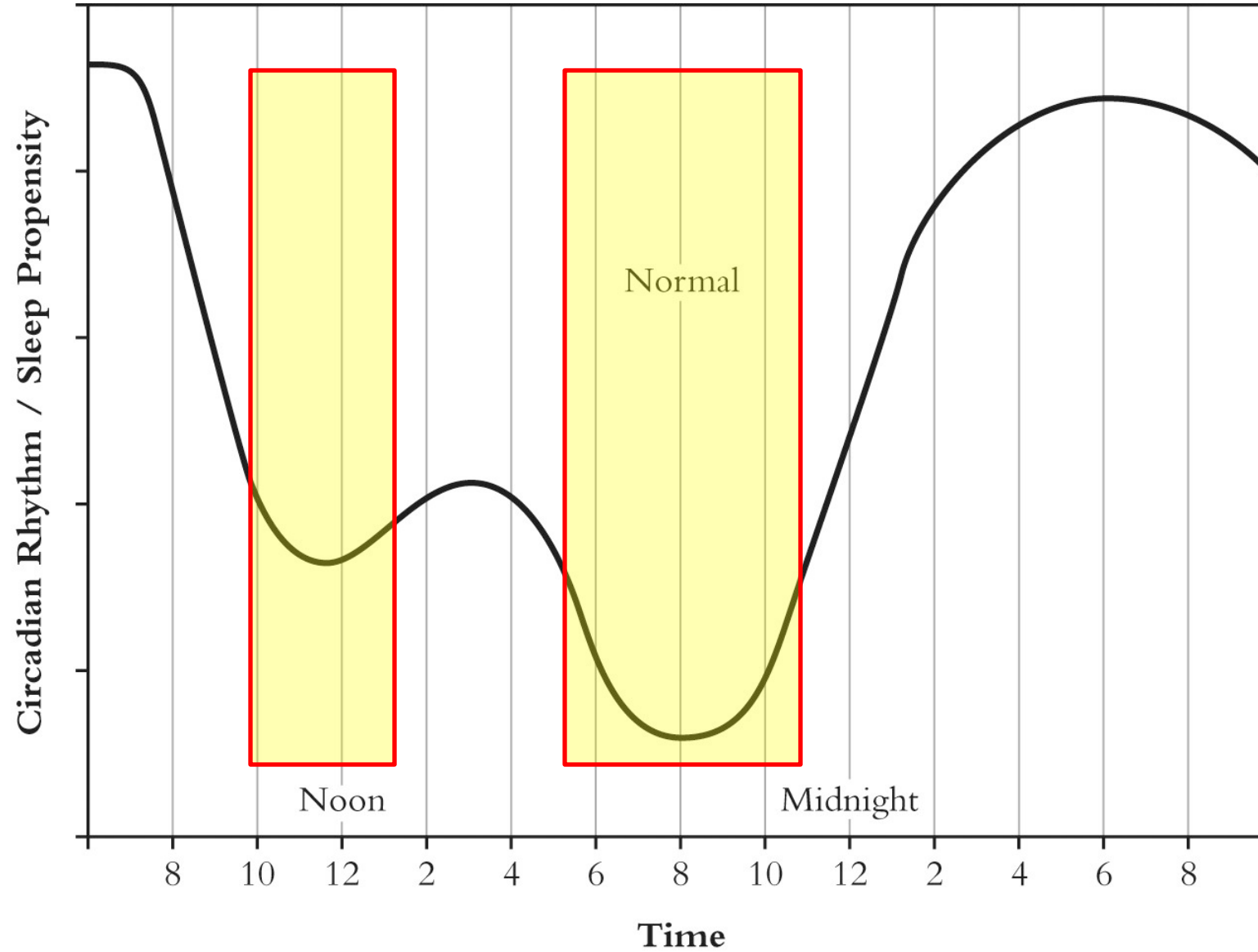
Systematic biases in data collection time may lead to spurious differences between conditions or groups!

Even a difference as small as 3-4 hours may lead to significant effects!

Solution

All subjects should be examined at around the same time of day (if possible in a time window not larger than 1-2 hours). It is good practice to report in your article at what time of day the data was collected and whether consistency across subjects and groups was ensured!

When data should be collected?



Implications for neuroimaging and behavioral studies

Systematic biases in vigilance levels caused by sleep restriction/deprivation, or by individual differences in circadian rhythms may lead to spurious differences between conditions or groups!

Example

In a study comparing university students and workers, students tend to sleep less during the week and wake-up later in the morning relative to employees.

Or

In a study comparing sighted and blind individuals, the latter have a free-running rhythm with longer-than-normal time constant.

Implications for neuroimaging and behavioral studies

Systematic biases in vigilance levels caused by sleep restriction/deprivation, or by individual differences in circadian rhythms may lead to spurious differences between conditions or groups!

Solution

Overall vigilance and sleep quality may be assessed using simple questionnaires. Actigraphic recordings may be used to ensure no between-group differences in sleep time before the experiment, or to exclude a lack of circadian alignment.

Implications for neuroimaging and behavioral studies



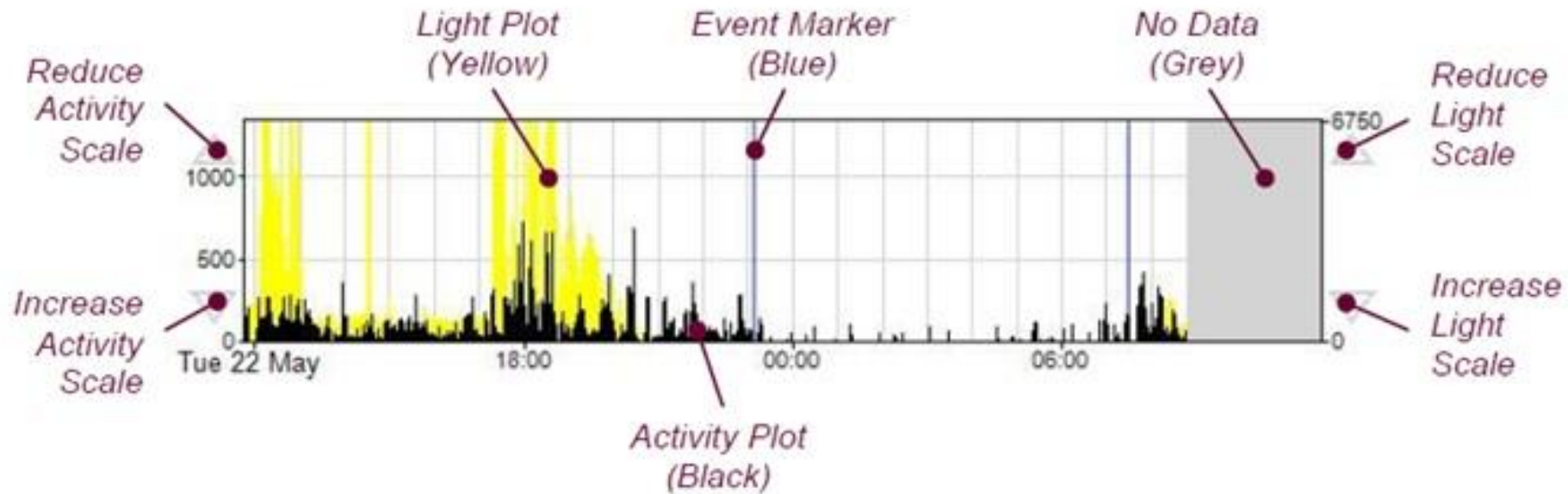
Questionnaires

- **Pittsburgh sleep quality index**: overall sleep quality
- **Epworth Sleepiness Scale**: sleepiness during the day
- **Morningness-Eveningness Questionnaire**: circadian rhythms
- **Stanford Sleepiness Scale**: current vigilance level
- **Karolinska Sleepiness Scale**: current vigilance level

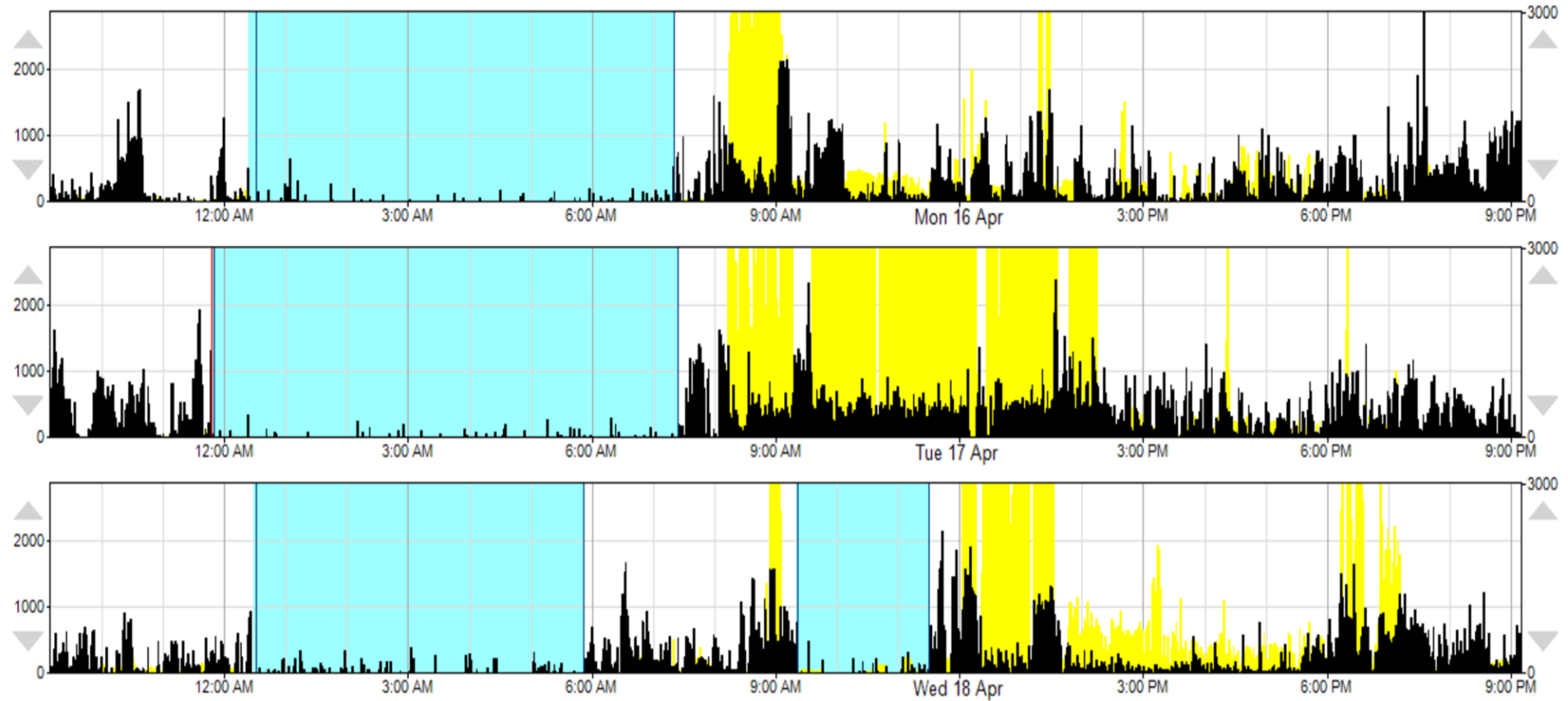
Implications for neuroimaging and behavioral studies



Actigraphy



Implications for neuroimaging and behavioral studies



Implications for neuroimaging and behavioral studies

Sleep Analysis

Start day	4/14/2018	4/14/2018	4/16/2018	4/16/2018	4/18/2018	4/18/2018
Day of week	Saturday	Saturday	Monday	Monday	Wednesday	Wednesday
End day	4/14/2018	4/15/2018	4/16/2018	4/17/2018	4/18/2018	4/18/2018
Lights out	12:30 AM	11:23 PM	12:24 AM	11:48 PM	12:29 AM	9:21 AM
Fell asleep	12:32 AM	11:23 PM	12:32 AM	11:50 PM	12:32 AM	9:22 AM
Woke up	7:19 AM	7:20 AM	7:21 AM	7:25 AM	5:52 AM	11:30 AM
Got up	7:19 AM	7:22 AM	7:21 AM	7:25 AM	5:52 AM	11:31 AM
Time in bed	06:49	07:59	06:57	07:37	05:23	02:10
Assumed sleep	06:47	07:57	06:49	07:35	05:20	02:08
Actual sleep time	05:39	06:37	05:25	06:20	03:29	01:26
Actual sleep (%)	83.3	83.2	79.5	83.5	65.3	67.2
Actual wake time	01:08	01:20	01:24	01:15	01:51	00:42
Actual wake (%)	16.7	16.8	20.5	16.5	34.7	32.8
Sleep efficiency (%)	82.9	82.9	77.9	83.2	64.7	66.2
Sleep latency	00:02	00:00	00:08	00:02	00:03	00:01
Sleep bouts	30	37	32	35	29	14
Wake bouts	29	38	33	35	29	15
Mean sleep bout	00:11:18	00:10:44	00:10:09	00:10:51	00:07:12	00:06:09
Mean wake bout	00:02:21	00:02:06	00:02:33	00:02:09	00:03:50	00:02:48
Immobile mins	368	422	355	406	232	103
Immobile time (%)	90.4	88.5	86.8	89.2	72.5	80.5
Mobile mins	39	55	54	49	88	25
Mobile time (%)	9.6	11.5	13.2	10.8	27.5	19.5
Immobile bouts	37	46	43	42	43	20
Mean immobile bout	00:09:57	00:09:10	00:08:15	00:09:40	00:05:24	00:05:09

Implications for neuroimaging and behavioral studies

In resting-state studies, systematic differences in sleep propensity may lead to spurious differences between conditions or groups!

Example

In a study investigating differences subjects with social anxiety and control subjects, patients have higher levels of anxiety and reduced sleepiness during the study, relative to control subjects.

Implications for neuroimaging and behavioral studies

In resting-state studies, systematic differences in sleep propensity may lead to spurious differences between conditions or groups!

Solution

Ask the subjects if they fell asleep during the scan session?

Implications for neuroimaging and behavioral studies

In resting-state studies, systematic differences in sleep propensity may lead to spurious differences between conditions or groups!

Solution

Ask the subjects if they fell asleep during the scan session?

NOT RELIABLE

Sleep Misperception!

Implications for neuroimaging and behavioral studies

In resting-state studies, systematic differences in sleep propensity may lead to spurious differences between conditions or groups!

Solution

If group-level differences cannot be excluded, the best practice is to monitor the state of vigilance during the MRI session and discard subjects or segments including behavioral sleep.

Implications for neuroimaging and behavioral studies

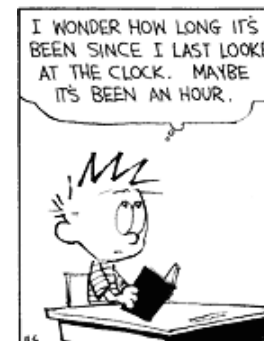
- Eyes-Open resting state paradigm (plus eye-tracking)
Not feasible for paradigms requiring eyes closure



- EEG-fMRI co-registration for sleep scoring
Other physiological parameters (HR variability)



- Shorter fMRI scan sessions
Or long acquisitions but small time-windows



Suggested Readings

Tononi G & Siclari F. (2015). Sleep and Dreaming. The Neurology of Consciousness 2nd Edition. Academic Press. Scientific American, 89-105.

Lewis PA. (2013). The secret world of sleep. Palgrave Macmillan. New York, US

Strata G. (2017). Dormire, forse sognare: Sonno e sogno nelle neuroscienze. Carocci Editore. Roma, IT (*In Italian*)

Haba-Rubio J & Heinzer R. (2016). Je rêve de dormir. Favre. Lausanne, CH (*In French*)

Arnulf I. (2014). Une fenêtre sur les rêves: neurologie et pathologies du sommeil. Odile Jacob. Paris, FR (*In French*)

Sarasso S, Pigorini A, Proserpio P, Gibbs SA, Massimini M, Nobili L. (2014). Fluid boundaries between wake and sleep: experimental evidence from Stereo-EEG recordings. Archives italiennes de biologie, 152(2/3), 169-177.

